

Gert v2 Design Document

Gert Team

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Chapter 1

Overview

1.1 Problem Statement

Operational procedures are the connective tissue of production engineering. When a service degrades, an incident fires, or a change must be applied under pressure, engineers rely on runbooks—step-by-step guides that encode institutional knowledge about how to respond. The research literature is unambiguous: *unstructured runbooks fail*. Static documents drift from reality, skip governance checks, and produce no audit evidence. Teams that operate at scale cannot afford a response that depends entirely on whether the on-call engineer remembers to follow the wiki [9, 22].

The core operational problem is threefold. First, runbooks must be **executable**: a document that cannot be run is a suggestion, not a procedure. Second, runbooks must be **governed**: every action that touches infrastructure must be bounded by policy—command allowlists, environment variable controls, output redaction, and role-based approval gates. Third, runbooks must be **traceable**: every execution must produce an immutable, auditable record that satisfies SOC 2, ISO 27001, and HIPAA audit requirements [23, 10, 26] without manual effort. Without all three properties, organizations face audit failures, runbook drift, and uncontrolled blast radius during incidents.

Gert is designed to solve exactly this problem: a governed, executable, debuggable runbook engine that produces full traceability and evidence by construction.

1.2 Why v2: What v1 Got Right, and What Constrained It

Gert v1 validated the core thesis. It delivered governed execution with command allowlists, environment variable blocking, output redaction, and per-step approval gates. It captured evidence—text, checklists, attachments with SHA256 hashes—and wrote an append-only

JSONL trace for every run. It provided deterministic replay via recorded scenarios, a full terminal UI (Bubble Tea), a VS Code extension over JSON-RPC 2.0, interactive debugging with a step-through REPL, and a `gert test` scenario runner. These are production-grade features that must be preserved in v2.

However, v1 accumulated architectural constraints that limit its evolution. The serve layer (`gert serve`) is tightly coupled to the runtime, making it difficult to add new adapters without modifying core logic. Schema versioning is brittle: `runbook/v0` and `runbook/v1` diverge with no formal compatibility contract. The tool and provider subsystems are functional but have no stable extension contract, making third-party tool development fragile. There is no saga or compensation pattern for rollback on failure, no parallel fan-out execution, no policy-as-code integration, and no OpenTelemetry support for distributed tracing—all of which are now industry-standard in comparable workflow orchestration systems [25, 4, 21].

These constraints are architectural, not cosmetic. A clean-break v2 is warranted.

1.3 Scope

Gert v2 is: a local-first, governed runbook execution engine with a stable component architecture, versioned extension contracts, a first-class governance and policy layer, full traceability by construction, and defined adapter interfaces for TUI, VS Code, and web clients. It ships as a single Go binary with an optional out-of-process extension runtime.

Gert v2 is not: a managed cloud execution service, a general-purpose workflow orchestrator, a CI/CD pipeline system, a SOAR platform, or a replacement for systems like Temporal, Argo, or PagerDuty Runbook Automation [25, 4, 19]. It does not provide a central policy server, a shared execution registry, or multi-tenant execution isolation.

1.4 Design Priorities

- **A core runtime isolated from presentation adapters.** The execution engine—parser, planner, state machine, governance enforcer, trace writer—must have no dependency on any presentation layer. TUI, VS Code, and web clients attach via a stable event subscription API; they observe execution but never own it. This isolation is the prerequisite for every other architectural goal.
- **Explicit and versioned contracts for schema, tools, and events.** Every interface that crosses a component boundary must be named, versioned, and documented. The

runbook schema carries an `apiVersion` field with formal compatibility guarantees. Tool definitions carry a version. The runtime event stream has a defined schema with ordered delivery guarantees. Breaking a contract requires a version increment, not a silent edit.

- **A secure extension runtime using out-of-process plugins.** Extensions run in isolated processes and are granted capabilities explicitly. No extension receives ambient authority. The capability model—what an extension may register, observe, or invoke—is a named, versioned list that the host enforces at admission time. This pattern follows the principle of least privilege endorsed by NIST SP 800-53 [14] and applied in production by systems such as Kubernetes admission controllers [8] and OPA Gatekeeper [15].

1.5 Document Roadmap

This document is organized as follows. §2 defines measurable goals and explicit non-goals, establishing scope boundaries for implementors. §3 describes the five primary components and the dependency rules between them. §4 specifies the v2 runbook schema, including versioning strategy, field inventory, and migration path from v1. §5 defines the extension runtime: capability taxonomy, handshake protocol, and trust model. §6 specifies the tool runtime: discovery, invocation, transport contracts, and governance integration. §7 defines the runtime event model: event schema, ordering guarantees, delivery semantics, and replay contract. §8 specifies the security and trust model: threat model, policy enforcement, supply chain controls, and audit requirements. §9 defines the testing and acceptance strategy: contract test categories, coverage requirements, and acceptance criteria. §10 documents open architectural questions with options and resolution criteria.

1.6 Success Definition

Gert v2 is complete when: (1) a v1 runbook can be migrated to v2 schema via `gert migrate` and executed without manual fixup; (2) the TUI, VS Code extension, and a third-party adapter can all be built against the published event and API contracts without importing internal packages; (3) the governance layer—allowlists, denylists, env blocking, redaction, approval gates—is enforced for every execution mode including extension-invoked runs; (4) every execution produces an append-only, crash-safe JSONL trace that is structurally valid

against the v2 trace schema; and (5) **gert test** passes a scenario suite covering all step types, governance rules, and evidence capture cases.

Chapter 2

Goals and Non-Goals

2.1 Goals

The following goals are the binding commitments of the v2 design. Where possible, each goal is stated as a verifiable criterion rather than an aspiration.

- G1. Preserve and formalize the governance layer.** Every capability that made v1 operationally useful—command allowlists and denylists, environment variable blocking, output redaction with regex patterns, and per-step role-based approval gates—must be present in v2 with identical or stronger semantics. Governance is gert’s core differentiator [19, 2]: approval gates integrated directly into step definitions. v2 must additionally support RBAC scoping on runbook execution: which identities may run which runbooks, and under what conditions.
- G2. Preserve the append-only, crash-safe, tamper-evident trace.** The v2 execution trace must be an append-only JSONL file. Each line is a self-describing JSON object with a unique event ID (UUID), ISO 8601 UTC timestamp [12], actor identity, step ID, and structured payload. The file must be crash-safe (partial writes must not corrupt prior entries) and must support optional cryptographic signing to satisfy SOC 2 and ISO 27001 (A.12.4) audit requirements [23, 10]. This is a launch blocker: no trace, no compliance.
- G3. Preserve evidence capture, deterministic replay, TUI, VS Code extension, and gert test.** The following v1 capabilities must be functionally preserved in v2: evidence capture with text, checklists, and SHA256-hashed attachments; scenario-based deterministic replay (`-mode replay`); the Bubble Tea terminal UI [6]; the VS Code extension powered by the JSON-RPC server [11, 13]; and the `gert test` scenario

runner with its assertion engine. These features represent validated operational value. Their absence in v2 is a regression, not a simplification.

- G4. Establish stable, versioned extension contracts.** Tool definitions (`.tool.yaml`), input providers (`.provider.yaml`), and out-of-process extensions must bind to named, versioned contracts. A host version bump that breaks an extension contract requires a major version increment and a migration path. The capability taxonomy—what an extension may register, observe, and invoke—must be a published, enumerated list, not an implicit set of permissions inferred from code. This pattern is established by VS Code’s extension API [13] and Kubernetes admission controller design [8].
- G5. Introduce saga and compensation for multi-step rollback.** v2 must support explicit compensation registration: when a forward step completes, the runbook may declare a compensating action to execute in reverse order on failure. This is the industry-standard saga pattern, implemented in production by Temporal [25] and AWS Step Functions [1]. Without compensation, partially-executed runbooks that modify infrastructure leave systems in undefined intermediate states.
- G6. Integrate OpenTelemetry for distributed tracing.** Every runbook execution must emit OpenTelemetry spans [18]: one root span per run, one child span per step, with W3C Trace Context [27] (`traceparent` / `tracestate`) propagated to tool invocations. Correlation IDs must appear in both the JSONL trace and the OTel span attributes, so that run traces can be joined with infrastructure observability data in Jaeger [7], Zipkin [29], or any OTLP-compatible backend. This is the current industry standard for operational visibility.
- G7. Decouple adapters from core via a typed event API.** The TUI, VS Code extension, and any future web adapter must attach to the core exclusively via a typed event subscription interface. No adapter may import internal runtime packages or call runtime methods directly. The event schema must be versioned: adapters declare which schema version they consume, and the host rejects incompatible connections. This constraint is necessary for gert to support concurrent adapters (e.g., TUI and VS Code running simultaneously against a single run).
- G8. Establish migration compatibility for v1 runbooks.** The command `gert migrate` (v2 edition) must produce a valid `runbook/v2` document from any well-formed `runbook/v0` or `runbook/v1` input with no manual fixup required for standard runbooks. The v2 runtime should additionally support a compatibility execution

mode that accepts v1 runbooks without prior migration, so that existing users are not blocked from adopting v2 incrementally.

2.2 Non-Goals

The following are explicitly outside the scope of gert v2. These boundaries are feature-scoped, not process-scoped, and are intended to prevent scope creep during implementation.

- NG1. Gert v2 does not provide a managed cloud execution service.** There is no gert-as-a-service, no central execution registry, and no SaaS offering. Gert v2 is a local-first binary. Cloud scheduling, multi-tenant isolation, and remote execution orchestration are concerns for a hypothetical v3 or a separate product.
- NG2. Gert v2 does not implement a general-purpose workflow orchestrator.** Gert is not Temporal, Argo, or Prefect [25, 4, 21]. It does not support distributed workers, fan-out parallelism across remote nodes, or arbitrary DAG execution. Parallel execution within a single local run is a stretch goal; cross-host parallelism is explicitly out of scope.
- NG3. Gert v2 does not ship a built-in policy server.** While v2 defines governance semantics and may integrate policy-as-code hooks [16], it does not bundle or operate an OPA server. Policy evaluation happens at the host process level. Central policy distribution, policy bundles over HTTP, and multi-host policy synchronization are out of scope.
- NG4. Gert v2 does not define an extension marketplace or signing authority.** v2 will define an extension signing *mechanism* (a verifiable public key model), but it will not operate a registry, curate extensions, or act as a certificate authority. Extension distribution is the responsibility of the extension author.
- NG5. Gert v2 does not guarantee binary backward compatibility with v1 internal packages.** v2 is a clean-break architecture. The internal Go package structure, private interfaces, and runtime internals are not subject to backward compatibility guarantees. Only the documented public contracts—the runbook schema, the tool schema, the JSON-RPC API surface, the extension protocol, and the JSONL trace schema—carry compatibility commitments.

Chapter 3

Architecture

This chapter defines the v2 system architecture: the five primary components, their interfaces and invariants, the complete data flow from runbook invocation to trace archive, the execution lifecycle, the concurrency model, and how the `gert serve` JSON-RPC layer integrates as a first-class adapter.

The single most important architectural rule is: **dependencies flow inward toward the Core Domain**. The Runtime Core knows nothing about TUI, VS Code, or JSON-RPC. Adapters know nothing about execution semantics. Every component boundary is enforced by a Go interface; no component imports a package that is “above” it in the dependency graph.

3.1 Primary Components

3.1.1 Parser

Responsibility. The Parser owns the transformation of a raw YAML document into a validated, semantically-resolved `ParsedRunbook` value. It does *not* own execution, planning, tool discovery, or input resolution. It validates schema structure and surface-level semantic rules (duplicate step IDs, unknown step types, malformed Go template expressions) but intentionally defers deep semantic validation (e.g., whether a `from:` binding can actually be resolved) to later pipeline stages.

Interface.

```
// Parser transforms raw YAML bytes into a validated ParsedRunbook.
type Parser interface {
    // Parse reads YAML from r, validates against the v2 JSON Schema,
```

```

    // applies structural semantic checks, and returns a ParsedRunbook.
    // Errors carry structured diagnostics (file, line, column, message).
    Parse(ctx context.Context, r io.Reader, opts ParseOptions) (*ParsedRunbook, error)
}

type ParseOptions struct {
    // SchemaVersion overrides the apiVersion field for compatibility shims.
    SchemaVersion string
    // StrictMode causes deprecation warnings to be surfaced as errors.
    StrictMode bool
}

// ParsedRunbook is an immutable, fully-validated representation of a runbook.
// All fields have been structurally validated. Templates are not yet evaluated.
type ParsedRunbook struct {
    Meta          RunbookMeta
    Governance GovernancePolicy // nil if no governance block
    Steps         []ParsedStep
    Imports       map[string]string // alias -> path (not yet resolved)
    Tools         []string          // tool names (not yet discovered)
    Schema        SchemaRef          // version and $schema URL
}

```

Dependencies. The Parser depends only on the JSON Schema artifact (a static embedded file) and the expression evaluator (for template syntax validation). It has no runtime dependencies.

Key Invariants.

- A `ParsedRunbook` returned without error is structurally valid against the v2 JSON Schema.
- All step IDs are unique within a `ParsedRunbook`.
- No I/O occurs during parsing (file imports are recorded as paths, not resolved).
- Parsing is pure and deterministic: the same input always produces the same output.

3.1.2 Planner

Responsibility. The Planner transforms a `ParsedRunbook` into an `ExecutionPlan`: a resolved, ordered, dependency-annotated representation of every step the runtime will execute. The Planner resolves imports (recursively parsing and inlining invoked runbooks), discovers and validates tool definitions, resolves input provider bindings to their declared sources, and produces the static step ordering. It does *not* evaluate dynamic conditions (branch predicates, iterate count expressions) — those are runtime concerns.

What is an `ExecutionPlan`? An `ExecutionPlan` is a *flat, ordered list* of resolved steps with pre-computed metadata, not a DAG. Branch logic and iteration are left as runtime decisions because they depend on captured variables evaluated during execution. The plan encodes *which steps exist* and *what preconditions they carry*, not which steps will actually run. This design keeps the planner pure and the runtime authoritative over control flow.

```
type Planner interface {
    // Plan resolves imports, discovers tools, validates bindings,
    // and returns a fully-resolved ExecutionPlan ready for the runtime.
    Plan(ctx context.Context, rb *ParsedRunbook, opts PlanOptions) (*ExecutionPlan, error)
}

type PlanOptions struct {
    BaseDir      string           // working directory for import resolution
    ToolDirs     []string         // directories to search for .tool.yaml files
    ProviderDir  string          // directory for .provider.yaml files
    Vars         map[string]string // CLI-supplied variable overrides
}

// ExecutionPlan is an immutable, fully-resolved plan for a single run.
type ExecutionPlan struct {
    RunID        string
    RunbookPath  string
    Steps        []ResolvedStep // ordered; includes inlined invoke steps
    Tools        map[string]*ToolDef // name -> resolved tool definition
    Providers    map[string]*ProviderDef // binding name -> provider
    Governance   *GovernancePolicy
    Metadata     PlanMetadata
}
```



```

}

// ResolvedStep is a single step in the plan, fully resolved.
type ResolvedStep struct {
    ID      string
    Kind     StepKind      // cli, choice, decision, collector, tool, invoke, branch, iterate
    Spec     StepSpec        // kind-specific parameters (resolved, not templated)
    Contract *contract.Contract // effects/reads/writes/idempotent/deterministic
    Depth    int             // chain depth (0 = root runbook)
    Origin   string         // source runbook path (for inlined invoke steps)
}

```

Dependencies. The Planner depends on the Parser (for import resolution), the Tool Registry (for `.tool.yaml` discovery), and the Provider Registry (for `.provider.yaml` discovery). It must not depend on the Runtime Core, Extension Host, or any adapter.

Key Invariants.

- An `ExecutionPlan` returned without error contains only resolvable steps.
- All tool names referenced by steps appear in `ExecutionPlan.Tools`.
- Import graphs are acyclic; the Planner detects and rejects cycles.
- All `from:` bindings that declare a provider appear in `ExecutionPlan.Providers`.
- The Planner does not execute or side-effect; it is safe to plan without running.

3.1.3 Runtime Core

Responsibility. The Runtime Core owns the execution loop: advancing through an `ExecutionPlan` step by step, evaluating templates and branch predicates, dispatching tool invocations, collecting evidence, writing the append-only trace, enforcing governance pre-flight checks, emitting structured events, and managing the mutable run state. It does *not* own user presentation, serialization formats, or network transport.

Interface.

```

type Runtime interface {
    // Start initializes a run from a plan and returns a RunHandle.
    // The run does not advance until Next is called.
    Start(ctx context.Context, plan *ExecutionPlan, opts RunOptions) (RunHandle, error)

    // Resume restores a previously checkpointed run from the run store.
    Resume(ctx context.Context, runID string, opts RunOptions) (RunHandle, error)
}

type RunHandle interface {
    // Next advances the run by one step. Blocks until the step completes
    // (or is paused waiting for evidence/approval).
    // Returns io.EOF when the run has no more steps.
    Next(ctx context.Context) (*StepResult, error)

    // Approve records approval for the current step's approval gate.
    Approve(ctx context.Context, decision ApprovalDecision) error

    // SubmitEvidence records evidence for interactive steps (choice, decision, collector).
    // For choice: stores selected option. For decision: stores route selection.
    // For collector: stores multi-field form data and artifact attachments.
    SubmitEvidence(ctx context.Context, stepID string, ev map[string]*EvidenceValue) error

    // Cancel requests a graceful stop of the run.
    Cancel(ctx context.Context, reason string) error

    // State returns the current mutable run state (read-only snapshot).
    State() RunState

    // Events returns a channel of structured events emitted during execution.
    Events() <-chan Event
}

type RunOptions struct {

```

```

Mode      RunMode      // real, dry-run, replay
Actor     string        // --as identity
Vars      map[string]string // variable overrides
ScenarioDir string      // replay scenario directory
Store     RunStore      // durable store for checkpointing
OnEvent   func(Event)   // synchronous event hook (nil = use Events channel)
}

```

Dependencies. The Runtime Core depends on the Governance Engine, Tool Runtime (for tool invocation), Expression Evaluator (for templates and branch predicates), Evidence store, and Trace Writer. It does not depend on the Extension Host, API/Adapter Layer, or any adapter.

Key Invariants.

- The trace is append-only: no event is ever deleted or modified after being written.
- Governance pre-flight is enforced before every `cli` and `tool` step dispatch.
- Redaction is applied to all captured output before it is written to the trace.
- The run state is checkpointed after every step completion.
- Events are emitted in a total order that matches step execution order.
- The runtime is re-entrant per run: multiple callers may call `Next` sequentially (but not concurrently) on the same `RunHandle`.

3.1.4 Extension Host

Responsibility. The Extension Host manages the lifecycle of out-of-process extensions: discovery, launch, capability negotiation, RPC dispatch, sandboxing, and graceful shutdown. It exposes extension-contributed tools and providers to the Tool Runtime and Provider Registry. It does *not* execute runbook steps directly and does not own the extension communication protocol (that belongs to the extension RPC layer, defined in §04).

Interface.

```
type ExtensionHost interface {  
    // Load discovers and launches all extensions declared in the project manifest.  
    Load(ctx context.Context, manifest *ProjectManifest) error  
  
    // Shutdown gracefully stops all running extensions.  
    Shutdown(ctx context.Context) error  
  
    // ContributedTools returns all tools contributed by loaded extensions.  
    ContributedTools() []*ToolDef  
  
    // ContributedProviders returns all providers contributed by loaded extensions.  
    ContributedProviders() []*ProviderDef  
  
    // ContributedPolicyRules returns governance rules contributed by extensions.  
    // These compose with host policy; host policy takes precedence on conflict.  
    ContributedPolicyRules() []PolicyRule  
}
```

Dependencies. The Extension Host depends on the extension manifest schema and the extension RPC transport (stdio JSON-RPC, as specified in §04). It does not depend on the Runtime Core or any adapter.

Key Invariants.

- No extension operates without an explicit capability grant from the host.
- Extension crashes do not crash the host process.
- The Extension Host enforces timeout and cancellation on all RPC calls to extensions.
- Extension-contributed policy rules are additive; they can only *restrict*, not *relax*, host-defined policy.

3.1.5 API / Adapter Layer

Responsibility. The API/Adapter Layer is the boundary between the Runtime Core and all external consumers (TUI, VS Code extension, web clients, CI pipelines). It translates

the runtime's event stream and `RunHandle` contract into transport-specific protocols. Each adapter is a thin, stateless renderer: it subscribes to the event stream, presents state to the user, and forwards user input (approvals, evidence, cancellation) back to the `RunHandle`. Adapters own *no* execution logic.

Interface.

```
// Adapter is the interface every consumer of the Runtime Core must implement.
type Adapter interface {
    // Attach binds the adapter to a RunHandle before the first Next call.
    Attach(handle RunHandle) error

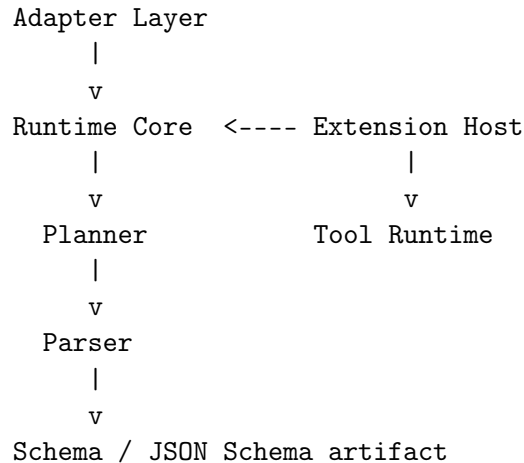
    // Run blocks until the run completes or is cancelled. It is responsible
    // for draining Events(), rendering output, and forwarding user input.
    Run(ctx context.Context) error
}

// The gert serve JSON-RPC layer implements Adapter over stdio.
// The TUI (Bubble Tea) implements Adapter over a terminal.
// Future: a web adapter implements Adapter over WebSocket.
```

Dependencies. Adapters depend on the `RunHandle` and `Event` types from the Runtime Core. They must not import any package from the Parser, Planner, or Extension Host.

Key Invariants.

- An adapter may not call `Next` more than once concurrently.
- An adapter may not modify run state directly; all state changes go through `RunHandle`.
- Adapters must handle the case where the event channel is closed (run ended) without blocking.
- An adapter failure must not corrupt the run trace or run state.

Dependency Direction Summary.

No upward arrow exists. The Parser knows nothing about the Runtime. The Runtime knows nothing about adapters. The Extension Host is a peer of the Runtime Core, contributing tools and policy rules but not owning execution.

3.2 Data Flow

This section traces the complete path from runbook invocation to trace archive, identifying exactly where each responsibility is exercised.

3.2.1 Phase 1: Parse

The user invokes `gert exec runbook.yaml -as eng@corp.com`. The CLI layer reads the YAML file and calls `Parser.Parse()`. The Parser validates the document against the embedded v2 JSON Schema, checks structural semantic rules, and returns a `ParsedRunbook`. If validation fails, structured diagnostics are returned and execution does not proceed.

3.2.2 Phase 2: Plan

The `ParsedRunbook` is passed to `Planner.Plan()`. The Planner:

1. Resolves all `imports`: recursively (parse $\rightarrow$ plan each imported runbook, inline its steps with incremented `Depth`).

2. Discovers `.tool.yaml` files for each named tool.
3. Resolves `from:` input bindings to their declared providers.
4. Validates that all step references (branch targets, iterate bodies) are valid step IDs.
5. Constructs and returns an `ExecutionPlan`.

Governance policy from the runbook's `meta.governance` block is extracted and embedded in the `ExecutionPlan` for runtime enforcement.

3.2.3 Phase 3: Run Initialization

`Runtime.Start(plan, opts)` is called. The runtime:

1. Generates a `RunID` (format: `YYYYMMDDTHHmss-xxxxxxx`).
2. Creates the run directory: `.runbook/runs/<runID>/`.
3. Opens the append-only trace file: `.runbook/runs/<runID>/trace.jsonl`.
4. Writes the `run/started` trace event (runbook path, actor, mode, plan hash).
5. Initializes the `GovernanceEngine` from the plan's governance policy.
6. Compiles redaction rules from the governance policy.
7. Returns a `RunHandle`; the run is now paused at step 0.

3.2.4 Phase 4: Step Execution Loop

The adapter calls `RunHandle.Next(ctx)` in a loop until `io.EOF`. For each step, the runtime executes the following sequence:

Step Pre-flight (all step types)

1. Write `step/started` trace event.
2. Emit `StepStarted` runtime event to the event channel.
3. Evaluate template expressions in the step spec against current run variables.

Governance Pre-flight (cli and tool steps)

1. **Command check:** `GovernanceEngine.CheckCommand(argv[0])` evaluates the denylist (deny takes precedence) and then the allowlist. A violation writes a `governance/commandBlocked` trace event and returns an error; the step fails.
2. **Env var check:** `GovernanceEngine.FilterEnvVars()` removes variables matching `deny_env_vars` patterns. Blocked variable names are written to the `governance/envVarBlocked` trace event.
3. **Contract check:** The step's `Contract.Risk()` is evaluated. Steps with `RiskCritical` that do not have an explicit approval gate declaration in the plan emit a `governance/contractWarning` event (not a hard failure in v2.0; upgraded to error in strict mode).
4. **Approval gate:** If the step declares `approvals.min > 0`, the runtime pauses and emits an `ApprovalRequired` event. Execution blocks at this step until `RunHandle.Approve()` is called with sufficient approvals. See §12.3 (Governance chapter) for the full approval protocol.

Dispatch (step-type-specific)

- **cli:** Fork a subprocess. Capture stdout/stderr. Apply redaction rules to captured output before storing in run variables or writing to trace.
- **tool:** Call `ToolRuntime.Invoke(toolName, input)`. The tool runtime dispatches to the appropriate transport (stdio, JSON-RPC, MCP). Redaction applied to returned output.
- **choice:** Emit `ChoiceRequired` event with option list. Block until `SubmitEvidence()` is called with selected option. Store result as named variable. No artifacts involved.
- **decision:** Emit `DecisionRequired` event with route map. Block until `SubmitEvidence()` is called with route selection. Alter execution flow by jumping to the selected runbook or label. Optionally store selection in variable for audit trail.
- **collector:** Emit `CollectorRequired` event with field schema. Block until `SubmitEvidence()` is called with multi-field form data. SHA256-hash each file/image attachment. Store text fields as variables, artifacts in evidence store.

- **invoke:** Inlined at plan time; at runtime the invoke body steps are already present in the plan at `Depth+1`. No special dispatch required.
- **branch:** Evaluate branch predicates (expression evaluator) against current variables. Select the matching branch arm. Write `event/branchResolved` trace event. Advance step cursor to the first step of the resolved arm.
- **iterate:** Evaluate the iteration expression. Write `event/iteratePassStart`. Execute body steps. Write `event/iteratePassEnd`. Repeat until condition is false.
- **wait_for_event:** Serialise run state to the run store. Register an event listener with the Event Dispatcher (source, event ID, filter predicate, and timeout). Set run state to `WAITING` and return to caller — execution is suspended. When the matching event arrives, the Event Dispatcher rehydrates the run, validates the payload against `event.payload_schema`, applies `event.filter` conditions, writes captured variables via `capture` mappings, and resumes at the next step. If `timeout` elapses before an event arrives the `on_timeout` policy governs the outcome (see §3.3.4). This dispatch path is only available in `gert serve` mode; `gert run` MUST reject the step with error: "step type 'wait_for_event' requires gert serve mode".

Step Type Classification

All step types fall into one of three categories based on runtime behavior:

Category	Step Types	Runtime Behavior
Execution	cli, tool	Execute external commands or tools. Require governance pre-flight (allowlist, denylist, redaction). Capture output and exit codes.
Interactive	choice, decision, collector	Block execution, emit event requiring user input. choice stores selected option as variable. decision alters control flow by routing to a runbook or label. collector captures multi-field form data and artifacts (files, images). All three write evidence to trace.
Control Flow	branch, iterate, invoke	Alter execution path or loop body. branch evaluates predicates at runtime. iterate repeats a body until a condition is false. invoke is resolved at plan time; body steps appear inline with incremented depth.
Synchronisation	wait_for_event	Suspend the run until an external event arrives via webhook, message queue, OS signal, or named in-process channel. Persists full run state to durable storage; resumes when the Event Dispatcher delivers a matching, validated event. Valid only in gert serve mode — rejected at runtime in gert run (local CLI) mode.

Note that **invoke** is somewhat special: it is a control flow step conceptually, but its inlining happens during the Plan phase, so at runtime it does not require special dispatch logic.

Step Post-flight (all step types)

1. Apply outcome evaluation: test **when:** expressions; record **OutcomeRecord**.
2. Write captured variables to run state.
3. Checkpoint run state to `.runbook/runs/<runID>/snapshots/<stepID>.json`.

4. Write `step/completed` trace event (outcome, captures, elapsed time).
5. Emit `StepCompleted` runtime event.

3.2.5 Phase 5: Run Completion

When `Next` returns `io.EOF`:

1. Write `run/completed` trace event (final outcome, step counts, elapsed time).
2. Flush and sync the trace file.
3. Compute SHA256 of the trace file; write `run/evidenceHash` record.
4. Optionally archive the run directory (compress to `.runbook/archive/`).
5. Close the event channel.

3.3 Lifecycle Model

3.3.1 Run Start

A run starts in one of two ways:

1. **Direct invocation:** The CLI or test harness calls `Runtime.Start()` directly. This is the path for `gert exec`, `gert tui`, `gert debug`, and `gert test`.
2. **RPC invocation:** A client sends `exec/start` to the JSON-RPC server (`gert serve`). The server calls `Runtime.Start()` on behalf of the client and returns the `RunID` in the response. Subsequent step advancement is driven by `exec/next` RPC calls.

3.3.2 Step Advancement

Step advancement is *always client-driven*. The runtime does not auto-advance. After `Start`, the client calls `Next` to execute each step in sequence. This applies to all adapters, including headless CI mode. The only exception is the `-auto` flag, which causes the CLI adapter to call `Next` in a tight loop, producing the appearance of automatic advancement while retaining the single-step contract.

3.3.3 Pause and Resume

A run is *paused* when `Next` returns a non-terminal result that requires external input: interactive steps (`choice`, `decision`, `collector`) or approval gates. The run remains paused until the required input is provided via `SubmitEvidence` or `Approve`. The adapter is responsible for presenting the pause to the user.

The checkpoint written after each completed step is sufficient to resume a run after a process restart. The checkpoint records: completed step IDs, current variable bindings, current evidence values, and the trace sequence number. Resumption is initiated by calling `Runtime.Resume(runID)` from any adapter.

3.3.4 Wait-for-Event Executor Contract

WAITING Run State

The `wait_for_event` step introduces a new **WAITING** run state that is distinct from the existing user-initiated **PAUSED** state:

State	Trigger	Exit Condition
RUNNING	<code>exec/start</code> or resume	Active step execution loop.
PAUSED	User-initiated pause; interactive / approval step	<code>SubmitEvidence</code> or <code>Approve</code> RPC call from adapter.
WAITING	<code>wait_for_event</code> step reached	Event Dispatcher delivers a matching event (<code>WAITING→RUNNING</code>); or timeout expires with <code>on_timeout: fail</code> (<code>WAITING→FAILED</code>) or timeout expires with <code>on_timeout branch:<id></code> (<code>WAITING→RUNNING</code> , jump to branch target).
COMPLETED	All steps finished	Terminal.
FAILED	Unhandled error or <code>on_timeout: fail</code>	Terminal (unless compensated in v2.1).

WAITING is a *system-driven* suspension: the run is blocked on an external signal, not on user input. Clients **MUST** differentiate the two states in UX (e.g. “awaiting webhook” vs. “awaiting approval”) and **MUST NOT** send `exec/next` while the run is in **WAITING** state.

Pause / Resume Protocol

When the step execution loop reaches a `wait_for_event` step it **MUST**:

1. **Serialise run state** — write the full run snapshot (step index, all variable bindings, call stack, active listener registration) to the run store (embedded BoltDB / SQLite, same store used for checkpoint snapshots).
2. **Register event listener** — call `EventDispatcher.Register(runID, eventID, source, filter, payloadSchema, timeout)`. The Dispatcher returns a registration token.
3. **Suspend** — set `run.state = WAITING`, write a `step/waiting` trace event (includes `event.source`, `event.id`, timeout deadline), and return from the execution goroutine. The run goroutine is now idle; no `Next` call is outstanding.

When the Event Dispatcher delivers a matching event to a `WAITING` run it **MUST**:

1. Load the serialised run state from the run store via `store.Load(runID)`.
2. Validate the payload against `step.event.payload_schema` (JSON Schema Draft 2020-12). Validation failure → `FAILED` with schema violation detail.
3. Apply `step.event.filter` key-value predicates against the payload. Mismatch is treated as a non-matching event (listener remains registered; the event is discarded).
4. Apply `step.capture` mappings: extract `json.<path>` fields from the payload and write them into `run.variables`.
5. Write a `step/eventReceived` trace event (event source, sanitised payload summary, captured variable names).
6. Set `run.state = RUNNING` and call `ExecuteNextStep(run)`.

Executor Pseudocode

```
// Suspend path
ExecuteStep(step: WaitForEvent, run: Run):
    store.Persist(run)                // 1. durable checkpoint
    token = dispatcher.Register(      // 2. register listener
        run.id, step.event.id, step.event.source,
        step.event.filter, step.event.payload_schema,
        step.timeout)
    run.variables["gert.event."+step.event.id+".token"] = token // HMAC token
```

```

run.state = WAITING
emit(step/waiting, {event_id: step.event.id, deadline: now()+step.timeout})
return (suspended)                                // 3. goroutine exits

// Resume path
OnEventArrived(run_id, event_id, payload):
    run = store.Load(run_id)                        // 1. rehydrate
    step = run.currentStep()                        // (WaitForEvent step)
    ValidatePayload(payload, step.event.payload_schema) // 2. schema check
    if not MatchFilter(payload, step.event.filter):    // 3. filter
        return // discard; listener stays registered
    ApplyCapture(payload, step.capture, run.variables) // 4. capture
    emit(step/eventReceived, {run_id, event_id})      // 5. trace
    run.state = RUNNING
    ExecuteNextStep(run)                            // 6. advance

// Timeout path
OnTimeout(run_id, event_id):
    run = store.Load(run_id)
    step = run.currentStep()
    switch step.on_timeout:
        case "fail":
            run.state = FAILED
            emit(step/failed, {reason: "wait_for_event timeout"})
        case "continue":
            run.state = RUNNING
            ExecuteNextStep(run)
        case "branch:<step-id>":
            run.state = RUNNING
            run.stepCursor = resolve(step-id)
            ExecuteNextStep(run)

```

Run Persistence Requirement

Runs in WAITING state MUST survive a `gert serve` process restart. The serialised run snapshot written in step 1 above MUST include:

- Current step index and depth.
- All variable bindings (including `gert.event.*` tokens).
- Call stack (for `invoke-inlined` steps).
- The active event listener registration (run ID, event ID, source, filter, timeout deadline as an absolute timestamp).

On `gert serve` restart, the server MUST re-register all persisted `WAITING` listener records with the Event Dispatcher before accepting new connections. Listeners whose timeout deadline has already passed are immediately moved to the timeout resolution path.

Serve-Only Constraint

`wait_for_event` is only valid when running under `gert serve`. If a runbook containing a `wait_for_event` step is executed via `gert run` (local CLI, non-server mode), the Runtime Core MUST fail at dispatch time with:

```
step type 'wait_for_event' requires gert serve mode
```

The Planner MAY emit a validation warning for `wait_for_event` steps in non-serve context (aiding the user before execution begins), but the enforcement point is the Runtime Core dispatch.

Security: HMAC Token for Webhook Transport

When `event.source: webhook`, each `wait_for_event` registration generates a cryptographically random one-time HMAC signing secret. The Event Dispatcher:

1. Generates 32 random bytes via `crypto/rand`.
2. Stores the secret alongside the listener registration in the run store.
3. Exposes the HTTP endpoint `POST /events/{run-id}/{event-id}` and validates incoming requests using HMAC-SHA256 over the raw request body (header: `X-Gert-Signature: sha256=<hex>`).

4. Injects the secret into the run variable `gert.event.{event-id}.token` so the runbook author can communicate it to the external system (e.g. via a preceding `cli` step that calls `curl -data-urlencode`).
5. Rejects requests with invalid or missing signatures with HTTP 401.
6. Invalidates the secret after the first accepted delivery (one-time use).

The token is intentionally injected as a run variable (not hard-coded in the runbook YAML) so it is unique per run and never committed to source control.

3.3.5 Cancellation

`RunHandle.Cancel(ctx, reason)` initiates graceful cancellation:

1. The currently-executing subprocess (if any) receives `SIGTERM` followed by `SIGKILL` after a 5-second grace period.
2. The runtime writes a `run/cancelled` trace event with the reason and actor.
3. Subsequent `Next` calls return `ErrRunCancelled`.
4. The run state is checkpointed as `cancelled` (resumable).

The cancellation contract does not invoke compensation handlers in v2.0. Saga/compensation (stepping backward through a compensation chain on failure) is deferred to v2.1 as specified in §09 (Open Questions).

3.3.6 Crash Recovery

If the `gert` process crashes mid-run, the trace and checkpoint files remain on disk in a consistent state due to the append-only, sync-after-write trace discipline. On restart, the user (or the VS Code extension) can call `Runtime.Resume(runID)` to continue from the last completed step. The runtime reads the checkpoint to reconstruct run state. Steps already recorded as completed in the trace are not re-executed.

3.4 Concurrency Model

3.4.1 Per-Process Run Isolation

The v2 runtime is **single-run-per-process**. Each `gert exec` invocation is a separate OS process. `gert serve` is the exception: it is a long-running process that may host multiple sequential runs (one at a time) for a single VS Code extension session. Concurrent runs within a single `gert serve` instance are explicitly out of scope for v2.0 (see §09).

Rationale. Governance enforcement, file-level trace writes, and subprocess management are dramatically simpler with one run per process. Process isolation also provides a natural crash boundary: a failed run cannot corrupt the state of a concurrent run.

3.4.2 Goroutine Model

Within a single run, the concurrency model is:

- **One goroutine per run:** the step execution loop runs on a single goroutine. Steps execute sequentially; there is no goroutine-per-step parallelism in v2.0.
- **Event fan-out goroutine:** a second goroutine reads from an internal event buffer and writes to the `Events()` channel. This decouples event emission (which must be synchronous with step execution) from event consumption (which may be slow, e.g., a WebSocket write).
- **Subprocess goroutine:** when a `cli` step spawns a subprocess, a goroutine is used to drain stdout/stderr concurrently, preventing deadlock on full pipe buffers.
- **Tool RPC goroutine:** tool invocations over JSON-RPC or MCP use a goroutine per call, managed by the Tool Runtime's connection pool.

3.4.3 Thread Safety at Component Boundaries

- `RunHandle` is **not safe for concurrent use**. The adapter must serialize all calls. `gert serve` enforces this with a per-session mutex.
- The `Events()` channel is safe for a single consumer goroutine.
- The `GovernanceEngine` is **read-only after construction** and therefore safe for concurrent read access (relevant if future versions support parallel steps).

- The Trace Writer is protected by an internal mutex; write calls are safe to issue from any goroutine within the run.

3.5 gert serve Integration

`gert serve` is the JSON-RPC 2.0 server that powers the VS Code extension, the MCP tools layer, and any future web client. In the v2 architecture it is classified as an **adapter in the API/Adapter Layer**.

3.5.1 Architectural Position

`gert serve` is a process that:

1. Starts a long-lived stdio JSON-RPC 2.0 listener (newline-delimited JSON over std-in/stdout).
2. On receiving `exec/start`, calls `Runtime.Start()` and stores the resulting `RunHandle` in a session map (keyed by `RunID`).
3. On receiving `exec/next`, calls `RunHandle.Next()` and returns the `StepResult` as the RPC response.
4. Drains the `Events()` channel in a background goroutine and forwards each event as a JSON-RPC notification to the client.
5. On receiving `exec/approve`, `exec/submitEvidence`, or `exec/cancel`, dispatches to the corresponding `RunHandle` method.

3.5.2 Event Forwarding

Every event emitted by the Runtime Core is forwarded to the client as a JSON-RPC *notification* (no `id` field) with method `event/eventType`. The payload is the event's JSON representation. The VS Code extension subscribes to these notifications to update its UI without polling.

Events forwarded over WebSocket (for a future web adapter) follow the same schema; only the transport changes.

3.5.3 Crash Recovery in `gert serve`

If the VS Code extension reconnects after a crash (e.g., the `gert` process was killed), `gert serve` exposes `exec/list` and `exec/resume` methods. The extension discovers the interrupted run, calls `exec/resume`, and receives a new `RunHandle` over the same session. The trace and checkpoint on disk ensure no steps are lost.

3.5.4 Event Dispatcher

The **Event Dispatcher** is a new runtime component that lives exclusively inside the `gert serve` process. It is not instantiated by `gert run`.

Responsibilities

- **Listener lifecycle** — maintain an in-memory registry of active event listeners (indexed by `run_id + event_id`). Each entry records: source type, filter predicates, payload schema URL, timeout deadline, and HMAC secret (webhook source only). Persist the registry to the run store so it survives process restart.
- **Webhook transport** — expose `POST /events/{run-id}/{event-id}` authenticated with HMAC-SHA256 (see security note in §3.3.4). Parse the request body as JSON and dispatch to the matching listener. Return `HTTP 202 Accepted` on success; `404` if no listener is registered; `401` on signature failure; `410 Gone` if the run is no longer in `WAITING` state.
- **Channel transport** — maintain an in-process `map[string]chan Payload` keyed by `event.id`. Any component within the same `gert serve` process can post to a named channel via `EventDispatcher.Send(eventID, payload)`.
- **Message broker transport** — delegate to a pluggable `MessageBroker` interface (implementation-defined for v2.0; a no-op stub ships by default). The interface follows the same pattern as the `ToolTransport` plug-in model: a registered factory function keyed by broker type string.
- **Signal transport** — register `os/signal` handlers scoped to a specific run. Signal handling is multiplexed: the same OS signal may be registered by multiple active listeners (e.g. `SIGUSR1` on two concurrent waiting runs). Each listener receives its own notification; signal handlers are deregistered on listener expiry or run completion.

- **Timeout enforcement** — maintain a min-heap of timeout deadlines. A background goroutine fires on the earliest deadline and invokes `OnTimeout(runID, eventID)`.

Go Interface

```
type EventDispatcher interface {
    // Register creates a listener for an inbound event on a waiting run.
    // Returns a one-time HMAC token (non-empty for webhook source only).
    Register(ctx context.Context, reg ListenerRegistration) (token string, err error)

    // Deregister removes a listener (called on run completion or cancellation).
    Deregister(runID, eventID string) error

    // Send posts a payload to a named in-process channel listener.
    Send(eventID string, payload json.RawMessage) error

    // RestoreFromStore re-registers all WAITING listeners after process restart.
    RestoreFromStore(store RunStore) error
}

type ListenerRegistration struct {
    RunID      string
    EventID    string
    Source     EventSource      // webhook | message | signal | channel
    Filter     map[string]string
    PayloadSchema string          // JSON Schema URL, may be empty
    TimeoutAt  time.Time
    OnTimeout  TimeoutPolicy    // fail | continue | branch:<step-id>
}
```

Availability Constraint

The Event Dispatcher is instantiated by `gert serve` only. It is **not** part of the `Runtime` interface exposed to step executors directly; instead, the Runtime Core receives a `DispatcherHandle` at construction time (nil in `gert run` mode). The `wait_for_event` dispatch path checks for a nil handle and fails immediately with the serve-mode error before attempting any state serialisation.

3.5.5 RPC Method Summary

<code>exec/start</code>	→ Start a new run (returns RunID, first StepResult)
<code>exec/next</code>	→ Advance run by one step (returns StepResult)
<code>exec/approve</code>	→ Submit approval decision for current gate
<code>exec/submitEvidence</code>	→ Submit evidence for interactive steps (choice/decision/collector)
<code>exec/cancel</code>	→ Cancel the active run
<code>exec/list</code>	→ List all runs for this workspace (active + completed)
<code>exec/resume</code>	→ Resume a previously interrupted run
<code>runbook/validate</code>	→ Validate a runbook YAML (parse + plan, no exec)
<code>runbook/diagram</code>	→ Generate a Mermaid or ASCII diagram
<code>event/<type></code>	→ Notification from server to client (not a request)

// HTTP endpoints (gert serve --http, Event Dispatcher)

POST /events/{run-id}/{event-id} → Deliver a webhook event to a waiting run
(HMAC-SHA256 authenticated)

Chapter 4

Schema vNext

This chapter is the normative specification for all gert v2 data structures: runbook documents, tool definitions, and provider definitions. It supersedes all v1 schema documentation and defines the contracts that Brian’s parser, Ken’s runtime, and the VS Code extension must implement.

The design philosophy is *minimal core, extensible periphery*: the core language fields that the runtime must understand are small, typed, and stable; everything else is declared as an extension with an explicit namespace. Where v1 accumulated surface area organically, v2 draws a hard boundary and enforces it at parse time.

4.1 Schema Identity and Versioning

4.1.1 apiVersion Values

Every gert document carries a mandatory **apiVersion** field that identifies both the document kind and the schema version. The v2 values are:

Document kind	v1 value	v2 value
Runbook	runbook/v1	runbook/v2
Tool definition	tool/v0	tool/v2
Provider definition	provider/v0	provider/v2

The format is `<kind>/<major>`. Minor and patch changes do not alter the **apiVersion** string; they are tracked by the schema artifact version (see §4.9).

4.1.2 Self-Description via \$schema

Following JSON Schema [28] best practice and the pattern used by GitHub Actions, Argo Workflows, and Kubernetes CRDs, every v2 document *should* carry a `$schema` field pointing to the canonical JSON Schema URL:

```
$schema: "https://schemas.gert.dev/runbook/v2.json"
apiVersion: runbook/v2
id: service-health
name: Service Health Check
...
```

The `$schema` field is optional at runtime (the runtime uses `apiVersion` for dispatch) but is *strongly recommended* because:

- VS Code and JetBrains IDEs provide schema-backed auto-completion and inline validation when `$schema` is present.
- The schema URL carries an implicit version contract: updating the URL updates the validation rules visible to tooling.
- It enables offline validation with any JSON Schema Draft 2020-12 validator (ajv, jsonschema, check-jsonschema).

Schema URLs for each document kind:

```
https://schemas.gert.dev/runbook/v2.json
https://schemas.gert.dev/tool/v2.json
https://schemas.gert.dev/provider/v2.json
```

4.1.3 Versioning Strategy

gert schema versioning follows a simplified semantic versioning model applied to the `apiVersion` major component:

- **Major increment** (e.g. `runbook/v1` → `runbook/v2`): A breaking change. The parser *must not* silently accept the old format. Automated migration is provided via `gert migrate`.

- **Minor/patch changes within a major:** Additive, backward-compatible. New optional fields may be added. Existing fields may not be removed or renamed. The JSON Schema artifact version tracks these with a semver suffix (e.g. `runbook/v2.json` at artifact version `2.1.0`).

What constitutes a breaking change A breaking change is any of:

- Renaming or removing a required field.
- Narrowing an existing enum (removing a valid value).
- Changing a field type (e.g. `string` $\rightarrow$ `object`).
- Changing the meaning of a field such that existing runbooks produce different runtime behavior.
- Removing a step type.

Adding new optional fields, adding new enum values, and loosening existing constraints are *not* breaking changes.

4.1.4 Compatibility Contract: v1 → v2

v1 field	v2 status	Notes
apiVersion: runbook/v0	Removed	Handled by <code>gert migrate</code>
apiVersion: runbook/v1	Deprecated	Compatibility mode in v2 parser
meta.name	Promoted	Becomes top-level <code>name</code>
meta.kind	Promoted	Becomes top-level <code>kind</code>
meta.description	Promoted	Becomes top-level <code>description</code>
meta.vars	Retained	Moved to top-level <code>vars</code>
meta.inputs	Retained	Moved to top-level <code>inputs</code>
meta.governance	Retained	Moved to top-level <code>governance</code>
meta.prose	Retained	Moved to top-level <code>prose</code>
meta.defaults	Retained	Moved to top-level <code>defaults</code>
imports	Retained	Unchanged
tools	Replaced	Replaced by <code>toolRefs</code> (typed)
flow	Retained	Canonical execution root
tree	Removed	Use <code>flow</code>
steps	Removed	Flat step list was never canonical
step.type: collector	Renamed	Split into <code>choice</code> , <code>decision</code> , <code>collector</code>
step.type: router	Renamed	Becomes <code>type: end</code>
step.type: choice	New	User selects from options; stores result
step.type: decision	New	User picks runbook path; routes execution
step.type: noop	Removed	Use <code>when</code> : guard on any step
step.type: extension	Retained	Unchanged
step.with.argv	Deprecated	Use <code>type: cli</code> with <code>command</code> :
step.assertions	Deprecated	Use <code>step.type: assert</code>

The v2 parser reads `runbook/v1` documents in **compatibility mode**: all v1 field paths are accepted and silently normalized to their v2 equivalents before validation. Running `gert migrate` permanently rewrites the file to v2 canonical form. v2 will not support `runbook/v0`; documents at that version must be migrated to v1 first or directly to v2 via a two-stage migration.

4.1.5 gert migrate Behavior

```
# Dry-run preview
gert migrate --dry-run runbook.yaml
```

```
# In-place rewrite
gert migrate runbook.yaml
```

```
# Migrate a whole directory
gert migrate ./runbooks/
```

Migration rules applied automatically:

1. Rewrite `apiVersion` to `runbook/v2`.
2. Add `$schema` field.
3. Promote `meta.name`, `meta.kind`, `meta.description` to top-level.
4. Move remaining `meta.*` fields to their v2 top-level equivalents.
5. Rename `tree:` to `flow:`.
6. Replace generic `step.type: manual` from early v2 drafts with three precise types: `choice`, `decision`, `collector`.
7. Rename `step.type: router` → `end`.
8. Rewrite `tools: [name]` shorthand to `toolRefs:` typed declarations.
9. Add `id:` to runbook if absent (derived from `name`).

4.2 Core Runbook Fields (v2)

4.2.1 Top-Level Field Inventory

The v2 runbook document has the following top-level fields:

Field	Type	Required	Description
<code>\$schema</code>	string	No	JSON Schema URL (recommended)
<code>apiVersion</code>	string	Yes	Must be <code>runbook/v2</code>
<code>id</code>	string	Yes	Unique machine identifier (slug)
<code>name</code>	string	Yes	Human-readable display name
<code>kind</code>	enum	No	<code>mitigation</code> <code>reference</code> <code>composable</code> <code>rca</code>
<code>description</code>	string	No	Short prose summary
<code>vars</code>	map	No	Default variable values
<code>inputs</code>	map	No	Input declarations (§4.3)
<code>outputs</code>	map	No	Output declarations (§4.3)
<code>toolRefs</code>	array	No	Tool references (§4.2.2)
<code>imports</code>	map	No	Named runbook imports
<code>defaults</code>	object	No	Default step settings
<code>governance</code>	object	No	Governance policy
<code>prose</code>	object	No	Human-readable TSG sections
<code>metadata</code>	map	No	Arbitrary key-value annotations
<code>flow</code>	array	Yes	Execution tree (TreeNode array)

id field The `id` is a slug: lowercase alphanumeric with hyphens, matching `^[a-z][a-z0-9-]*[a-z0-9]$`. It uniquely identifies a runbook within a project and is used by the `invoke` step and the trace record. It must be stable across renames; changing it is a breaking change for any runbook that invokes it by `id`.

kind values

- **mitigation** — Incident response procedure; governance defaults to `require-approval` for destructive effects.
- **reference** — Documentation-first runbook; dry-run by default.
- **composable** — Sub-runbook intended for invoke from other runbooks; not directly executable from CLI.
- **rca** — Root-cause analysis template.

4.2.2 toolRefs

In v1, tools were declared as a flat string array (`tools: [curl, nslookup]`). In v2, `toolRefs` is a typed array that allows aliasing and explicit path overrides:

```
toolRefs:
  - name: curl
  - name: nslookup
  - name: k8s
    path: ./tools/k8s.tool.yaml # override default discovery
    alias: kubect1-wrapper      # optional local alias
```

The default discovery path for `name: foo` is `tools/foo.tool.yaml` relative to the runbook file. The `path` field overrides this. The `alias` field lets the runbook reference the tool by a local name that differs from the tool's declared `meta.name`.

4.2.3 Minimal Valid v2 Runbook

```
$schema: "https://schemas.gert.dev/runbook/v2.json"
apiVersion: runbook/v2
id: hello-world
name: Hello World
```

```
flow:
  - step:
      id: greet
      type: cli
      title: Print greeting
      command: echo
      args: ["Hello, world!"]
      capture:
        output: stdout
  - step:
      id: done
      type: end
      title: Complete
      outcome:
```

```
category: resolved
code: success
```

4.2.4 Complete Annotated Example

```
$schema: "https://schemas.gert.dev/runbook/v2.json"
apiVersion: runbook/v2
id: service-health-check
name: Service Health Check
kind: mitigation
description: |
  Validate DNS resolution and HTTP endpoint reachability
  for a named service. Escalates on failure.

vars:
  default_timeout: "30s"

inputs:
  hostname:
    type: string
    required: true
    description: "Hostname to check"
    from: prompt
  incident_id:
    type: string
    required: false
    description: "Incident tracking ID"
    from: incident.id      # provider-resolved

outputs:
  health_status:
    type: string
    description: "Final health assessment"
    from: captures.http_result

toolRefs:
```

```
- name: curl
- name: nslookup

governance:
  rules:
    - effects: ["network.probe"]
      action: allow
    - effects: ["service.restart"]
      action: require-approval
      min_approvers: 2
  redact:
    - pattern: "(?i)Authorization:\\s*\\S+"
      replace: "Authorization: <redacted>"

defaults:
  timeout: "60s"

prose:
  safety: |
    Run this runbook only during an active incident. Do not
    loop - each execution creates a trace record.
  prerequisites: "Requires curl >= 7.80 and nslookup on PATH."

metadata:
  team: platform-sre
  runbook-url: "https://wiki.example.com/runbooks/service-health"
  severity: P2

flow:
  - step:
      id: check_dns
      type: tool
      title: "Resolve DNS: {{ .hostname }}"
      tool:
        name: nslookup
        action: lookup
```

```
    args:
      host: "{{ .hostname }}"
    capture:
      dns_result: stdout
    contract:
      effects: ["network.probe"]
      reads: ["hostname"]
      writes: ["dns_result"]

- step:
  id: check_http
  type: tool
  title: "HTTP health check: {{ .hostname }}"
  tool:
    name: curl
    action: head
    args:
      url: "https://{{ .hostname }}/healthz"
  capture:
    http_result: stdout
  timeout: "{{ .default_timeout }}"

- step:
  id: evaluate
  type: branch
  title: Evaluate health results
  branches:
    - condition: '{{ contains .http_result "200" }}'
      label: Healthy
      steps:
        - step:
            id: resolved
            type: end
            title: Service is healthy
            outcome:
              category: resolved
```

```

        code: http_ok
- condition: '{{ not (contains .http_result "200") }}'
  label: Unhealthy
  steps:
    - step:
      id: escalate
      type: end
      title: Escalate to on-call
      outcome:
        category: escalated
        code: http_fail

```

4.3 Input and Output Declarations

4.3.1 Input Declarations

Inputs declare variables that must be resolved before execution begins. In v2 the `inputs` map is a top-level field (promoted from `meta.inputs`).

```

inputs:
  <name>:
    type: string | number | boolean | secret
    required: true | false           # default: true
    description: "..."
    default: "..."                 # valid only when required: false
    example: "..."
    pattern: "[a-z]+"               # regex validation (string only)
    from: <binding>                 # resolution source

```

type field The `type` field (new in v2) declares the value's type. Supported values:

- `string` — Default. UTF-8 text.
- `number` — Integer or float, passed as string in templates.
- `boolean` — `true` or `false`.
- `secret` — String, redacted in traces and log output.

from: binding syntax The **from** field specifies how the value is resolved. v2 binding forms:

Binding	Behavior
<code>prompt</code>	Ask the operator interactively at runtime
<code>env.<NAME></code>	Read from environment variable NAME
<code>file.<path></code>	Read first line from a local file
<code><provider>.<field></code>	Delegate to a registered input provider (e.g. <code>incident.id</code>)
<code>var.<name></code>	Copy from another variable or capture

If **from** is omitted, the field must have a **default** value or the runtime will reject the runbook at semantic validation time.

4.3.2 Output Declarations

Outputs declare values that a composable runbook exposes to its caller via the **invoke** step's output extraction.

outputs:

```
<name>:
  type: string | number | boolean | secret
  description: "..."
```

from: `captures.<captureName>` # must reference a capture variable

The **from: captures.*** binding is the only supported form in v2. The named capture must be written by at least one step in the runbook; the semantic validator checks this.

4.3.3 Expression Language

v2 retains **Go templates** (`text/template`) as the expression language for all interpolated fields (`title`, `args`, `when`, `condition`, `instructions`, etc.). The decision to retain Go templates is grounded in two facts:

1. The v1 template engine is battle-tested and broadly used in v1 runbooks.
2. Go templates' pipelines and custom function maps provide sufficient expressiveness for operational conditionals.

The template data context is the *run variable map*: all `vars`, resolved `inputs`, and accumulated `captures` merged into a single flat `map[string]string`. Access by dot notation: `{{ .hostname }}`.

Built-in template functions (v2 additions) v2 adds a standard function library accessible in all templates:

Function	Signature
<code>contains</code>	<code>(s, substr string) bool</code>
<code>hasPrefix</code>	<code>(s, prefix string) bool</code>
<code>hasSuffix</code>	<code>(s, suffix string) bool</code>
<code>trim</code>	<code>(s string) string</code>
<code>toLowerCase</code>	<code>(s string) string</code>
<code>toUpperCase</code>	<code>(s string) string</code>
<code>split</code>	<code>(s, sep string) []string</code>
<code>join</code>	<code>(elems []string, sep string) string</code>
<code>env</code>	<code>(name string) string</code>
<code>now</code>	<code>() string (RFC3339)</code>
<code>runID</code>	<code>() string</code>
<code>not</code>	<code>(v bool) bool</code>
<code>eq, ne, lt, gt, le, ge</code>	Standard comparison

All functions from `text/template` (built-ins: `and`, `or`, `index`, `len`, `printf`, etc.) remain available.

Template errors In v2, a template evaluation error in a `condition` or `when` field is a hard step failure (not a soft skip). The step transitions to `failed` and execution halts unless `continue_on_fail: true` is set on the step.

4.4 Step Types (v2 Inventory)

Each step in flow is a `TreeNode`: either a `step` object (with optional `branches` and `iterate` siblings) or a standalone `iterate` block.

v2 defines thirteen step types:

Type	Purpose
<code>cli</code>	Execute a local command (§4.4.2)
<code>choice</code>	Operator selects from predefined options (§4.4.3)
<code>decision</code>	Operator selects an execution path (§4.4.4)
<code>collector</code>	Operator provides unstructured input/files (§4.4.5)
<code>tool</code>	Invoke a declared tool action (§4.4.6)
<code>wait_for_event</code>	Pause until an inbound event arrives (§4.4.7)
<code>invoke</code>	Call another runbook as a sub-procedure (§4.4.8)
<code>branch</code>	Conditional fan-out to labelled step sequences (§4.4.9)
<code>iterate</code>	Loop over a collection (§4.4.10)
<code>parallel</code>	Fan-out concurrent branches with join semantics (§4.4.11)
<code>assert</code>	Evaluate a guard expression and halt on failure (§4.4.12)
<code>compensate</code>	Declare a saga compensation action (§4.4.13)
<code>end</code>	Declare a terminal outcome (§4.4.14)

4.4.1 Common Step Fields

The following fields are available on every step type:

Field	Type	Required	Description
id	string	Yes	Unique within the runbook
type	enum	Yes	Step type (see below)
title	string	No	Template; displayed in TUI/trace
subtitle	string	No	Secondary template annotation
when	string	No	Guard expression; skips step if false
timeout	string	No	Duration (e.g. 30s); overrides default
delay	string	No	Pre-execution pause (e.g. 5s)
retry	object	No	Retry config (§4.4.1)
continue_on_fail	bool	No	Continue execution on failure
scope	string	No	Named scope for variable isolation
export	string array	No	Variables to export from scope
capture	map	No	Output capture bindings
contract	object	No	Behavioral contract for governance

Retry configuration

```

retry:
  max: 3          # maximum retry attempts (required)
  interval: "5s"  # initial wait between attempts
  backoff: linear  # linear | exponential (default: linear)

```

Step contract The `contract` field annotates a step with its behavioral properties for governance policy evaluation:

```

contract:
  effects:
    - network.probe
    - service.restart
  reads:  [hostname, port]
  writes: [http_result]
  deterministic: true
  idempotent: false
  secrets: [auth_token]

```

Effects are free-form strings matched against `governance.rules[].effects`. The `secrets` list names variables whose values must be redacted in traces.

4.4.2 Step Type: cli

Executes a local command directly. In the v2 runtime `cli` is the canonical command step; the v1 `with.argv` form is accepted in compatibility mode and rewritten to `cli` by the migration tool.

Field	Type	Required	Description
<code>command</code>	string	Yes	Executable name or full path
<code>args</code>	string array	No	Arguments (templates expanded)
<code>env</code>	map	No	Additional environment variables
<code>workdir</code>	string	No	Working directory
<code>stdin</code>	string	No	Template string piped to stdin
<code>capture</code>	map	No	<code>stdout</code> <code>stderr</code> <code>exit_code</code> → var

Governance checks against the `allowed_commands` / `denied_commands` lists and any matching `governance.rules[].effects` are evaluated before the command executes. If the rule action is `require-approval`, the step pauses for human approval before proceeding.

```
- step:
  id: tail_logs
  type: cli
  title: "Tail last 100 lines from {{ .log_file }}"
  command: tail
  args: ["-n", "100", "{{ .log_file }}"]
  capture:
    output: stdout
    exit: exit_code
  contract:
    effects: [filesystem.read]
    reads: [log_file]
    writes: [output]
```

4.4.3 Step Type: choice

Prompts the user to select from a set of predefined options and stores the result in a variable. This is a value-capture primitive, analogous to a form field or radio-button group. The selected value is available to subsequent steps via variable substitution.

Field	Type	Required	Description
<code>prompt</code>	string	Yes	Question or prompt displayed to user
<code>options</code>	array	Yes	List of option objects
<code>variable</code>	string	Yes	Variable name to store selected value
<code>default</code>	string	No	Default option value if user skips

Option structure Each option is an object with:

Field	Type	Required	Description
<code>label</code>	string	Yes	Display text shown to user
<code>value</code>	string	Yes	Value stored in variable when selected
<code>hint</code>	string	No	Additional context or help text

Constraints

- At least 2 options must be defined.
- All `option.value` entries must be unique within a step.
- If `default` is provided, it must match one of the option values.
- The `variable` name follows standard identifier rules (alphanumeric + underscore).

Example: Environment selection

```
- step:
  id: select_environment
  type: choice
```

```

title: Select deployment target
prompt: Which environment are you deploying to?
options:
  - label: Staging
    value: staging
    hint: Non-production environment for validation
  - label: Production
    value: production
    hint: Live customer-facing environment
  - label: Canary (10%)
    value: canary
    hint: Gradual rollout to 10% of production traffic
variable: deploy_env
default: staging

```

The value `staging`, `production`, or `canary` is stored in `{{ .deploy_env }}` and can be referenced by subsequent steps.

4.4.4 Step Type: decision

Presents the user with 2 or more execution paths and routes the flow to the selected one. This is a control-flow primitive: the user picks a branch, and execution forks to a different runbook or labeled section. Unlike `choice`, which merely stores a value, `decision` directs the execution path itself.

Also acceptable name: `router` (alias for `decision`).

Field	Type	Required	Description
<code>prompt</code>	string	Yes	Question or scenario description
<code>routes</code>	array	Yes	List of route objects (minimum 2)
<code>variable</code>	string	No	Variable to store selected route label (audit)

Route structure Each route is an object with:

Field	Type	Required	Description
label	string	Yes	Display text shown to user
runbook	string	No	Runbook ID to invoke (mutually exclusive with <code>goto</code>)
goto	string	No	Step ID to jump to within current runbook
hint	string	No	Additional context or help text

Constraints

- At least 2 routes must be defined.
- Each route must specify exactly one of `runbook` or `goto`, not both.
- All `route.label` entries must be unique within a step.
- If `variable` is provided, it stores the selected route's `label` for audit trails.

Example: Incident response fork

```
- step:
  id: triage_decision
  type: decision
  title: Choose incident response path
  prompt: |
    Based on the error rate and customer impact, which response
    path should we take?
  routes:
    - label: Standard mitigation
      runbook: incident-standard-mitigation
      hint: Error rate < 5%, no customer escalations
    - label: Emergency rollback
      runbook: incident-emergency-rollback
      hint: Error rate > 10% or critical customer impact
    - label: Escalate to on-call
      goto: escalate_to_oncall
      hint: Unclear root cause; needs senior diagnosis
  variable: chosen_path
```


After the user selects a route, execution continues at the target runbook or step. The `chosen_path` variable contains the selected label ("Standard mitigation", "Emergency rollback", or "Escalate to on-call") for audit purposes.

4.4.5 Step Type: collector

Prompts the user to provide unstructured input: free-form text, file attachments, screenshots, or multi-field forms. The collected data is stored as named variables or artifacts for downstream steps or evidence capture. This replaces the overly-generic `manual` step type from early v2 drafts.

Field	Type	Required	Description
<code>prompt</code>	string	Yes	Markdown instructions displayed to user
<code>fields</code>	array	Yes	List of field objects to collect
<code>approvals</code>	object	No	Approval gate config (§4.4.5)

Field structure Each field is an object with:

Field	Type	Required	Description
<code>name</code>	string	Yes	Variable name to store collected value
<code>type</code>	enum	Yes	<code>text</code> , <code>multiline</code> , <code>file</code> , <code>image</code> , <code>url</code>
<code>label</code>	string	Yes	Prompt or label text shown to user
<code>required</code>	bool	No	If true, user must provide a value (default: false)
<code>hint</code>	string	No	Additional context or validation help

Field types

- `text`: Single-line text input.
- `multiline`: Multi-line text area (e.g., for notes or summaries).

- **file**: File attachment; stored as artifact with SHA256 hash.
- **image**: Image/screenshot attachment; stored as artifact.
- **url**: URL input with basic validation (must be valid URL).

Approval gate If the **approvals** field is present, the step requires explicit approval from designated roles before proceeding.

Field	Type	Required	Description
min	integer	Yes	Minimum approvals required
roles	array	Yes	List of role names authorized to approve
timeout	string	No	Duration before timeout (e.g. 4h)
on_timeout	enum	No	escalate , fail , skip
escalate_to	array	No	Roles to escalate to if timeout occurs

The **timeout** and **on_timeout** fields are new in v2, addressing the SLA enforcement gap identified in the research brief.

Example: Incident evidence collection

```
- step:
  id: collect_evidence
  type: collector
  title: Collect incident evidence
  prompt: |
    Gather all relevant evidence for the incident investigation.
    Include screenshots, error logs, and a summary of impact.
  fields:
    - name: incident_summary
      type: multiline
      label: Incident summary
      required: true
```

```
    hint: Describe what happened, when, and what services were affected
  - name: error_log
    type: file
    label: Error log file
    required: true
    hint: Attach the relevant error log from the affected service
  - name: dashboard_screenshot
    type: image
    label: Monitoring dashboard screenshot
    required: false
    hint: Screenshot showing metrics during the incident
  - name: related_ticket
    type: url
    label: Related ticket or alert URL
    required: false
approvals:
  min: 1
  roles: [DRI, incident-commander]
  timeout: 30m
  on_timeout: fail
```

After the user provides the requested evidence and approval is granted, the collected values are available as `{{ .incident_summary }}`, `{{ .error_log }}`, etc. File and image attachments are stored as artifacts in the execution trace with SHA256 hashes for tamper-evidence.

4.4.6 Step Type: tool

Invokes a named tool action. The tool must be declared in `toolRefs`. The runtime dispatches to the tool's configured transport (stdio, jsonrpc, or mcp).

Field	Type	Required	Description
<code>tool.name</code>	string	Yes	Tool name (must match tool-Refs entry)
<code>tool.action</code>	string	Yes	Action declared in tool definition
<code>tool.args</code>	map	No	Argument key-value map (templates expanded)
<code>tool.version</code>	string	No	Optional version pin
<code>capture</code>	map	No	<code>stdout stderr json.<path> → var</code>

Output capture in v2 adds a `json.<path>` form for JSON-structured tool outputs, eliminating the need for follow-up `cli` steps with `jq`:

```
capture:
  pod_name: json.items[0].metadata.name
  status:   json.items[0].status.phase

- step:
  id: list_pods
  type: tool
  title: "List pods in {{ .namespace }}"
  tool:
    name: kubectl
    action: get-pods
    args:
      namespace: "{{ .namespace }}"
      output: json
  capture:
    pod_count: json.items | length
    first_pod: json.items[0].metadata.name
  contract:
    effects: [k8s.read]
    reads: [namespace]
    writes: [pod_count, first_pod]
```

4.4.7 Step Type: `wait_for_event`

Pauses runbook execution until an inbound event arrives on a declared source. Use this step type whenever a runbook must suspend and wait for an external signal before proceeding — a Prometheus alert webhook, an approval callback from a ticketing system, a SIEM event, a message-queue message, or an OS/process signal.

When the runtime reaches a `wait_for_event` step it:

1. Opens the declared inbound endpoint (HTTP listener, queue consumer, or internal channel) for the lifetime of the wait.
2. Suspends the run record in state `waiting_for_event`.
3. When a matching event arrives and passes any declared `event.filter` and `event.payload_schema` checks, it captures payload fields into run variables (via `capture`) and resumes execution at the next step.
4. If the `timeout` expires before a matching event arrives, the step behaves according to `on_timeout: fail` (default), `continue` (skip and proceed), or `branch:<step-id>` (jump to the named step).

Field	Type	Required	Description
<code>event.source</code>	string	Yes	Event source type: <code>webhook</code> , <code>message</code> , <code>signal</code> , or <code>channel</code>
<code>event.id</code>	string	No	Logical event name or channel (e.g. <code>prometheus.alert.firing</code>); templates expanded
<code>event.filter</code>	map	No	Key-value pairs that must match in the incoming event payload
<code>event.payload_schema</code>	string	No	JSON Schema URL to validate the incoming payload before capture
<code>capture</code>	map	No	<code>json.<path></code> → variable; maps event payload fields to run variables
<code>timeout</code>	string	No	Wall-clock duration (e.g. <code>30m</code> , <code>2h</code>); default: no timeout
<code>on_timeout</code>	string	No	<code>fail</code> (default), <code>continue</code> , or <code>branch:<step-id></code>

Example 1 — Prometheus alert webhook

```
- step:
  id: wait_prometheus_alert
  type: wait_for_event
  title: "Wait for Prometheus alert to fire"
  event:
    source: webhook
    id: prometheus.alert.firing
    filter:
      alertname: "HighErrorRate"
      severity: critical
  capture:
    alert_value: json.commonAnnotations.value
    alert_labels: json.commonLabels
```

```
timeout: 30m
on_timeout: fail
contract:
  effects: []
  reads: []
  writes: [alert_value, alert_labels]
```

Example 2 — Named channel / approval callback

```
- step:
  id: wait_approval_callback
  type: wait_for_event
  title: "Wait for external approval system callback"
  event:
    source: channel
    id: "approval.{{ .request_id }}"
  capture:
    approved_by: json.approver
    decision: json.decision
  timeout: 48h
  on_timeout: branch:escalate_approval
  contract:
    effects: []
    reads: [request_id]
    writes: [approved_by, decision]
```

Transport Note The runtime is responsible for exposing the inbound endpoint; the runbook schema only declares intent. The specific endpoint URL or address is runtime-determined and available to subsequent steps as `{{ .gert.event.<id>.endpoint }}`. Event source semantics:

- **webhook** — runtime opens an HTTP POST listener; the sender POSTs a JSON body.
- **message** — runtime subscribes to a message-queue topic or queue name given by `event.id`.
- **signal** — runtime registers a named OS or process signal handler.

- **channel** — runtime uses a gert-internal named channel, enabling runbook-to-runbook or tool-to-runbook event delivery.

Security Note Webhook endpoints are authenticated per run. The runtime generates a one-time HMAC token when the endpoint is opened; the token is available as `{{ .gert.event.<id>.token }}` for use in configuring the sending system (e.g. embedding in a Prometheus alertmanager receiver URL). The token is single-use and expires when the endpoint closes (on resume or timeout).

4.4.8 Step Type: invoke

Invokes another runbook inline as a sub-procedure. The invoked runbook's outputs are mapped into the caller's variable space.

Field	Type	Required	Description
<code>invoke.runbook</code>	string	Yes	Import alias or relative path
<code>invoke.inputs</code>	map	No	Input bindings (template-expanded values)
<code>invoke.outputs</code>	map	No	Output extraction: local-var → output-name
<code>invoke.gate</code>	object	No	Stop-if conditions on child outcome

```
imports:
  connectivity: ./connectivity-check.runbook.yaml

flow:
  - step:
      id: run_connectivity
      type: invoke
      title: Run connectivity check
      invoke:
        runbook: connectivity
        inputs:
          service_name: "{{ .hostname }}"
          incident_id:  "{{ .incident_id }}"
        outputs:
```



```

    conn_status: health_status    # local var ← child output
  gate:
    stop_if: [escalated]

```

4.4.9 Step Type: branch

Evaluates a list of conditions and executes the first matching arm. In v2, **branch** is an explicit first-class step type (as well as the inline **branches:** sibling on any step node).

Field	Type	Required	Description
branches	array	Yes	List of Branch objects
branches[].condition	string	Yes	Go template evaluating to true/false
branches[].label	string	No	Display label for diagram/-trace
branches[].steps	array	No	Steps to execute if condition matches

At most one branch arm executes. If no arm matches and no arm has an **else: true** marker, execution continues past the branch node with no steps run. The **else: true** marker (new in v2) designates a catch-all arm:

```

- step:
  id: route_on_env
  type: branch
  title: Route by environment
  branches:
    - condition: '{{ eq .environment "production" }}'
      label: Production path
      steps:
        - step:
          id: require_approval
          type: manual
          title: Require change-manager approval
          approvals:
            min: 1
            roles: [change-manager]

```

```
- condition: '{{ eq .environment "staging" }}'
  label: Staging path
  steps:
    - step:
      id: log_staging
      type: cli
      title: Log staging deploy
      command: echo
      args: ["Deploying to staging"]
- else: true
  label: Unknown environment - abort
  steps:
    - step:
      id: abort
      type: end
      title: Unknown environment
      outcome:
        category: escalated
        code: unknown_env
```

4.4.10 Step Type: iterate

Repeats a block of steps over a collection or until a convergence condition is met. In v2 **iterate** is a `TreeNode` sibling (not a step type in the `step.type` enum) — it appears at the tree level alongside `step` nodes.

Field	Type	Required	Description
<code>iterate.over</code>	string	*	Template resolving to comma-separated list
<code>iterate.as</code>	string	No	Loop variable name (default: <code>item</code>)
<code>iterate.max</code>	int	*	Max iterations (convergence mode)
<code>iterate.until</code>	string	*	Convergence condition expression
<code>iterate.collect</code>	map	No	Accumulate captures across passes
<code>iterate.steps</code>	array	Yes	Steps to execute each pass

* `over` or (`max` + `until`) required; mutually exclusive.

The loop variable `{{ .iteration }}` (0-indexed) is always available. In list mode, `{{ .item }}` (or the `as` name) holds the current element.

```
# List iteration
- iterate:
  id: check_all_services
  over: "{{ .service_list }}" # e.g. "api,worker,scheduler"
  as: svc
  collect:
    results: "{{ .svc }}:{{ .http_result }}"
  steps:
    - step:
      id: probe
      type: tool
      title: "Check {{ .svc }}"
      tool:
        name: curl
        action: head
        args:
          url: "https://{{ .svc }}.internal/healthz"
      capture:
        http_result: stdout
```

```
# Convergence iteration
- iterate:
  id: wait_for_deploy
  max: 10
  until: '{{ eq .deploy_status "Running" }}'
  steps:
    - step:
      id: poll_deploy
      type: tool
      title: "Poll deployment (pass {{ .iteration }})"
      tool:
        name: kubectl
        action: get-deploy-status
        args:
          name: "{{ .deployment }}"
      capture:
        deploy_status: json.status.phase
    - step:
      id: wait
      type: cli
      title: Wait 15s
      command: sleep
      args: ["15"]
```

4.4.11 Step Type: parallel

Executes a set of independent step groups concurrently, then joins. New in v2, addressing the fan-out/fan-in gap identified in Dennis's research brief.

Field	Type	Required	Description
branches	array	Yes	Each branch is an independent group
join.wait_for	enum	No	all any majority (default: all)
join.on_failure	enum	No	fail continue (default: fail)

Parallel branches are isolated from each other's variable writes during execution. After the join, all `capture` values from all branches are merged into the caller's variable map. Write conflicts (two branches writing the same variable) are resolved by last-writer-wins and a warning is emitted.

```
- step:
  id: parallel_checks
  type: parallel
  title: Run DNS and HTTP checks concurrently
  branches:
    - label: DNS check
      steps:
        - step:
            id: dns_check
            type: tool
            tool:
              name: nslookup
              action: lookup
              args:
                host: "{{ .hostname }}"
            capture:
              dns_result: stdout
    - label: HTTP check
      steps:
        - step:
            id: http_check
            type: tool
            tool:
              name: curl
              action: head
              args:
                url: "https://{{ .hostname }}/healthz"
            capture:
              http_result: stdout
  join:
    wait_for: all
```

```
on_failure: fail
```

4.4.12 Step Type: assert

Evaluates a list of assertions against captured variables. Fails the run if any assertion does not pass.

```
- step:
  id: verify_deployment
  type: assert
  title: Verify deployment succeeded
  assert:
    - type: contains
      subject: "{{ .deploy_output }}"
      expected: "successfully rolled out"
    - type: eq
      subject: "{{ .exit_code }}"
      expected: "0"
    - type: json_path
      subject: "{{ .api_response }}"
      path: "$.status"
      expected: "active"
```

Assertion types: contains, not_contains, matches (regex), eq, ne, lt, gt, exit_code, json_path.

4.4.13 Step Type: compensate

New in v2. Registers a compensation (rollback) action that is executed in reverse order if the run fails or is cancelled after this step. Implements the saga pattern for long-running operational transactions.

Field	Type	Required	Description
compensate.on	enum	No	failure cancel any (default: failure)
compensate.steps	array	Yes	Steps to execute as compensation

The compensation steps execute in LIFO order relative to the sequence of `compensate` nodes encountered during the forward execution pass. Compensation steps share the same variable context as the point at which the `compensate` node was registered.

```
- step:
  id: scale_down
  type: tool
  title: Scale down service
  tool:
    name: kubectl
    action: scale
    args:
      deployment: "{{ .deployment }}"
      replicas: "0"

- step:
  id: register_scale_up
  type: compensate
  title: Compensation - scale back up on failure
  compensate:
    on: failure
    steps:
      - step:
        id: scale_up_comp
        type: tool
        title: Restore service scale
        tool:
          name: kubectl
          action: scale
          args:
            deployment: "{{ .deployment }}"
            replicas: "{{ .original_replicas }}"
```

4.4.14 Step Type: end

Declares a terminal outcome and halts execution. Renamed from `router` in v1.

```
- step:
  id: resolved
  type: end
  title: Incident resolved
  outcome:
    category: resolved      # resolved | escalated | no_action | needs_rca
    code: dns_healthy
    meta:
      resolution_time: "{{ .elapsed }}"
```

4.5 Tool Definition Schema (v2)

Tool definitions describe callable tools: their actions, argument schemas, transport, and behavioral contract. In v2 the `tool/v0` version is replaced by `tool/v2`.

4.5.1 Tool Definition Field Inventory

```
$schema: "https://schemas.gert.dev/tool/v2.json"
apiVersion: tool/v2
```

```
meta:
  name: kubectl
  version: "1.2"
  description: Kubernetes CLI wrapper
  binary: kubectl
```

```
transport:
  mode: stdio      # stdio | jsonrpc | mcp
  # jsonrpc options:
  # start: ["node", "server.js"]
  # port: 3000
  # mcp options:
  # start: ["python", "mcp_server.py"]
```

```
actions:
  get-pods:
```



```
description: List pods in a namespace
argv: ["get", "pods", "-n", "{{ .namespace }}", "-o", "json"]
args:
  namespace:
    type: string
    required: true
    description: Kubernetes namespace
  output:
    type: string
    required: false
    default: json
capture:
  stdout:
    format: json    # json | text | lines
contract:
  effects: [k8s.read]
  reads: [namespace]
governance:
  action: allow
```

New in v2

- `capture.stdout.format` — declares the output format; enables structured `json.<path>` capture in runbook steps.
- `actions.<name>.contract` — per-action behavioral contract automatically merged with any step-level contract override.
- `actions.<name>.governance` — per-action governance policy; overrides the runbook-level rules for this action only.
- `meta.version` — explicit tool version for auditing and pinning.

4.6 Provider Definition Schema (v2)

Provider definitions declare external input resolution services. In v2 the `provider/v0` version is replaced by `provider/v2`.

```
$schema: "https://schemas.gert.dev/provider/v2.json"
apiVersion: provider/v2

meta:
  name: incident
  version: "1.0"
  description: PagerDuty incident context provider
  kind: input-provider      # input-provider | enrichment-provider

transport:
  mode: jsonrpc
  start: ["/providers/pagerduty-provider"]

fields:
  id:
    type: string
    description: Active incident ID
  title:
    type: string
    description: Incident title
  severity:
    type: string
    enum: [P1, P2, P3, P4]
  affected_service:
    type: string
    description: Primary impacted service name

capabilities:
  - input-resolution
```

The fields map defines the provider's schema. Each field name corresponds to a **from:** `<provider>.<field>` binding in a runbook input. The semantic validator checks that all **from:** `<provider>.*` bindings reference a field declared in a loaded provider definition.

4.7 Namespace Convention for Extensions

4.7.1 The Extension Field Problem

Strict JSON Schema validation (`additionalProperties: false`) is a core v2 design constraint: unknown fields in a runbook must be rejected, not silently ignored. At the same time, extensions must be able to attach metadata to steps, runbooks, and tool definitions without forking the core schema.

4.7.2 Convention: `x-<namespace>`: at the Field Level

v2 adopts the `x-<namespace>` prefix convention used by OpenAPI and GitHub Actions. Extension fields must:

1. Start with `x-`.
2. Be followed by a namespace component (lowercase alphanumeric and hyphens, matching `[a-z][a-z0-9-]+`).
3. Be declared in an **extension manifest** registered via `gert extension register` or in `gert.yaml`.

At the runbook level

```
$schema: "https://schemas.gert.dev/runbook/v2.json"
apiVersion: runbook/v2
id: deploy-service
name: Deploy Service

x-myorg-cost-center: "engineering-ops"
x-myorg-change-id: "CHG-20260418-001"
x-otel-service-name: "payment-service"
```

At the step level

```
- step:
  id: deploy
  type: tool
```

```

title: Deploy to production
x-myorg-risk-level: high
x-myorg-rollback-id: "{{ .deploy_id }}"

```

4.7.3 Extension Validation

1. **Unregistered extension fields** — The JSON Schema validator accepts any field matching `^x-[a-z][a-z0-9-]+$` without requiring a registered schema. This enables zero-friction annotation. The `gert validate` command emits a `WARNING` for each unregistered extension field.
2. **Registered extension fields** — An extension may ship a JSON Schema fragment that constrains its field values. Once registered, the validator applies the fragment and reports violations as errors, not warnings.
3. **Unknown non-extension fields** — Any field that does not match a core field name and does not start with `x-` is a hard validation error (`additionalProperties: false`).

4.7.4 Extension Registration Format

```

# .gert/extensions/myorg.yaml
apiVersion: extension/v1
namespace: myorg
description: MyOrg internal runbook annotations

fields:
  cost-center:
    type: string
    description: Cost center code for billing
    pattern: "^[a-z]+-[a-z]+"
  risk-level:
    type: string
    enum: [low, medium, high, critical]
  change-id:
    type: string
    pattern: "^CHG-[0-9]{8}-[0-9]{3}$"

```

```
rollback-id:
  type: string

applies-to: [runbook, step]
```

The extension schema is merged into the live JSON Schema when the extension is registered, enabling IDE auto-completion for extension fields as well.

4.8 Structural vs. Semantic Validation

v2 validation runs in two sequential phases. Both must pass for a runbook to be considered valid.

4.8.1 Phase 1: Structural Validation

Structural validation is pure JSON Schema Draft 2020-12 evaluation. It checks:

- **Required fields** present (`id`, `name`, `apiVersion`, `flow`).
- **Field types** correct (string vs. integer vs. object vs. array).
- **Enum values** valid (e.g. `kind` must be one of the four defined values).
- **Pattern constraints** satisfied (e.g. `id` must match slug pattern; `timeout` must match `[0-9]+(s|m|h)`).
- **minItems** / **minLength** constraints met (e.g. `flow` must have at least one node; `choices.options` must have at least two options).
- **additionalProperties** — no unknown core fields present.

Structural validation is performed by the generated JSON Schema artifact (see §4.9). It is stateless: it does not load tool definitions, resolve imports, or evaluate templates.

Implementation note for Brian The Go runtime uses the `invopop/jsonschema` library to generate the JSON Schema from struct tags at startup (no static file required), then validates the parsed YAML document against it. The `yaml.v3 KnownFields(true)` call provides an additional pre-JSON-Schema pass that rejects unknown fields before the struct is even unmarshalled.

4.8.2 Phase 2: Semantic Validation

Semantic validation enforces domain rules that cannot be expressed in JSON Schema. It runs after structural validation and requires the full runbook AST and loaded tool/provider definitions. Semantic checks include:

1. **Step ID uniqueness** — All step IDs in the flat walk of the tree must be unique within the runbook.
2. **Invoke target resolution** — Every `invoke.runbook` value must resolve to a declared `imports` alias or a relative file path that exists on disk.
3. **Branch condition variable references** — Each `condition` template references only variables that are in scope at that point in the execution tree (declared as `inputs`, `vars`, or captured by an earlier step on a reachable path).
4. **Tool reference resolution** — Every `tool.name` in a tool step must match a declared `toolRefs` entry, and the named `.tool.yaml` file must be loadable and structurally valid.
5. **Tool action existence** — The `tool.action` must be declared in the referenced tool definition.
6. **Provider binding resolution** — Every `inputs.*.from: <provider>.<field>` must resolve to a loaded provider whose field manifest declares that field name.
7. **Circular invoke detection** — The invoke graph must be a DAG; any cycle is a semantic error.
8. **Capture write-before-read** — A step that reads a captured variable via a template must have a reachable predecessor step that writes it; a warning is emitted if the write is only on some branches.
9. **Governance policy pre-check** — The static governance rules are evaluated against the step contracts declared in the runbook. Any step with an effect that would trigger `deny` is flagged as a semantic error. Steps that would trigger `require-approval` are flagged as warnings.
10. **Output declaration completeness** — Every declared `output` must reference a capture name that is written by at least one step.

11. **Compensate ordering** — Each `compensate` step must appear after the step it is compensating for in tree walk order.

Error reporting Both validation phases produce structured error reports with:

- Error code (e.g. SCH-001, SEM-007)
- File path and YAML line/column (where available)
- Human-readable message
- Suggested fix (where computable)

```
$ gert validate service-health.runbook.yaml
[SEM-003] step "check_http": tool action "head" not declared in
          tools/curl.tool.yaml (declared actions: get, post, download)
          --> service-health.runbook.yaml:34:7
[SEM-007] step "eval": template references variable "dns_result" which
          is only written on branch "DNS check" - may be unbound
          --> service-health.runbook.yaml:51:12
          hint: add a default value in vars: or guard with {{ if .dns_result }}
```

4.9 Schema Artifacts

4.9.1 Shipped Artifacts

`gert v2` ships the following schema artifacts:

Artifact	Description
<code>runbook.v2.schema.json</code>	JSON Schema Draft 2020-12 for runbook documents
<code>tool.v2.schema.json</code>	JSON Schema for tool definitions
<code>provider.v2.schema.json</code>	JSON Schema for provider definitions

All three schemas are generated at runtime from Go struct tags via `invopop/jsonschema` and served by `gert schema export`. They are also embedded in the binary and served at <https://schemas.gert.dev/> for IDE tooling.

4.9.2 gert schema export

```
# Export runbook schema (default)
gert schema export

# Export specific schema
gert schema export --kind runbook    # runbook | tool | provider | bundle

# Export full bundle (all three)
gert schema export --kind bundle > gert-v2-schemas.json

# Validate a document against the exported schema
gert schema export | check-jsonschema --schemafile - runbook.yaml
```

The bundle format is a JSON object with three keys: `runbook`, `tool`, and `provider`.

4.9.3 VS Code Integration

The gert VS Code extension configures schema-backed YAML validation via the `yaml-language-server` extension:

```
# .vscode/settings.json (auto-generated by gert vscode init)
{
  "yaml.schemas": {
    "https://schemas.gert.dev/runbook/v2.json": "**/*.runbook.yaml",
    "https://schemas.gert.dev/tool/v2.json":    "**/*.tool.yaml",
    "https://schemas.gert.dev/provider/v2.json": "**/*.provider.yaml"
  }
}
```

Because the schema URL matches the `$schema` field embedded in each document, IDEs with built-in JSON Schema support (VS Code 1.80+, IntelliJ 2024+) pick up validation and auto-completion without any per-project configuration.

The gert VS Code extension additionally provides:

- Step-type-aware field completion via the `schema/stepFields` JSON-RPC method (returns the field set for a given `type` value).

- Tool argument completion via `schema/toolArgs` (returns the `args` schema for a given tool+action pair from the live tool catalog).
- Inline governance warnings derived from `exec/dryRun` pre-check.

4.9.4 Schema Version Manifest

Each schema artifact carries a version manifest in its `$id` and a `x-gert-version` annotation:

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.gert.dev/runbook/v2.json",
  "x-gert-version": "2.0.0",
  "x-gert-released": "2026-04-18",
  "title": "gert runbook v2",
  ...
}
```

Automated compatibility testing (part of the gert CI pipeline) runs every v1 example runbook through the v2 compatibility parser and asserts no regression, and runs every v2 example through the exported JSON Schema validator.

Chapter 5

Extension Runtime

The gert v2 extension runtime defines how external packages contribute tools, schema fields, governance policies, and input providers to a running gert host without requiring changes to the host binary. Extensions run as child processes; they communicate with the host over JSON-RPC 2.0 and are governed by an explicit capability grant model.

A normative reference specification is maintained at `specs/002-extension-runtime-v0/spec.md` in the repository. This chapter derives from and supersedes that document for design purposes; in any conflict between the two, the specification takes precedence for implementation.

5.1 Architecture Overview

1. The gert host discovers extension manifests on startup.
2. The host evaluates compatibility and capability policy before launching any process.
3. Each accepted extension is started as a child process; its executable communicates with the host over its declared transport.
4. The host and extension perform a versioned handshake. The host sends the set of capabilities it is willing to grant; the extension confirms receipt.
5. The extension registers its contributions (tools, schema fields, policies, providers).
6. For the remainder of the run the host dispatches invocations to the extension and enforces all capability constraints.
7. At run completion the host sends a graceful shutdown signal.

5.2 Capability Taxonomy

Every operation an extension may perform is gated by a named capability. Capabilities are requested in the manifest and granted (or denied) by host policy before the process is started. The initial capability set is given in Table 5.1.

The capability string format is `capability/<name>`. Future versions may introduce scoped variants using path notation, e.g. `capability/file-read:logs/**`.

5.3 Extension Manifest Format

Each extension ships a `gert-extension.yaml` file alongside its executable. The manifest declares identity, requested capabilities, transport, and compatibility constraints.

```
# gert-extension.yaml - Extension Manifest
apiVersion: extension/v2

meta:
  name: acme.incident-pack           # Reverse-DNS namespaced identifier
  version: "1.2.0"                  # Semver
  description: "Acme incident resolution tools and providers"
  author: "Acme Platform Team <platform@acme.example>"

compatibility:
  api-version: ">=2.0.0 <3.0.0"      # Host API version range (semver)
  protocol-version: "2.0"           # Extension protocol version

entry-point:
  executable: "./bin/acme-extension"
  args: ["--config", "acme.yaml"]

transport: stdio-jsonrpc             # stdio-jsonrpc | grpc | mcp

capabilities:
  - capability/tool-registration
  - capability/provider-registration
  - capability/schema-extension
```

```
- capability/network
- capability/file-read

# Required when capability/file-read is declared:
file-read-paths:
  - "./runbooks/**"
  - "./data/**"

# Required when capability/network is declared:
network-hosts:
  - "api.pagerduty.com"
  - "*.acme.example"

# Required when capability/env-read is declared:
env-read-patterns:
  - "ACME_API_*"
  - "PD_TOKEN"

platform:                                # Optional OS/arch constraints
  os: ["linux", "darwin", "windows"]
  arch: ["amd64", "arm64"]
```

Manifest validation is performed by the host before the process is started. An extension with an unsatisfied `compatibility.api-version` range is rejected before any subprocess is created.

5.4 Extension Discovery

The host discovers extensions using the following resolution order, with later entries taking precedence:

1. **Built-in extensions:** compiled into the host binary; always available; no manifest required.
2. **Workspace config:** extensions declared in `.gert/extensions.yaml` at the workspace root.

3. **Runbook declaration:** extensions listed in the runbook's top-level `extensions:` field.
4. **CLI override:** `-extension <path>` flags provided at invocation time.

The `.gert/extensions.yaml` workspace config format:

```
# .gert/extensions.yaml
extensions:
  - path: "./extensions/acme-pack"    # directory containing gert-extension.yaml
  - path: "/opt/gert-extensions/audit-pack"
```

The runbook `extensions:` field:

```
apiVersion: runbook/v2
meta:
  name: incident-triage
extensions:
  - path: "./extensions/acme-pack"
```

When the same extension appears in multiple discovery sources, the host deduplicates by `meta.name`. If two distinct extensions share the same name, the one with the higher resolution precedence wins; the lower one is logged as a warning and skipped.

5.5 Handshake and Lifecycle Protocol

All messages use JSON-RPC 2.0. The host is always the initiator; the extension never sends unsolicited requests.

5.5.1 extension/initialize (host → extension)

Sent immediately after the extension process starts. The host communicates the capabilities it is willing to grant and its own version identity.

Request params:

```
{
  "jsonrpc": "2.0",
```

```
"id": "1",
"method": "extension/initialize",
"params": {
  "host": {
    "name": "gert",
    "version": "2.0.0",
    "apiVersion": "2.0"
  },
  "protocolVersion": "2.0",
  "grantedCapabilities": [
    "capability/tool-registration",
    "capability/network"
  ],
  "runContext": {
    "runId": "run-abc123",
    "mode": "real"
  }
}
}
```

Success response:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "result": {
    "extension": {
      "name": "acme.incident-pack",
      "version": "1.2.0"
    },
    "protocolVersion": "2.0",
    "acknowledgedCapabilities": [
      "capability/tool-registration",
      "capability/network"
    ]
  }
}
```

Error codes for this method:

- -32001: Protocol version not supported by extension.
- -32002: Required capability not granted (extension considers the capability non-negotiable).
- -32003: Host API version outside extension compatibility range.

5.5.2 contributions/list (host → extension)

Sent after a successful `extension/initialized` notification. The extension returns its contribution manifest, filtered to the granted capability set.

```
{
  "jsonrpc": "2.0",
  "id": "2",
  "method": "contributions/list",
  "params": {}
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "id": "2",
  "result": {
    "tools": [ { "name": "acme.pd-lookup", ... } ],
    "providers": [ { "name": "pd", "prefixes": ["pd."] } ],
    "schemaExtensions": [],
    "policyContributions": []
  }
}
```

5.5.3 extension/ping (host → extension)

Periodic health check. The extension **MUST** respond within the host-configured ping timeout (default 5 seconds). A non-response triggers a crash-recovery cycle (Section 5.7.3).

```
{ "jsonrpc": "2.0", "id": "42", "method": "extension/ping", "params": {} }
```

Response:

```
{ "jsonrpc": "2.0", "id": "42", "result": { "status": "ok" } }
```

5.5.4 extension/shutdown (host → extension)

Graceful termination. The extension SHOULD drain in-flight requests and release resources within the shutdown timeout (default 10 seconds).

```
{ "jsonrpc": "2.0", "id": "99", "method": "extension/shutdown", "params": {} }
```

Response (best-effort; host may not wait):

```
{ "jsonrpc": "2.0", "id": "99", "result": { "status": "ok" } }
```

If the extension does not exit within the shutdown timeout, the host sends SIGTERM followed by SIGKILL after a grace period.

5.5.5 Lifecycle State Machine

1. **discovered**: manifest found and parsed; no process yet.
2. **verified**: manifest validated; capabilities evaluated against policy.
3. **starting**: executable launched; stdin/stdout pipes attached.
4. **initializing**: `extension/initialize` sent; awaiting response.
5. **ready**: `contributions/list` completed; accepting invocations.
6. **draining**: `extension/shutdown` sent; waiting for in-flight completions.
7. **stopped**: process has exited cleanly.
8. **crashed**: process exited unexpectedly (see Section 5.7.3).

5.6 Versioning and Compatibility

The host maintains a monotonically increasing API version string following semver. The extension manifest declares a semver range in `compatibility.api-version`. Compatibility is evaluated before any process is started:

- If the host API version falls outside the declared range, the extension is rejected with a structured error logged to the run trace. The run continues with the remaining available extensions; a warning is surfaced to the operator.
- During `extension/initialize`, the host also sends its `protocolVersion`. If the extension does not support that protocol version it MUST return error code `-32001`.
- If an extension requests a capability that the host version does not implement (e.g., a capability introduced in a later host version), the capability is silently absent from `grantedCapabilities`. The extension SHOULD treat absent non-required capabilities as optional degradation rather than a fatal error.

Host API versions are tracked in `specs/api-versions.md`. Each new capability addition constitutes a minor version bump; breaking changes to existing method contracts constitute a major version bump.

5.7 Sandboxing and Process Isolation

5.7.1 Process Isolation

Extensions always run as child processes of the gert host. They are never loaded as shared libraries or plug-ins executed in-process. The host:

- Sets a restricted environment: only env vars matching `env-read-patterns` from the manifest are forwarded.
- Attaches the extension's stderr to a structured log collector; raw stderr lines are stored in the run trace under `extension.<name>.stderr`.
- Does not forward the host process's file descriptor table beyond stdin/stdout/stderr.

5.7.2 Capability Enforcement

Every RPC method the extension may call is guarded by a capability check. The host maintains a granted-capability set per extension process. Before dispatching any extension-initiated request (e.g., a tool registration during `contributions/list`) the host verifies the required capability is in the granted set. Violations return:

```
{
  "jsonrpc": "2.0",
  "id": "...",
  "error": {
    "code": -32010,
    "message": "capability not granted",
    "data": {
      "required": "capability/file-write",
      "granted": ["capability/file-read", "capability/network"]
    }
  }
}
```

File-path and network-host constraints are enforced by the host at the system call boundary using OS-level mechanisms where available (e.g., `pledge/unveil` on OpenBSD, `seccomp` on Linux) and by host-side validation of all path and URL arguments on other platforms.

5.7.3 Timeout and Crash Handling

- **Per-invocation timeout:** each `tools/invoke` call carries a `timeoutMs` field. If the extension does not respond within that deadline, the host sends a `tools/cancel` request and, if still unresponsive after a grace period, terminates the process.
- **Ping timeout:** if an extension fails to respond to `extension/ping` within 5 seconds it is considered unresponsive. The host logs a structured warning and terminates the process.
- **Unexpected exit:** if the extension process exits with a non-zero status or is killed by a signal, the host marks the extension as **crashed**, emits a `extension.crashed` trace event, and cancels all in-flight invocations with a structured `-32099` (extension crashed) error.

- **Host isolation:** extension crashes never crash the host. A run that loses an extension mid-execution surfaces an error on the affected step and allows the operator to decide whether to abort or continue with a degraded tool set.

Capability name	Allows	Excludes
capability/tool-registration	Register new tool definitions at runtime via <code>contributions/list</code> .	Does not allow the extension to invoke other tools or modify existing tool definitions.
capability/schema-extension	Contribute namespaced additional fields to the runbook schema (e.g. <code>x-acme.*</code> annotations).	Cannot modify or remove existing schema fields; cannot alter core validation rules.
capability/event-subscription	Subscribe to host runtime events (step-started, step-completed, run-ended, etc.) via the <code>events/subscribe</code> method.	Cannot emit synthetic events into the host event bus.
capability/policy-contribution	Register additional governance policy rules evaluated during pre-flight and approval checks.	Cannot override or disable built-in governance rules; cannot grant capabilities to other extensions.
capability/provider-registration	Register new input providers that handle <code>from:</code> prefix bindings.	Cannot intercept resolution requests destined for another registered provider.
capability/file-read	Read files within the path scope declared in the manifest (<code>file-read-paths</code>).	Cannot read files outside declared paths; cannot write or delete files.
capability/file-write	Write or create files within the path scope declared in the manifest (<code>file-write-paths</code>).	Cannot write outside declared paths; does not imply read access.
capability/network	Make outbound network calls to the hosts declared in the manifest (<code>network-hosts</code>).	Cannot connect to undeclared hosts; cannot listen for inbound connections.
capability/env-read	Read environment variables whose names match the declared <code>env-read-patterns</code> list.	Cannot read undeclared variables; cannot write or unset environment variables.
capability/run-state-read	Read current run state (step status, captured variables, trace events).	Cannot modify run state; does not imply the ability to emit output.

TABLE 5.1: Extension capability taxonomy (v2).

Chapter 6

Tool Runtime

The tool runtime is the subsystem responsible for resolving, validating, and invoking tool definitions at runbook execution time. Tools are the primary mechanism by which gert performs side-effecting work: executing binaries, calling JSON-RPC services [11], and delegating to MCP-hosted capabilities [3].

6.1 Tool Discovery

The host resolves tool references in three phases:

1. **Static discovery:** `.tool.yaml` files are located by scanning the following directories in order:
 - (a) The built-in tool registry compiled into the host binary.
 - (b) `<workspace-root>/tools/` — the conventional location for project-level tool definitions (`tools/<name>.tool.yaml`).
 - (c) Directories listed in the workspace config (`.gert/config.yaml`, field `tool-paths`).
 - (d) Required packages declared in the runbook's `requires:` field (each package's `tools/` subdirectory is scanned).
2. **Dynamic registration:** extensions with `capability/tool-registration` may register additional tool definitions during the `contributions/list` phase (see Section 5.5). Dynamically registered tools are merged into the resolved catalog before execution begins.

3. **MCP discovery**: when the runbook declares an MCP server under `tools.mcp-servers`, the host performs a `tools/list` call to that server at startup and registers every tool reported as a dynamically discovered tool (see Section 6.7).

6.1.1 Name Resolution Order

When a runbook step references a tool by name, the host resolves as follows:

1. Exact match in dynamically registered tools (MCP and extension, in registration order).
2. Exact match in project-level `tools/` directory.
3. Exact match in required-package tool directories (in `requires:` declaration order).
4. Exact match in the built-in registry.

If two tools share the same name, the first match wins and a warning is emitted. Tool authors SHOULD use reverse-DNS namespace prefixes to avoid collisions (e.g., `acme.pd-lookup`).

6.2 Tool Definition Schema (v2)

```
# tools/my-service.tool.yaml
apiVersion: tool/v2

meta:
  name: my-service           # Unique identifier within the workspace
  version: "2.0.0"           # Tool definition version (semver)
  description: "Calls the MyService REST API"
  binary: my-service-cli     # Executable name (looked up in PATH)

transport:
  mode: stdio-jsonrpc        # stdio | stdio-jsonrpc | grpc | mcp

# For stdio-jsonrpc and mcp: startup behaviour
startup:
  argv: ["--server-mode"]    # Extra args passed to the binary
```

```
ready-pattern: "ready"      # Regex matched against stdout/stderr to
                             # confirm the server is accepting requests
timeout: "10s"
shutdown-method: "shutdown" # JSON-RPC method to call on graceful stop

governance:
  requires-capabilities: []      # Host capabilities the tool needs, e.g.
                                # ["capability/network"]
  allowed-environments: ["real", "dry-run"]
  requires-approval: false      # Set true to gate invocation on operator OK

actions:
  lookup:
    description: "Look up an entity by ID"
    argv: ["lookup", "--id", "{{ .id }}"] # Only for stdio transport
    timeout: "30s"
    args:
      id:
        type: string
        required: true
        description: "Entity identifier"
        redact: false
    output:
      format: json             # text | json | jsonl
  capture:
    stdout:
      format: json
```

6.2.1 Transport-Specific Fields

stdio The binary is spawned once per invocation. The `argv` array in the action definition is rendered as a Go template with the resolved argument values, then appended to the binary name to form the complete command. The process's stdout is captured as the tool output.

stdio-jsonrpc The binary is spawned once per run (persistent process). The host keeps a single process alive across all action calls for that tool. Each action invocation sends a JSON-RPC request to the process's stdin and reads the response from stdout. The `startup.ready-pattern` confirms the server is accepting connections.

mcp The host spawns the binary and performs the MCP initialization handshake (Section 6.7). Actions map to MCP `tools/call` requests.

6.3 Transport Layer

6.3.1 stdio Transport

Process establishment. The host calls `exec.Command(binary, resolvedArgv...)` with a controlled environment. A per-invocation timeout context wraps the call.

Invocation envelope. For stdio tools the invocation is entirely encoded in the process arguments and environment; there is no envelope. The arguments are rendered from the action's `argv` template.

Response envelope. Stdout is captured in full after process exit. The exit code is checked: non-zero is mapped to a tool error.

Error envelope.

```
{
  "error": {
    "code": "TOOL_NONZERO_EXIT",
    "message": "tool exited with code 1",
    "details": {
      "exitCode": 1,
      "stderr": "connection refused\n",
      "durationMs": 320
    }
  }
}
```


6.3.2 stdio-jsonrpc Transport

Process establishment. A single process is spawned at first use and kept alive for the duration of the run. The host waits for `startup.ready-pattern` on stdout/stderr before sending the first request.

Invocation envelope.

```
{
  "jsonrpc": "2.0",
  "id": "<invocationId>",
  "method": "<actionName>",
  "params": {
    "args": { "id": "INC-12345" },
    "context": {
      "runId": "run-abc123",
      "stepId": "lookup_incident",
      "traceId": "trace-xyz",
      "userId": "ops@acme.example",
      "attempt": 1
    }
  }
}
```

Response envelope.

```
{
  "jsonrpc": "2.0",
  "id": "<invocationId>",
  "result": {
    "output": { ... },
    "telemetry": {
      "startedAt": "2026-04-18T10:00:00Z",
      "finishedAt": "2026-04-18T10:00:00.320Z",
      "durationMs": 320
    }
  }
}
```

Error envelope.

```
{
  "jsonrpc": "2.0",
  "id": "<invocationId>",
  "error": {
    "code": -32050,
    "message": "entity not found",
    "data": {
      "category": "runtime",
      "retryable": false,
      "details": { "entityId": "INC-12345" }
    }
  }
}
```

Error code ranges:

- -32700 to -32600: JSON-RPC parse and request errors.
- -32050: Application-level tool error (retryable per `data.retryable`).
- -32051: Capability not granted.
- -32052: Input validation failed.
- -32053: Output schema validation failed.
- -32054: Timeout.
- -32055: Cancelled by host.
- -32099: Tool process crashed.

6.3.3 MCP Transport

See Section 6.7 for the full MCP integration specification.

6.4 Invocation Context

Every tool invocation — regardless of transport — receives a structured context object. For `stdio` tools, relevant fields are injected as environment variables prefixed `GERT_`; for `stdio-jsonrpc` and `mcp` tools, the context is transmitted in the invocation envelope.

```
{
  "context": {
    "runId":      "run-abc123",          // Globally unique run identifier
    "stepId":     "lookup_incident",    // Step ID within the runbook
    "traceId":    "trace-xyz789",       // Distributed trace correlation ID
    "userId":     "ops@acme.example",   // Identity executing the run
    "mode":       "real",               // real | dry-run | replay
    "attempt":    1,                   // Retry attempt number (1-based)
    "capabilities": [                  // Effective host capability set
      "capability/network",
      "capability/file-read"
    ]
  }
}
```

6.5 Capability Gate

Before dispatching any tool invocation the host checks two conditions:

1. The tool's `governance.requires-capabilities` list is a subset of the current run's granted capability set. If not, the step fails immediately with a `CAPABILITY_DENIED` error; the run does not attempt to start the tool process.
2. The tool's `governance.allowed-environments` list includes the current execution mode. A tool not listed for `dry-run` will be skipped (with a warning) when the run mode is `dry-run`.

6.6 Output Capture

Tool output is mapped to named step captures using the `capture:` declaration in the action definition.

```
capture:
  dns_output: stdout          # Capture stdout as variable dns_output
  error_log:  stderr          # Capture stderr as variable error_log
  exit_code:  exitCode        # Capture numeric exit code
```

For json output format, capture paths may use dot notation to extract nested fields:

```
capture:
  incident_id: "stdout.incident.id"
  severity:    "stdout.incident.severity"
```

Redaction. Before any captured value is stored in the run state or emitted to the trace, the governance redaction pipeline is applied. Redaction rules are declared in the runbook's `meta.governance.redact` block and in the tool action's per-argument `redact: true` flag. Redacted values are replaced with `<redacted>` in all stored and displayed output.

6.7 MCP Tool Integration

MCP (Model Context Protocol) servers are long-lived processes that expose a dynamic catalog of tools. Gert treats an MCP server as a tool source, not as a single tool.

6.7.1 MCP Server Declaration

```
# In runbook meta or .gert/config.yaml
tools:
  mcp-servers:
    - name: acme-mcp
      binary: acme-mcp-server
      args: ["--port", "stdio"]
      transport: mcp
      startup:
        timeout: "15s"
```

6.7.2 MCP Discovery

On run startup the host:

1. Spawns the MCP server binary.
2. Sends the MCP `initialize` request (per MCP specification).
3. After receiving `initialized`, sends `tools/list`.
4. Registers each returned tool under the name `<server-name>/<tool-name>` in the dynamic tool catalog.

6.7.3 MCP Tool Invocation

When a runbook step calls an MCP-backed tool, the host sends:

```
{
  "jsonrpc": "2.0",
  "id": "5",
  "method": "tools/call",
  "params": {
    "name": "lookup_incident",
    "arguments": { "id": "INC-12345" },
    "_gert_context": {
      "runId": "run-abc123",
      "stepId": "triage_step",
      "traceId": "trace-xyz"
    }
  }
}
```

The `_gert_context` field is a gert extension to the MCP envelope. MCP servers that do not recognise it **MUST** ignore it. The response follows the standard MCP `tools/call` response format.

6.7.4 MCP vs stdio-jsonrpc Differences

- MCP servers expose multiple tools under a single process; stdio-jsonrpc tools are one tool per process.
- MCP uses a distinct initialization handshake (`initialize` / `initialized`) before tool registration.

- MCP tool names are discovered dynamically at runtime; stdio-jsonrpc action names are declared statically in the `.tool.yaml`.
- MCP responses use a `content` array; gert flattens text content to a single string for capture.

6.8 Cancellation and Timeout

Per-tool timeout. Declared in the action definition under `timeout:` (e.g., 30s). The default is the runbook's `meta.defaults.timeout`. The host wraps each invocation in a `context.WithTimeout`; when the deadline elapses the host sends a `tools/cancel` request (JSON-RPC transport) or sends SIGTERM (stdio transport), then forcibly terminates the process after a 5-second grace period.

Cancellation propagation. When a run is cancelled (via `Ctrl-C`, the debugger quit command, or an API cancellation request), the host:

1. Sets the run-level cancel context.
2. Sends `tools/cancel` to all in-flight JSON-RPC and MCP invocations with a `reason: "run-cancelled"` field.
3. After a 5-second drain window, forcibly terminates any unresponsive tool processes.
4. Emits a `run.cancelled` trace event recording which steps were interrupted.

The `tools/cancel` request envelope:

```
{
  "jsonrpc": "2.0",
  "id": "cancel-5",
  "method": "tools/cancel",
  "params": {
    "invocationId": "5",
    "reason": "run-cancelled"
  }
}
```

6.9 Tool Versioning

Tool definitions carry a `meta.version` field. When multiple definitions for the same tool name are present (e.g., from a required package and from a workspace override), the host selects by resolution order (Section 6.1), not by version number. Explicit version pinning is supported via the runbook's `requires:` block:

```
requires:
- package: github.com/acme/gert-tools
  version: ">=1.2.0 <2.0.0"
```

Chapter 7

Runtime Events and Determinism

Gert v2 is an event-driven system. Every state change in the runtime—run initiation, step advancement, governance checks, tool invocations, human interactions—produces a structured event. Events serve three purposes: (1) adapters (TUI, Web, VS Code) subscribe to events to render UI state without coupling to execution logic, (2) events are appended to the JSONL trace file for audit and replay (§12), and (3) events propagate to observability systems for real-time monitoring and alerting.

This section defines the complete event system: the event envelope schema, delivery semantics, the normative event catalog, event channels, subscription API, wire format, and the required vs. optional event classification.

7.1 Event System Design

7.1.1 Role of Events

Events are **fan-out notifications to subscribers**, not a command or control channel. The Runtime Core emits events as a consequence of state transitions; events do not trigger state transitions. This design ensures that execution semantics remain centralized in the Runtime Core and adapters remain passive renderers.

Key principles:

- **One-way flow:** Events flow from the Runtime Core outward to subscribers. Subscribers never write events back to the runtime.
- **Non-blocking emission:** Event emission must not block step execution. The event bus uses a bounded channel with a discard-on-full policy for in-process subscribers to prevent slow consumers from stalling the runtime.

- **Immutable once emitted:** Events are value objects; they cannot be modified after emission.
- **Best-effort delivery to in-process subscribers:** If a subscriber channel is full, the event is dropped. The trace file write (§12) is the authoritative record.

7.1.2 Delivery Semantics

At-least-once delivery. The JSONL trace file receives every event via a synchronous write before the runtime proceeds to the next action. In-process subscribers (TUI, event forwarders) receive events on a best-effort basis; if the subscriber channel is full, the event is discarded without blocking the runtime.

In-order per run. All events within a single run are totally ordered by their **sequence** field. Events from different runs are not ordered relative to each other. Subscribers that process events asynchronously must tolerate out-of-order arrival from concurrent runs.

Replay determinism. Events written to the trace file are sufficient to reconstruct the execution timeline. Replay mode (§12.3) re-emits the same events in the same order, allowing adapters to render a historical run indistinguishably from a live run.

7.2 Event Envelope

Every event carries a mandatory envelope with the following fields. Consumers **MUST** tolerate unknown fields to support additive schema evolution.

Field	Type	Description
<code>event_id</code>	UUID v4	Globally unique event identifier
<code>run_id</code>	UUID	Execution run identifier (stable across pauses/resumes)
<code>runbook_id</code>	string	Runbook name or path from runbook YAML <code>meta.name</code>
<code>timestamp</code>	RFC3339	UTC timestamp with microsecond precision [12]
<code>kind</code>	string	Event type (e.g., <code>step/started</code> , <code>governance/approved</code>)
<code>sequence</code>	int64	Monotonically increasing integer within this run (starting from 0)
<code>payload</code>	object	Event-specific data (schema varies per kind)

Sequence numbers. The `sequence` field provides a total order over events in a single run. It is assigned by the `RunStore.WriteTrace` implementation and persists across pause/resume cycles. Sequence numbers are not globally unique; they are scoped to a single `run_id`.

Timestamps. The `timestamp` field uses RFC3339 format with at least microsecond precision (e.g., `2026-04-18T14:32:05.123456Z`). Timestamps are wall-clock times and must not be used for precise duration calculations (use event-specific `duration_ms` fields instead).

Event IDs. The `event_id` is a UUID v4 assigned at emission time. It is globally unique across all runs and all gert deployments. Event IDs are used for correlation in distributed tracing systems (§15) and for deduplication in event forwarding pipelines.

7.3 Event Catalog

The following subsections define every normative event kind in v2.0. Each definition includes the payload schema, emission semantics, and whether the event is required or optional.

7.3.1 Run Lifecycle Events

run/started

Emitted: Once, as the first event in every execution. Must be the first line in the trace file.

Payload:

```
{
  "runbook_id":    <string>, // from meta.name or file path
  "version":       <string>, // runbook schema version (e.g., "runbook/v2")
  "actor":         <string>, // identity (from --as flag or adapter session)
  "input_hash":    <string>, // SHA256 of sorted input key-value pairs
  "mode":          <string>, // "real" | "dry-run" | "replay"
  "client":        <string>, // "cli" | "tui" | "web" | "vscode" | "mcp"
  "parent_run_id": <string>  // omitted if not an invoke child
}
```

Semantics. The `input_hash` is the SHA256 hex digest of the JSON-serialized input dictionary with keys sorted lexicographically. This hash is used to detect input tampering in replay mode. The `parent_run_id` field is present only if this run was initiated by an invoke step in another run.

Required: YES. Every run MUST emit this event.

run/completed

Emitted: Once, as the terminal event for a successful run (all steps passed or skipped).

Payload:

```
{
  "outcome":      <string>, // "success" | "failure" | "cancelled"
  "step_count":   <int>,    // total steps in the plan
  "passed":       <int>,    // steps that completed successfully
  "failed":       <int>,    // steps that failed
  "skipped":      <int>,    // steps skipped (branch/iterate)
  "duration_ms":  <int64>   // wall-clock duration from run/started
}
```

Semantics. The `outcome` field distinguishes clean completion (`success`), error termination (`failure`), and operator-initiated halt (`cancelled`). If `outcome` is `failure`, the last event before this one should be a `step/failed` or `governance/blocked`.

Required: YES. Every run that reaches a terminal state MUST emit this event.

run/cancelled

Emitted: When a run is cancelled by operator action (SIGINT, API call, or approval rejection).

Payload:

```
{
  "reason":      <string>, // "user_interrupt" | "approval_rejected" | "timeout"
  "cancelled_by": <string>, // actor identity (omitted if SIGINT)
  "step_id":     <string>  // step that was active at cancellation time
}
```

Semantics. This event is followed immediately by run/completed with outcome: "cancelled". The reason field disambiguates graceful operator cancellation (user_interrupt) from governance-driven rejection (approval_rejected).

Required: YES, if a run is cancelled before natural completion.

7.3.2 Step Lifecycle Events

step/started

Emitted: Immediately before a step executes (after governance checks pass).

Payload:

```
{
  "step_id":    <string>, // stable step identifier from runbook YAML
  "step_type":  <string>, // "cli" | "manual" | "tool" | "invoke" | "branch"
                                // | "iterate" | "parallel" | "compensate"
  "index":     <int>      // zero-based position in the ExecutionPlan
}
```

Required: YES. Every step that begins execution MUST emit this event.

step/completed

Emitted: After a step exits successfully.

Payload:

```
{
  "step_id":      <string>,
  "outcome":      <string>, // "success" | "skipped"
  "duration_ms":  <int64>,
  "exit_code":    <int>      // for cli/tool steps; 0 for manual/invoke
}
```

Semantics. An outcome of `skipped` occurs for steps in non-taken branch paths or exhausted iterate loops. Exit code is present only for `cli` and `tool` steps; for other step types it is always 0.

Required: YES, for every step that completes without error.

`step/failed`

Emitted: When a step exits with a non-zero exit code or runtime error.

Payload:

```
{
  "step_id":      <string>,
  "error_code":    <string>, // "exit_nonzero" | "timeout" | "tool_error"
                                // | "governance_blocked" | "invoke_failed"
  "error_message": <string>, // human-readable (redacted)
  "retryable":     <bool>,    // true if retry policy allows retry
  "exit_code":     <int>      // for exit_nonzero; omitted otherwise
}
```

Semantics. The `retryable` field indicates whether the runtime will attempt a retry (requires a retry policy on the step). If `retryable` is true, this event will be followed by `step/retrying`.

Required: YES, for every step that fails.

`step/skipped`

Emitted: When a step is not executed due to control flow.

Payload:

```
{
  "step_id": <string>,
```

```

    "reason": <string> // "branch_not_taken" | "iterate_exhausted"
                  // | "dry_run" | "cancelled"
}

```

Required: YES, for every step that is planned but not executed.

step/retrying

Emitted: Before each retry attempt (v2.1 feature; requires retry policy).

Payload:

```

{
  "step_id":      <string>,
  "attempt":      <int>,      // 1 = first retry, 2 = second, etc.
  "max_attempts": <int>,
  "backoff_ms":   <int64>     // delay before this retry
}

```

Required: YES, if retry is attempted. Optional in v2.0 (retry not yet implemented).

7.3.3 Governance Events

All governance events are normatively defined in §11. The following is a summary for completeness.

governance/command_checked

Emitted: When a command is evaluated against allowlist and denylist.

Payload:

```

{
  "step_id":      <string>,
  "command":      <string>, // base name (argv[0] stripped of path)
  "verdict":      <string>, // "allowed" | "denied"
  "rule_matched": <string>  // "allowlist" | "denylist" | "default_allow"
}

```

Required: MAY be emitted. Recommended for audit trails but not required for basic operation.

governance/approval_requested

Emitted: When a step's approval gate is entered.

Payload:

```
{
  "step_id":      <string>,
  "approver_roles": [<string>], // roles that can approve
  "min_approvals": <int>,
  "timeout_seconds": <int>,      // 0 if no timeout
  "message":      <string>      // prompt text for approvers (redacted)
}
```

Required: YES, when approval gates are used.

governance/approval_received

Emitted: Each time an approver submits a decision.

Payload:

```
{
  "step_id": <string>,
  "approver": <string>, // actor identity
  "role": <string>,    // role the approver is acting in
  "decision": <string>, // "approved" | "rejected"
  "comment": <string>  // optional justification (redacted)
}
```

Required: YES, for every approval decision (including rejections).

governance/redaction_applied

Emitted: When redaction rules match and modify captured output.

Payload:

```
{
  "step_id":      <string>,
  "field_path":   <string>, // "stdout" | "stderr" | "capture.var_name"
  "pattern_count": <int>    // number of patterns that matched (not the patterns)
}
```

Semantics. The actual redaction patterns are not logged to avoid accidentally revealing what sensitive data looks like. Only the fact that redaction occurred is recorded.

Required: MAY be emitted. Useful for audit compliance.

7.3.4 Extension and Tool Events

extension/loaded

Emitted: When an extension successfully completes the handshake.

Payload:

```
{
  "extension_id": <string>, // from extension manifest
  "version":      <string>, // semver version
  "capabilities": [<string>] // list of capability names granted
}
```

Required: MAY be emitted. Useful for debugging extension issues.

extension/unloaded

Emitted: When an extension process exits or is terminated.

Payload:

```
{
  "extension_id": <string>,
  "reason":       <string> // "clean_exit" | "crash" | "timeout" | "killed"
}
```

Required: MAY be emitted.

tool/invoked

Emitted: When a tool call is dispatched (before the tool executes).

Payload:

```
{
  "step_id":      <string>,
  "tool_name":    <string>,
}
```



```

"transport":      <string>, // "stdio" | "jsonrpc" | "mcp"
"correlation_id": <string>  // UUID v4, links to tool/completed
}

```

Required: MAY be emitted. Recommended for observability.

tool/completed

Emitted: When a tool call returns.

Payload:

```

{
  "step_id":      <string>,
  "tool_name":    <string>,
  "correlation_id": <string>, // matches tool/invoked
  "exit_code":    <int>,
  "duration_ms":  <int64>
}

```

Required: MAY be emitted.

7.3.5 Human-in-the-Loop Events

input/prompted

Emitted: When a manual step or input provider displays a prompt.

Payload:

```

{
  "step_id":      <string>, // omitted if prompt is from provider
  "prompt_id":    <string>, // UUID v4 for correlation
  "input_type":   <string>, // "text" | "checklist" | "attachment"
  "prompt":      <string>  // prompt text (redacted)
}

```

Required: YES, for manual steps. MAY be emitted for provider prompts.

input/received

Emitted: When the operator submits a response.

Payload:

```
{
  "step_id":    <string>,
  "prompt_id": <string>, // matches input/prompted
  "redacted":  <bool>    // true if redaction rules matched the response
}
```

Semantics. The actual input value is not included in this event (it is recorded in the trace via `manual/completed` or `step/completed`). The `redacted` flag indicates whether the input passed through the redaction pipeline.

Required: YES, for manual steps.

7.3.6 Saga and Compensation Events (v2.1+)

The following events are reserved for the saga/compensation pattern, which is targeted for v2.1. They are included here for forward compatibility.

saga/compensation_triggered

Emitted: When a step fails and has registered compensation handlers.

Payload:

```
{
  "failed_step_id":    <string>,
  "compensation_steps": [<string>] // step IDs to execute in reverse order
}
```

Required: YES, if compensation is triggered (v2.1+).

saga/compensation_completed

Emitted: After all compensation steps complete.

Payload:

```
{
  "outcome": <string>  // "success" | "partial" | "failed"
}
```

Required: YES, if compensation is triggered (v2.1+).

7.4 Event Channels

Events flow through three channels, each with distinct delivery semantics and subscribers.

7.4.1 In-Process Event Bus

Purpose. The in-process event bus delivers events from the Runtime Core to adapters (TUI, `gert serve` session handlers) running in the same process.

Implementation. A Go `chan Event` with a fixed buffer size (default: 100 events). The Runtime Core writes to the channel; adapters read from it. If the channel is full, new events are discarded (the trace file is the authoritative record).

Subscriber API. Adapters call `RunHandle.Events()` to obtain a read-only event channel:

```
type RunHandle interface {
    // Events returns a read-only channel of events for this run.
    // The channel is closed when the run enters a terminal state.
    Events() <-chan Event
    // ... other methods ...
}
```

Lifecycle. The event channel is created at `Runtime.Start()` and closed when the run emits `run/completed` or `run/cancelled`. Subscribers must drain the channel to avoid blocking the runtime (though the runtime will discard events if the subscriber is too slow).

7.4.2 WebSocket Broadcast (HTTP Adapter)

Purpose. For `gert serve -http`, events are broadcast to WebSocket clients for real-time UI updates.

Implementation. The HTTP adapter subscribes to `RunHandle.Events()` and forwards events to all connected WebSocket clients for the given `run_id`. The `/events` endpoint (existing in v1) is upgraded to WebSocket in v2.

Wire format. JSON-encoded event envelopes, one per WebSocket message frame.

Delivery semantics. Best-effort. If a WebSocket client is slow, the adapter may drop events to prevent backpressure. Clients should use the `/runs/:id/trace` HTTP endpoint to fetch the authoritative trace file if they detect gaps.

7.4.3 JSONL Trace File

Purpose. The append-only JSONL trace file (§12) is the durable, authoritative record of all events. It is the only channel with guaranteed at-least-once delivery.

Implementation. The Runtime Core writes every event synchronously to `.runbook/runs/<run-id>/trace.jsonl` before proceeding to the next action. File writes use `O_APPEND` mode to ensure atomicity on Unix systems.

Format. One JSON object per line, with the envelope fields from §6.2. See §12.1 for the complete trace file specification.

7.5 Event Subscriptions

7.5.1 Programmatic Subscription API

Extensions and adapters can subscribe to events programmatically. The subscription API supports filtering by event kind prefix and by `run_id`.

```
// EventSubscriber is the interface for subscribing to runtime events.
type EventSubscriber interface {
    // Subscribe registers a subscription with optional filters.
    // Returns a channel that will receive matching events.
    Subscribe(ctx context.Context, opts SubscribeOptions) (<-chan Event, error)
}
```

```
type SubscribeOptions struct {
    // KindPrefix filters events to those matching the prefix.
    // Examples: "step/", "governance/", "run/started"
    KindPrefix string

    // RunID filters events to a specific run. If empty, all runs.
    RunID string

    // BufferSize is the channel buffer size. Default: 100.
    BufferSize int
}
```

Example use cases.

- A TUI adapter subscribes with KindPrefix: "step/" to render step progress.
- A governance dashboard subscribes with KindPrefix: "governance/" to display policy violations.
- An observability forwarder subscribes with no filters to send all events to an Open-Telemetry collector (§15).

7.6 Wire Format

Events are serialized as JSON. The wire format is identical for the JSONL trace file, WebSocket broadcasts, and JSON-RPC notifications.

7.6.1 Concrete Example: step/started

```
{
  "event_id":    "a3f9c1e2-8d4b-4e1f-9a2c-3d5e6f7a8b9c",
  "run_id":      "20260418T143022-a3f9c1",
  "runbook_id":  "deploy-frontend",
  "timestamp":   "2026-04-18T14:30:25.123456Z",
  "kind":        "step/started",
  "sequence":    3,
  "payload": {
```

```
    "step_id":    "build_assets",
    "step_type":  "cli",
    "index":      2
  }
}
```

7.6.2 Concrete Example: governance/approval_requested

```
{
  "event_id":    "b1c2d3e4-5f6a-7b8c-9d0e-1f2a3b4c5d6e",
  "run_id":      "20260418T143022-a3f9c1",
  "runbook_id":  "deploy-frontend",
  "timestamp":   "2026-04-18T14:32:10.456789Z",
  "kind":        "governance/approval_requested",
  "sequence":    12,
  "payload": {
    "step_id":    "deploy_to_prod",
    "approver_roles": ["SRE", "PM"],
    "min_approvals": 2,
    "timeout_seconds": 3600,
    "message":    "Deploying to production. Confirm change window."
  }
}
```

7.7 Required vs. Optional Events

The following table classifies every event as REQUIRED (MUST be emitted) or OPTIONAL (MAY be emitted). Required events form the minimal contract for adapter compatibility; optional events provide observability and debugging enhancements.

Event Kind	Status	Notes
run/started	REQUIRED	First event in every trace
run/completed	REQUIRED	Terminal event
run/cancelled	REQUIRED	If cancelled
step/started	REQUIRED	For every executed step
step/completed	REQUIRED	For every successful step
step/failed	REQUIRED	For every failed step
step/skipped	REQUIRED	For every skipped step
step/retrying	OPTIONAL	v2.1+ retry feature
governance/command_checked	OPTIONAL	Useful for audit trails
governance/approval_requested	REQUIRED	If approval gates used
governance/approval_received	REQUIRED	If approval gates used
governance/redaction_applied	OPTIONAL	Compliance logging
extension/loaded	OPTIONAL	Debugging
extension/unloaded	OPTIONAL	Debugging
tool/invoked	OPTIONAL	Observability
tool/completed	OPTIONAL	Observability
input/prompted	REQUIRED	For manual steps
input/received	REQUIRED	For manual steps
saga/compensation_triggered	REQUIRED	v2.1+ if saga used
saga/compensation_completed	REQUIRED	v2.1+ if saga used

Adapter compatibility. Adapters **MUST** handle all **REQUIRED** events. They **SHOULD** gracefully ignore unknown event kinds to support forward compatibility (e.g., a v2.0 adapter rendering a v2.1 trace with saga events).

7.8 Determinism and Replay Semantics

7.8.1 Replay Mode

In replay mode (§12.3), the Runtime Core reads events from a historical trace file and re-emits them in sequence order, respecting the original timing and causality. Adapters (TUI, Web, VS Code) subscribe to the replayed event stream and render the historical run indistinguishably from a live run.

Replay guarantees.

- Every event in the original trace is re-emitted with the same **sequence**, **timestamp**, **kind**, and **payload**.

- New `event_id` values are assigned (event IDs are not stable across replays).
- Step execution does not occur; the runtime only emits events. No subprocesses are forked, no tools are invoked, no governance checks are evaluated.

7.8.2 Non-determinism Boundaries

The following actions introduce non-determinism and are not captured in events:

- System time (`timestamp` fields use wall-clock time, not logical time).
- External network calls (DNS resolution, HTTP responses, database queries).
- File system state outside `.runbook/`.
- Random UUIDs (`event_id` and `correlation_id` fields are regenerated in replay).

To achieve full deterministic replay (for testing and debugging), use dry-run mode (§8.2) or snapshot the external environment.

Chapter 8

Security and Trust

Gert v2 is designed to operate in regulated, high-trust environments where operational actions must be auditable, accountable, and governed by explicit policy. Security is not an add-on; it is foundational to the system’s architecture. This chapter defines the threat model, trust boundaries, process isolation mechanisms, credential handling, transport security, RBAC model, audit non-repudiation, and supply chain security.

8.1 Threat Model

Gert v2 addresses the following threat categories, which are common in operational runbook systems and have real-world precedent in systems like AWS Systems Manager, PagerDuty Runbook Automation, and Rundeck [2, 19, 20].

This threat model is derived from NIST SP 800-53 security controls [14] and ISO/IEC 27001 guidance on privileged access management and audit logging [10].

8.2 Extension Trust Model

Extensions operate under one of three trust levels, declared explicitly in the extension manifest and evaluated by the host before any process is started.

8.2.1 Trust Levels

Local trusted. Installed by the machine owner (via package manager or explicit installation to a system-wide extension directory). These extensions run as child processes of the current user, inherit the user’s capability set (subject to manifest restriction), and

Threat	Mitigation	Section Ref
Runbook author executes unauthorized commands	Command allowlist/denylist enforced at runtime core	§8.1, §11
Malicious extension injects code into host process	Out-of-process extension isolation; cryptographic signature verification	§8.2, §4
Tool or extension leaks credentials into trace	Mandatory redaction for sensitive fields; <code>sensitive: true</code> input marking	§8.4
Unauthorized user triggers runbook execution	RBAC enforced at <code>gert serve</code> ingress	§8.6
Man-in-the-middle attack on JSON-RPC transport	TLS required for all non-localhost listeners	§8.5
Replay attack on approval decision	Approval tokens signed with Ed25519; <code>approvalId</code> is single-use	§8.6, §11
Compromised tool binary executes untrusted code	Tool binary checksum verification before invocation	§8.8

TABLE 8.1: Threat model and mitigations for gert v2.

have access to all host capabilities declared in their manifest. No signature verification is performed; trust is conferred by the installation act.

Example: an extension installed via `apt-get` or `brew` into `/usr/local/lib/gert-extensions/`.

Project-scoped. Declared in the runbook project directory (e.g., `.gert/extensions.yaml`) or inline in the runbook’s `extensions:` field. These extensions run in a restricted sandbox: only capabilities explicitly granted by the workspace governance policy are forwarded. File and network access are constrained to declared paths and hosts. Signature verification is *optional* but recommended.

Use case: a team-maintained extension for Acme-specific incident tooling, distributed via the runbook repository.

Remote/signed. Extensions distributed via package repository or downloaded from the internet. These extensions *must* carry a cryptographic signature verified against a trust

store. The signature covers the extension manifest and binary. If verification fails, the extension is rejected before any subprocess is started.

Signature format is defined in Section 8.2.2.

8.2.2 Extension Signature Format

Remote/signed extensions must include a `signature:` block in `gert-extension.yaml`:

```
# gert-extension.yaml - signed extension
apiVersion: extension/v2

meta:
  name: acme.incident-pack
  version: "1.2.0"

signature:
  algorithm: ed25519
  publicKeyFingerprint: "sha256:abcd1234..."
  signatureData: "base64-encoded-signature"
  signedFields: ["meta", "capabilities", "entry-point", "transport"]

trust_chain:
  - fingerprint: "sha256:abcd1234..."
    issuer: "Acme Platform Team CA"
    notBefore: "2024-01-01T00:00:00Z"
    notAfter: "2025-01-01T00:00:00Z"
```

Signature verification algorithm. The signature is computed over the canonical JSON representation of the fields listed in `signedFields`. The host:

1. Extracts the public key fingerprint from the `signature` block.
2. Looks up the corresponding public key in the trust store (`$HOME/.gert/trusted-keys/` or `/etc/gert/trusted-keys/`).
3. Reconstructs the canonical JSON of the signed fields (sorted keys, no whitespace).
4. Verifies the Ed25519 signature over the canonical JSON bytes.

5. Checks that the current time falls within the `notBefore` and `notAfter` window of the trust chain.

If any step fails, the extension is rejected with a `governance/extensionRejected` trace event and reason `signature_verification_failed`.

8.2.3 Extension Verification Command

The `gert extension verify` command allows operators to manually verify an extension before installation:

```
$ gert extension verify ./acme-extension/gert-extension.yaml \
  --trust-store /etc/gert/trusted-keys/
Extension: acme.incident-pack v1.2.0
Signature: valid (ed25519, expires 2025-01-01)
Issuer: Acme Platform Team CA
Capabilities: tool-registration, network
Trust level: remote/signed
Verification: PASSED
```

8.3 Process Isolation

Extensions and tools always run as separate OS processes. They are never loaded as shared libraries or executed in-process. This isolation boundary ensures that a crash, hang, or malicious extension cannot compromise the host process.

8.3.1 Isolation Mechanisms

Separate OS process. As defined in §4, extensions are forked as child processes with their own address space, file descriptor table, and signal handlers.

Filesystem access control. Extensions with `capability/file-read` or `capability/file-write` declare the set of filesystem paths they may access via `file-read-paths` and `file-write-paths` in the manifest. The host enforces these constraints:

- On OpenBSD: via `unveil(2)` to restrict visible paths.

- On Linux: via `seccomp-bpf` to filter `open(2)` and `openat(2)` syscalls.
- On all platforms: by validating all path arguments in `file-read` and `file-write` RPC methods before dispatch.

Attempts to access undeclared paths return a `capability-denied` error.

Network access control. Extensions with `capability/network` declare the set of hostnames they may connect to via `network-hosts`. The host validates outbound connections:

- Before dispatching an extension RPC that specifies a target URL, the host checks that the hostname matches one of the declared `network-hosts` patterns.
- Extensions may not listen for inbound connections.
- Wildcard patterns are supported: `*.acme.example` permits `api.acme.example`, `prod.acme.example`, etc.

Environment variable isolation. Extensions receive ONLY the environment variables explicitly forwarded by the runbook's `governance.allowed_env` list. Host environment variables that match `governance.deny_env_vars` patterns are never forwarded, even if they appear in `allowed_env`.

The denylist takes precedence: if `AWS_SECRET_ACCESS_KEY` is denied, it is stripped from the extension's environment regardless of other policy.

Crash isolation. If an extension process crashes (exits with non-zero status or is killed by a signal), the host:

1. Emits an `extension.crashed` trace event with exit code and signal number.
2. Marks the extension as unavailable for the remainder of the run.
3. Cancels all in-flight invocations to that extension with error code `-32099` (`extension_crashed`).
4. Continues the run: the host process does not terminate.

8.4 Credential and Secret Handling

Gert v2 does not store secrets. It acts as a pass-through: secrets are resolved from input providers at runtime, used for a single step invocation, and then discarded. Secrets never appear in trace logs or captured output.

8.4.1 Secret Resolution and Redaction

Input provider sensitivity marking. Input provider definitions declare which resolved values are sensitive via `sensitive: true` in the provider schema:

```
# vault.provider.yaml
outputs:
  credentials:
    type: object
    sensitive: true
```

When the host resolves `from: vault.credentials`, it marks the returned value as sensitive in memory. This flag is respected by all downstream consumers.

Tool sensitive inputs. Tool action definitions declare which input parameters accept sensitive data via `sensitive_inputs`:

```
# my-tool.tool.yaml
actions:
  deploy:
    args:
      api_key:
        type: string
        required: true
    sensitive_inputs: [api_key]
```

When the host prepares the tool invocation envelope, it marks the `api_key` field as sensitive. The tool runtime redacts this field in all trace events.

Mandatory redaction. All values marked `sensitive: true` are redacted before being written to the trace or displayed in any adapter UI. Redaction replaces the value with the string `<redacted>`.

In addition, the runbook's `governance.redact` block defines regex patterns applied to all captured output (stdout, stderr, JSON responses) before storage. See §11 for the full redaction specification.

8.4.2 Secret Storage Policy

Gert v2 does NOT:

- Store secrets in the runbook YAML.
- Write secrets to disk (trace files, state snapshots, etc.).
- Cache secrets across steps (unless explicitly declared by the input provider's `cache_scope`).
- Transmit secrets over unencrypted transports.

Secrets must be resolved via input providers that integrate with secret management systems (e.g., HashiCorp Vault, AWS Secrets Manager, 1Password).

8.5 Transport Security

Gert v2 supports two primary transport modes for JSON-RPC communication:

1. **stdio:** JSON-RPC over stdin/stdout of a child process. No network exposure; inherently local. No TLS required.
2. **HTTP/WebSocket:** JSON-RPC over HTTP for `gert serve` mode. Network exposure possible; TLS and authentication required for non-localhost.

8.5.1 TLS Requirements

Localhost exception. If `gert serve` binds to `127.0.0.1` or `localhost`, TLS is optional. Connections are assumed to be local and trusted.

Non-localhost listeners. If `gert serve` binds to `0.0.0.0`, a public IP, or a hostname, TLS is *mandatory*. The host will refuse to start without a valid TLS certificate and private key.

Configuration:

```
$ gert serve --http 0.0.0.0:8443 \  
  --tls-cert /etc/gert/tls/cert.pem \  
  --tls-key /etc/gert/tls/key.pem
```

The host performs standard TLS 1.3 handshake with the client. TLS 1.2 is supported for compatibility but TLS 1.1 and earlier are rejected.

8.5.2 Client Authentication

The host supports two authentication modes for `gert serve -http`:

Bearer token authentication. Clients include an `Authorization: Bearer <token>` header on all HTTP requests. The token is validated against a configured token store (file-based, environment variable, or integration with an IdP).

Example:

```
$ gert serve --http :8443 --auth-mode bearer \  
  --bearer-tokens /etc/gert/tokens.yaml
```

The `tokens.yaml` file maps tokens to actor identities:

```
# tokens.yaml  
tokens:  
  - token: "abc123..."  
    actor: "ops@acme.example"  
    roles: ["operator", "viewer"]
```

Mutual TLS (mTLS). The host requires clients to present a valid X.509 client certificate signed by a trusted CA. The client certificate's subject CN or SAN is used as the actor identity.

Configuration:


```
$ gert serve --http :8443 --auth-mode mtls \  
  --tls-client-ca /etc/gert/tls/client-ca.pem
```

mTLS is the recommended authentication mode for production deployments and enterprise environments.

8.5.3 WebSocket Security

WebSocket connections for the event stream (§13) inherit the security context of the HTTP server:

- If the HTTP server uses TLS, WebSocket connections are encrypted (wss://).
- Client authentication applies: bearer tokens or client certificates are validated before the WebSocket upgrade.
- Once upgraded, the WebSocket connection is bound to the authenticated actor identity for the lifetime of the connection.

8.5.4 Extension JSON-RPC Transport

Extensions communicate with the host via stdio JSON-RPC (§4). This transport is not exposed to the network and does not require TLS. The process boundary is the trust boundary: the host trusts that OS-level process isolation prevents interception of stdin/stdout pipes.

8.6 RBAC for gert serve

When gert runs in server mode (`gert serve -http`), it enforces role-based access control (RBAC) on all API operations. Actors are assigned roles, and each role grants a specific set of permissions.

8.6.1 Actor Identity

An **actor** is the authenticated identity making a request. Actor identity is derived from:

- The bearer token (mapped to an actor ID via the token store), or

- The client certificate subject CN/SAN (in mTLS mode).

The actor identity is recorded in all trace events where a human or external system initiates an action (run start, approval submission, cancellation).

8.6.2 Roles

Gert v2 defines three standard roles:

Viewer. Read-only access. Permissions:

- List runs (GET /runs)
- Get run details (GET /runs/:id)
- Subscribe to event stream (GET /ws or GET /events)
- Read trace files

Cannot start runs, approve, or cancel.

Operator. Execution permissions. Includes all **viewer** permissions plus:

- Start a run (POST /exec/start)
- Cancel a run (POST /exec/cancel)
- Submit approvals (POST /approvals)

Cannot modify governance policy or manage roles.

Admin. Full control. Includes all **operator** permissions plus:

- Modify workspace governance policy
- Manage role assignments (add/remove actors from roles)
- Configure extension trust store

8.6.3 Role Assignment

Roles are assigned in the workspace config (`gert-project.yaml`):

```
# gert-project.yaml
access:
  roles:
    - role: admin
      actors:
        - "alice@acme.example"
        - "bob@acme.example"
    - role: operator
      actors:
        - "ops-team@acme.example"
        - "oncall@acme.example"
    - role: viewer
      actors:
        - "*@acme.example" # All Acme employees
```

Actor patterns support wildcards for organization-wide grants.

8.6.4 Approval Authorizer Role

The `governance.require_approval` field in the runbook declares the role required to approve a step:

```
steps:
  - id: delete_database
    cli: "kubectl delete database prod-db"
    approvals:
      min: 1
      roles: ["dba", "admin"]
```

When the run reaches this step, it emits an `approval/required` trace event. The step is paused. Only actors with the `dba` or `admin` role may submit a valid approval.

The approval submission includes:

```
POST /approvals
{
  "runId": "run-abc123",
  "stepId": "delete_database",
  "decision": "approve",
  "actorId": "alice@acme.example",
  "signature": "<ed25519-signature-of-approval-payload>",
  "timestamp": "2026-04-18T12:00:00Z"
}
```

The host verifies:

1. The actor is authenticated and has one of the required roles.
2. The signature is valid (Ed25519 over the canonical JSON of `runId`, `stepId`, `decision`, `actorId`, `timestamp`).
3. The `timestamp` is within a 5-minute clock skew window.
4. The `runId` and `stepId` match an active approval gate.
5. This approval has not been submitted before (replay protection).

If all checks pass, the approval is recorded in the trace (`approval/received`) and the step proceeds if the minimum approval count is reached.

8.7 Audit Non-Repudiation

Non-repudiation ensures that governance decisions — especially approvals and rejections — cannot be denied by the actor who made them. This is critical for regulatory compliance (SOC2, ISO27001, HIPAA) [23, 10, 26].

8.7.1 Governance Decision Recording

Every governance decision event (`governance/approval_received`, `governance/approval_rejected`, `governance/commandBlocked`) must record:

- Actor identity (authenticated user or service principal)

- Timestamp (RFC 3339 with timezone) [12]
- Decision payload (what was approved/rejected)
- Signature (Ed25519 over the canonical decision payload)

Example trace event:

```
{
  "eventType": "governance/approval_received",
  "timestamp": "2026-04-18T12:00:00Z",
  "runId": "run-abc123",
  "stepId": "delete_database",
  "actor": "alice@acme.example",
  "decision": "approve",
  "signature": "ed25519:abcd1234...",
  "signatureAlgorithm": "ed25519",
  "signaturePublicKey": "sha256:pubkey-fingerprint"
}
```

The signature covers the fields `runId`, `stepId`, `actor`, `decision`, and `timestamp`. The public key is looked up from the actor's identity record or client certificate.

8.7.2 Trace File Integrity (HMAC Chaining)

The JSONL trace file (§12) is append-only, but this alone does not prevent tampering. To provide cryptographic integrity, the host optionally computes an HMAC chain over the trace:

1. At run start, initialize `H_0 = HMAC-SHA256(key, runId)`.
2. After writing each trace event `E_i`, compute `H_i = HMAC-SHA256(key, H_{i-1} || E_i)`.
3. Store `H_i` in the event as `integrityHash`.
4. At run completion, emit a final `run.completed` event with `finalHash: H_N`.

Verification: a verifier recomputes the HMAC chain from `H_0` forward. If any event is modified or deleted, the chain breaks.

The HMAC key is derived from the workspace master key or provided via `-integrity-key`.

8.7.3 SHA256 Evidence Capture

As defined in §12, gert captures SHA256 hashes of all evidence artifacts (attachments, screenshots, CLI output files). These hashes are written to the trace event immediately after the artifact is captured, providing a cryptographic anchor for the artifact's integrity.

The combination of HMAC trace chaining + SHA256 evidence hashes provides end-to-end non-repudiation for the entire run.

8.8 Supply Chain Security

Tools and extensions are executable code distributed via package managers, repositories, or direct download. Supply chain attacks (e.g., compromised binaries, backdoored dependencies) are a real threat in operational environments. Gert v2 provides checksum verification and signature validation to mitigate these risks.

8.8.1 Tool Binary Checksums

Tool definitions may declare a `checksum:` field in `.tool.yaml`:

```
# kubectl.tool.yaml
meta:
  name: kubectl
  binary: kubectl

checksum:
  algorithm: sha256
  digest: "a3b4c5d6e7f8..."
```

Before invoking the tool, the host:

1. Resolves the binary path (via `PATH` lookup or absolute path).
2. Computes the SHA256 hash of the binary file.
3. Compares the computed hash to `checksum.digest`.
4. If the hashes do not match, the tool invocation is blocked with a `governance/toolChecksumMismatch` trace event.

8.8.2 gert verify Command

The `gert verify` command validates all tools and extensions in the workspace before execution:

```
$ gert verify --workspace .
Verifying tools...
✓ kubect1: checksum verified (sha256:a3b4c5...)
✓ curl: no checksum declared (skipped)
× jq: checksum mismatch (expected sha256:abc123, got sha256:def456)

Verifying extensions...
✓ acme.incident-pack: signature valid (ed25519, expires 2025-01-01)
× debug-tools: no signature (required for remote extensions)

Verification: FAILED (2 errors)
```

This command should be run in CI/CD pipelines before deploying runbooks to production.

8.8.3 Extension Signature Verification

As defined in §8.2.2, remote extensions must carry a cryptographic signature. The signature verification process ensures that:

- The extension binary and manifest have not been tampered with.
- The extension was signed by a trusted authority (public key in the trust store).
- The signature has not expired (`notAfter` check).

If verification fails, the extension is rejected before any subprocess is started.

8.9 Security Recommendations

For production deployments, the following security practices are recommended:

1. **Enable TLS for all gert serve deployments** — even internal networks can be compromised.

2. **Use mTLS for client authentication** — bearer tokens are vulnerable to theft; client certificates provide stronger identity.
3. **Enable HMAC trace chaining** — provides cryptographic integrity for audit trails.
4. **Declare tool binary checksums** — prevents execution of tampered binaries.
5. **Require signatures for all remote extensions** — do not trust unsigned code from the internet.
6. **Run `gert verify in CI`** — catch integrity violations before production.
7. **Use least-privilege capability grants** — extensions should request only the capabilities they need; admins should audit and deny excessive requests.
8. **Regularly rotate approval keys** — the Ed25519 keys used for signing approvals should be rotated every 90 days.

These recommendations align with NIST SP 800-53 controls for privileged access management (AC-2, AC-6), audit and accountability (AU-2, AU-9), and system integrity (SI-7) [14].

Chapter 9

Testing and Acceptance

This section specifies the complete testing strategy for gert v2: the test framework, unit and contract test patterns, integration test strategy, the `gert test` feature, regression testing against the v1 runbook corpus, performance acceptance criteria, the extension test harness, and CI/CD gate requirements.

9.1 Test Framework

9.1.1 Standard Library and testify

All gert v2 tests are written in Go using the standard `testing` package as the test runner and github.com/stretchr/testify [24] for assertions:

- `testify/assert` — non-fatal assertions; test continues on failure.
- `testify/require` — fatal assertions; test halts on failure.
- `testify/mock` — interface mocking for unit tests (engine, event bus, trace writer).

Table-driven tests are the preferred pattern for all logic that has well-defined input/output pairs (e.g. expression evaluation, schema validation, governance rule matching):

```
func TestGovernanceAllowlist(t *testing.T) {
    cases := []struct {
        name    string
        cmd     []string
        allowed bool
    }{
```

```

    {"exact match", []string{"kubectl", "get", "pods"}, true},
    {"partial match", []string{"kubectl", "delete", "ns"}, false},
    {"empty allowlist", []string{"echo", "hi"}, false},
  }
  for _, tc := range cases {
    t.Run(tc.name, func(t *testing.T) {
      result := allowlist.Check(tc.cmd)
      assert.Equal(t, tc.allowed, result)
    })
  }
}

```

9.1.2 Test Packages

Each Go package in the v2 module ships a co-located `_test.go` file (white-box tests) and optionally an `<pkg>_integration_test.go` file (integration tests, gated by a build tag). The naming convention:

```

pkg/engine/engine.go
pkg/engine/engine_test.go           // white-box unit tests
pkg/engine/engine_integration_test.go // build tag: //go:build integration

```

9.2 The gert test Feature

9.2.1 Scenario Replay Testing

`gert test <runbook.yaml>` discovers and executes all scenario tests for a runbook. A scenario test consists of:

- `inputs.yaml` — resolved input values for the run.
- `scenario.yaml` (optional) — pre-recorded command responses and manual evidence.
- `test.yaml` — assertions over the resulting trace and captures.

Scenario directories are located by convention:

```
{runbook-dir}/scenarios/{runbook-name}/{scenario-name}/
  inputs.yaml
  scenario.yaml  # optional; omit for runbooks with no CLI/tool steps
  test.yaml
```

The test runner executes the runbook in replay mode using the scenario file as the command/evidence fixture, then evaluates assertions from `test.yaml`.

9.2.2 Test Assertions (`test.yaml`)

```
assertions:
  outcome: passed          # "passed" | "failed" | "skipped"
  steps_passed: 5
  steps_failed: 0

captures:
  db_host:
    equals: "prod-db.example.com"
  migration_status:
    contains: "complete"

trace:
  - type: step/completed
    step_id: run-migration
  - type: governance/blocked
    step_id: drop-table
    data.rule: denylist
```

The `trace` assertions match events in order of occurrence. The `data.*` dotted paths dereference into the event's `data` object.

9.2.3 v1 Scenario Compatibility

v1 scenario files (`commands/evidence` structure) are forward-compatible with v2 without modification. The v2 parser accepts both the v1 format and the extended v2 format (which adds the `assertions` block). Runbook authors can migrate incrementally by adding `test.yaml` alongside existing v1 scenarios.

9.3 Contract Tests

Contract tests verify the interface boundaries between v2's five components. Each contract test lives in the *consumer* package and uses a conformance helper that validates any implementation satisfying the interface.

9.3.1 Core Runtime Contract

The runtime `Executor` interface must satisfy:

- Emits a `run/started` event before executing any step.
- Emits a `step/started` before and `step/completed/step/failed` after each step.
- Emits `run/completed` or `run/failed` as the final event.
- Does not emit events after the terminal event.
- Does not call any adapter method directly (events only).

```
// ConformanceTest validates any Runtime implementation.
func ConformanceTest(t *testing.T, rt Runtime) {
    t.Helper()
    bus := newTestEventBus()
    err := rt.Execute(context.Background(), fixtureRunbook, bus)
    require.NoError(t, err)
    events := bus.Events()
    require.Equal(t, "run/started", events[0].Type)
    require.Equal(t, "run/completed", events[len(events)-1].Type)
    assertNoEventsAfterTerminal(t, events)
}
```

9.3.2 Schema Contract Tests

- Every `apiVersion`: `runbook/v2` document in `testdata/` must pass structural validation without errors.
- Every invalid fixture in `testdata/invalid/` must fail validation with a structured error (message + path).

- Structural and semantic validation are tested independently: a structurally valid document may fail semantic validation (e.g. referencing an undefined capture).

9.3.3 Extension Manifest and Handshake Tests

The extension host verifies:

- An extension process that responds to the initialisation handshake within the configured timeout is accepted.
- An extension process that fails to respond is terminated and its capabilities are not registered.
- Capability grants in the manifest are enforced: an extension without `cap:schema.register` cannot register schema.
- A manifest with missing required fields is rejected at load time with a structured error.

9.3.4 Tool Invocation Envelope Tests

- Tool invocations carry the correct `run_id` and `step_id` correlation fields.
- Tool outputs are captured and available as step captures after the step completes.
- A tool that exits with non-zero triggers `step/failed` with `error_type: tool_error`.
- Timeout cancellation: a tool that exceeds its timeout receives a `context.DeadlineExceeded` and emits `step/failed` with `error_type: timeout`.

9.3.5 Event Bus Determinism Tests

- Events delivered to the event bus maintain the `seq` ordering guaranteed by `RunStore.WriteTrace`.
- Slow adapter consumers do not cause the engine's execution loop to block (bounded channel or dropping policy required).
- Event replay (`ReadTraceSince`) returns events in ascending `seq` order.

9.4 Integration Test Strategy

9.4.1 End-to-End Fixture Runs

Integration tests spin up the gert runtime against fixture runbooks stored in `testdata/runbooks/` and assert on the resulting trace files. They are gated by the `integration` build tag and run in CI separately from unit tests.

```
//go:build integration
```

```
func TestE2E_BranchingRunbook(t *testing.T) {
    dir := t.TempDir()
    runID, err := exec.RunFixture(dir, "testdata/runbooks/branch.runbook.yaml",
        WithInputs(map[string]string{"env": "prod"}),
        WithScenario("testdata/scenarios/branch-prod.yaml"),
    )
    require.NoError(t, err)
    trace := readTrace(t, dir, runID)
    assertEvent(t, trace, "step/completed", map[string]any{
        "data.step_id": "deploy-prod",
    })
    assertNoEvent(t, trace, "step/completed", map[string]any{
        "data.step_id": "deploy-staging",
    })
}
```

9.4.2 Fixture Runbook Corpus

The integration test corpus includes fixtures covering:

- Linear runbook (all step types: cli, manual, tool, invoke).
- Branching runbook (both branch arms; governance-blocked branch).
- Iterate runbook (list mode, convergence mode, parallel concurrency).
- Nested invoke (parent + two child runbooks).
- Retry policy (transient failure followed by success).

- Resumption (crash after step 2, resume, verify trace continuity).
- Governance: allowlist violation, denylist violation, approval gate.

9.4.3 Trace Assertion Helpers

A shared test package `pkg/testutil` provides helpers:

```
package testutil

// ReadTrace reads and parses all events from a run's trace.jsonl.
func ReadTrace(t *testing.T, runDir, runID string) []engine.DurableEvent

// AssertEvent asserts that at least one event matches type and data predicates.
func AssertEvent(t *testing.T, events []engine.DurableEvent,
    eventType string, predicates map[string]any)

// AssertEventOrder asserts events appear in the given type order.
func AssertEventOrder(t *testing.T, events []engine.DurableEvent,
    types ...string)

// AssertNoEvent asserts no event matches the given type and predicates.
func AssertNoEvent(t *testing.T, events []engine.DurableEvent,
    eventType string, predicates map[string]any)
```

9.5 Regression Testing

9.5.1 v1 Runbook Corpus as Fixtures

The v1 runbook corpus (located in `examples/` and operator-contributed runbooks) serves as a regression suite for v2 compatibility. The regression test runner:

1. Reads each v1 runbook from the corpus.
2. Runs `gert validate` (v2 schema) and asserts it passes (or passes after migration).
3. For runbooks with associated v1 scenario files, runs `gert test` and asserts the outcome matches the v1 golden output.

4. Fails the build if any previously passing corpus runbook regresses.

Migration compatibility is measured as: the percentage of v1 corpus runbooks that pass v2 `gert validate` without modification. The v2 launch target is $\geq 95\%$.

9.5.2 Golden Trace Comparisons

For deterministic scenarios, the expected trace can be stored as a golden file. The golden comparison test:

1. Runs the scenario.
2. Normalises timestamps and run IDs (replaces with fixed values).
3. Compares the normalised JSON-lines output to `testdata/golden/<scenario>.jsonl`.
4. Fails if any event differs. Golden files are updated by running with `-update` flag.

9.6 Performance Acceptance Criteria

The following latency budgets must pass on a reference machine (4-core, 8 GB RAM, SSD) with the integration test suite:

Operation	p50 target	p99 target
<code>gert validate</code> (single runbook, <50 steps)	<50 ms	<200 ms
Step startup (cli step)	<10 ms	<50 ms
Event delivery to adapter (local channel)	<1 ms	<5 ms
Trace event write + fsync	<5 ms	<20 ms
Snapshot write (100 steps history)	<20 ms	<80 ms
<code>gert test</code> (10-step replay scenario)	<500 ms	<1 s
Tool invocation round-trip (stdio, local process)	<50 ms	<200 ms

Performance benchmarks are written using Go's `testing.B` and run in CI weekly (not on every PR). A regression of $>2\times$ on any metric blocks the nightly build.

9.7 Extension Test Harness

9.7.1 Overview

Extension developers can test their extension logic without spinning up the full gert runtime. The `pkg/exttest` package provides a test harness:

```
import "github.com/ormasoftchile/gert/v2/pkg/exttest"

func TestMyExtension(t *testing.T) {
    h := exttest.NewHarness(t,
        exttest.WithManifest("testdata/my-ext.manifest.yaml"),
        exttest.WithCapabilities("cap:schema.register", "cap:tool.register"),
    )
    defer h.Close()

    // Start the extension process
    h.Start("./my-extension")

    // Assert it registered expected tools
    tools := h.RegisteredTools()
    require.Contains(t, tools, "my-tool")

    // Invoke a tool and assert the response
    resp := h.InvokeTool("my-tool", map[string]any{"arg1": "value"})
    assert.Equal(t, 0, resp.ExitCode)
    assert.Contains(t, resp.Stdout, "expected output")
}
```

9.7.2 Harness Capabilities

The test harness provides:

- A mock extension host that implements the full handshake protocol.
- Assertion helpers for registered tools, schema registrations, and event subscriptions.
- A `FakeRunContext` that provides a realistic run context without executing a runbook.

- Automatic process cleanup on test completion.
- Configurable capability grants (to test capability enforcement).

9.7.3 Mocking the Extension Transport

For pure unit tests of extension logic (without spawning a subprocess), the transport layer is mockable:

```
// InProcessTransport runs the extension handler in-process for unit testing.
type InProcessTransport struct {
    Handler ExtensionHandler
}
```

9.8 Acceptance Criteria

The following acceptance criteria must be satisfied before a v2 release is considered shippable:

1. Core runtime has no direct adapter dependencies — verified by the `no_import` contract test.
2. Extensions register schema and tools through declared contracts — verified by the extension handshake conformance suite.
3. Capability violations fail fast with structured errors — verified by the governance contract test for each capability type.
4. Adapters integrate using the same stable core contracts — verified by running each adapter’s integration test against a mock core.
5. Every step type (`cli`, `manual`, `tool`, `invoke`, `branch`, `iterate`) has ≥ 1 end-to-end integration test covering the happy path and at least one failure path.
6. The v1 runbook corpus passes `gert validate` at $\geq 95\%$ without modification.
7. All performance targets in § 8.6 pass on the reference machine.
8. `gert test` successfully replays all scenario fixtures in `testdata/scenarios/`.

9.9 CI/CD Integration

9.9.1 PR Gate (every pull request)

1. **Unit tests:** `go test ./...` (no build tags). Target: <60 s.
2. **Schema validation:** validate all YAML fixtures in `testdata/` against v2 JSON Schema. Target: <10 s.
3. **Contract tests:** `go test -run Contract ./...` Target: <30 s.
4. **Linters:** `golangci-lint` run with the project's `.golangci.yaml` config.
5. **Vet:** `go vet ./...`

All five gates must pass for a PR to be mergeable.

9.9.2 Integration Gate (merge to main)

1. **Integration tests:** `go test -tags=integration ./...` Target: <5 min.
2. **Regression corpus:** `gert test` against v1 corpus. Target: <2 min.
3. **Golden trace comparison:** normalised trace comparison for all deterministic fixtures.

9.9.3 Nightly Gate

1. **Performance benchmarks:** `go test -bench=. -benchtime=10s ./...` — compare against stored baseline; fail on >2× regression.
2. **Race detector:** `go test -race ./...`
3. **Extension conformance suite:** run all built-in extensions through the conformance harness.

Chapter 10

Open Questions

The questions below are unresolved architectural decisions. Each entry states **(a)** why the question matters for the design, **(b)** the options under consideration, and **(c)** what additional information is needed to close it. Decisions must be recorded in `.squad/decisions.md` before core contract freeze.

Q1. Can the v2 runtime execute v1 runbooks without prior migration?

Why it matters: Existing gert users have production runbooks in `runbook/v0` and `runbook/v1` format. If v2 can only execute v2-schema runbooks, adoption requires a migration step that may be blocked by organizational change control. A compatibility execution mode (“v1 shim”) removes this barrier but adds permanent maintenance cost to the parser.

Options: (a) Full compatibility mode: v2 runtime transparently executes v1 runbooks by internally upgrading them to v2 schema at load time, with no schema file written. (b) Migration gate: v2 refuses to run v1 runbooks and provides a clear error pointing to `gert migrate`. (c) Opt-in compatibility flag: `-compat-v1` enables the shim; production is v2-only.

Information needed: A survey of how many v1-format runbooks exist in representative deployments, and whether `gert migrate` can be made fully lossless for all v1 constructs.

Q2. What is the concurrency model for the core runtime?

Why it matters: The v1 runtime is single-run-per-process. Whether v2 supports multiple concurrent runs in a single process affects the state isolation model, the trace writer design, the governance enforcer (is it per-run or global?), and whether `gert serve` can multiplex concurrent VS Code sessions.

Options: (a) Single-run-per-process, matching v1 semantics; each `gert exec` or `gert tui` is a separate OS process. (b) Multiple concurrent runs in-process, with run-scoped state isolation via Go context and per-run goroutine groups. (c) Multiple runs via a long-lived daemon (`gert serve` as the host), with runs as logical sessions.

Information needed: Whether the VS Code extension use case requires concurrent runs in a single `gert serve` process (e.g., multiple open runbook panels), and what the goroutine safety requirements are for the trace writer and governance layer.

Q3. Is the v2 trace format backward-compatible with the v1 JSONL trace, or is it a new format?

Why it matters: Operators and compliance tools may have parsers, SIEM integrations, or dashboards built against the v1 JSONL event schema. A breaking change in the trace format requires those integrations to be updated before v2 can be rolled out. A backward-compatible superset format avoids this but constrains the v2 event schema design.

Options: (a) New format: v2 defines a clean JSONL schema with a required `schema_version` field; v1 trace parsers are unsupported. (b) Backward-compatible superset: v2 trace events include all v1 fields plus new fields; existing parsers continue to work on the subset they understand. (c) Parallel trace output: v2 writes both a v2 trace and an optional v1 compatibility trace (flag-enabled).

Information needed: An inventory of known downstream trace consumers and whether the v1 format has fields that would be structurally incompatible with the v2 event model.

Q4. How do extension-contributed policies compose with host-level governance policies?

Why it matters: The host enforces a governance policy (command allowlist, env blocking, redaction). An extension may also wish to contribute policies—e.g., a security extension that adds additional command denylists or redaction patterns. The composition model determines whether extensions can weaken host policies (prohibited), extend them (likely desired), or operate in a separate policy namespace (safest but most limited).

Options: (a) Additive composition only: extensions may only add restrictions (denylist entries, redaction patterns); they may never remove host-level restrictions. (b) Policy namespacing: each extension has an isolated policy scope; the host applies all active

policies via intersection. (c) Policy delegation: the host can explicitly delegate a policy domain to an extension (e.g., redaction rules owned by a data-classification extension).

Information needed: A threat model analysis of what happens when a malicious or compromised extension contributes a policy that attempts to weaken host restrictions. This should feed into §8.

Q5. Does the `.provider.yaml` input provider framework survive unchanged in v2?

Why it matters: v1's input provider model (`.provider.yaml`, JSON-RPC resolution, `from:` bindings, workspace config) is a complete and working framework. It must either be carried forward as-is, redesigned to fit the v2 component model, or replaced by an extension-based resolution mechanism. The choice affects the schema (§4), the extension runtime (§5), and any operator who has a custom provider binary.

Options: (a) Preserve as-is: `.provider.yaml` format and JSON-RPC protocol are unchanged; providers are a stable first-class concept. (b) Redesign as extension capability: providers become a named capability that extensions may implement; the `.provider.yaml` format is deprecated. (c) Hybrid: `.provider.yaml` providers continue to work as a built-in extension type, with a migration path to the full extension protocol.

Information needed: Whether any v1 provider binaries are in active production use that would be broken by a protocol change, and whether the extension handshake protocol can be made a strict superset of the provider JSON-RPC protocol.

Q6. Is the `gert serve` JSON-RPC contract a versioned breaking change or an evolutionary extension?

Why it matters: The VS Code extension consumes the `gert serve` JSON-RPC 2.0 API over stdio. If v2 changes the method names, parameter shapes, or notification payloads, the extension must be updated simultaneously. A versioned negotiation protocol (client declares supported API version at handshake) allows the extension to evolve independently of the binary.

Options: (a) Hard break: v2 defines a new JSON-RPC method set; the VS Code extension is updated as part of the v2 launch. (b) Versioned negotiation: the `initialize` handshake includes a protocol version field; the host advertises capabilities and the client selects a compatible version. (c) No change: the v1 JSON-RPC surface is preserved unchanged in v2.

Information needed: A complete method inventory of the v1 JSON-RPC API surface (all methods and notification shapes currently used by the VS Code extension), and a determination of which methods require changes to expose v2-specific features (e.g., saga events, OTel trace IDs).

Q7. What is the extension supply chain and signing model?

Why it matters: Extensions run in separate processes but are granted capabilities that include tool registration, schema field contribution, and event subscription. A malicious or tampered extension binary could exfiltrate captured secrets, inject commands, or suppress governance events. An extension signing model (verified publisher key) raises the bar for supply chain attacks, but adds operational friction for local development.

Options: (a) No signing required; extensions are trusted if they are present in the declared extension directory. (b) Optional signing: extensions may carry a detached signature; the host warns but does not block unsigned extensions. (c) Mandatory signing for capability classes above a threshold (e.g., extensions that request `governance.policy` capability must be signed). (d) Keyless signing via Sigstore/cosign for developer convenience.

Information needed: A threat model for the extension trust boundary (see Q4 above), and a survey of whether the target operator community has PKI infrastructure that could issue extension signing keys.

Q8. Is a gRPC transport for the VS Code extension viable for v0, or should it be deferred?

Why it matters: The current JSON-RPC 2.0 over stdio transport works but limits the VS Code extension to a single active connection and requires the extension to manage the gert process lifecycle. A gRPC transport would enable streaming (server-push events without polling), multiple concurrent clients, and connection multiplexing. However, it adds protocol complexity and requires a port allocation model.

Options: (a) Retain JSON-RPC stdio as the sole transport in v2; gRPC deferred to a future version. (b) Add gRPC as an opt-in alternative transport alongside JSON-RPC stdio. (c) Replace JSON-RPC stdio with gRPC as the primary VS Code transport; provide a backward-compatible shim for the transition period.

Information needed: Benchmarking of event delivery latency over JSON-RPC stdio versus gRPC under realistic VS Code workloads (step execution at 100ms/step, TUI

refresh at 60fps equivalent). Also requires a determination of whether VS Code's Node.js extension host supports gRPC client libraries without native module issues.

Q9. What is the parallel execution model for independent steps within a single run?

Why it matters: v1 executes all steps sequentially. Many real runbooks contain independent diagnostic steps that could execute in parallel, reducing total wall-clock time. However, parallelism complicates the trace ordering guarantee, the governance enforcer (can two parallel steps both consume the same approval token?), and the TUI rendering model.

Options: (a) No parallelism in v2; fan-out deferred to a future version. (b) Explicit parallel blocks via a new `parallel:` step type that joins on completion of all children. (c) Implicit parallelism inferred from the step dependency graph (steps with no data dependency on preceding steps execute concurrently).

Information needed: Whether any v1 runbooks in active use have independent steps that would benefit from parallelism, and what the governance implications are for concurrent step execution (e.g., two CLI steps executing simultaneously under the same allowlist).

Q10. How should the saga compensation registry interact with the trace and approval model?

Why it matters: A compensation registered for a step must itself be subject to governance (it may execute privileged commands), must appear in the trace (compensations are audit events), and may require its own approval gate if the original step had one. The interaction between the compensation executor and the governance layer is non-trivial and must be specified before implementation.

Options: (a) Compensations execute with the same governance context as the original step (same allowlist, same approver roles). (b) Compensations are exempt from approval gates (compensations are emergency rollback actions; blocking them on approval would defeat their purpose). (c) Compensation governance is separately configurable per compensation definition.

Information needed: Industry precedent from Temporal (compensation activities run in the same workflow context) and AWS Step Functions [1] (catch/retry blocks have no separate IAM policy). A decision here feeds into §4 (compensation field design) and §8 (governance scope).

Q11. What is the schema namespace convention for extension-contributed fields, and how are they validated?

Why it matters: The v2 schema design (§4) states that extension fields are namespaced, but the namespace convention and validation mechanism are undefined. If two extensions contribute fields under the same namespace, or if the host does not validate extension-contributed fields against a schema, runbooks using those fields will have unpredictable behavior at runtime.

Options: (a) Extension ID as namespace prefix (`x-myext.fieldName`); validated against a JSON Schema fragment published by the extension at registration time. (b) Dot-notation namespace under a reserved top-level key (`extensions.myext.fieldName`); host validates at parse time. (c) Extension fields are opaque to the host; the extension validates its own fields at execution time.

Information needed: Whether JSON Schema Draft 2020-12 supports dynamic `$ref` resolution at validation time, which would allow the host to assemble a composite schema from the host schema plus all registered extension schema fragments.

Q12. What is the timeout and escalation model for human approval steps?

Why it matters: v1 approval gates block indefinitely waiting for a role-holder to approve. In production, a stalled approval can block an incident response indefinitely. SLA enforcement—timeout after N minutes, escalate to the next role, optionally auto-approve or auto-reject—is a standard pattern in ITIL emergency change processes and is implemented by PagerDuty [19] and AWS Step Functions [1] human approval tasks. Without it, gert is not usable for SLA-bound incident response (*Dennis Research Brief §5.2*).

Options: (a) No timeout in v2; approval escalation deferred. (b) Optional `timeout` field on approval steps; on expiry, run is suspended and operator is notified. (c) Full escalation chain: `timeout` triggers escalation to a secondary role list; if that also times out, the step enters a configurable terminal state (auto-approve, auto-reject, or suspend-for-manual-review).

Information needed: Whether the v2 runtime concurrency model (Q2 above) supports timer-based step suspension without blocking the process, and what notification mechanism is available to alert escalation targets (stdout, webhook, or an extension-contributed notifier).

Chapter 11

Migration and Compatibility

Gert v2 is a breaking redesign from v1. This section specifies the migration path from `runbook/v0` and `runbook/v1` to the v2 schema, defines what `gert migrate` (v2 version) must accomplish, and establishes compatibility guarantees for existing runbooks, tools, providers, and integrations.

Migration is a launch blocker for adoption: every operator with a v1 runbook must understand what breaks, what is automatically migrated, and what requires manual effort. This chapter provides a concrete, actionable roadmap for teams moving from v1 to v2.

11.1 v1 → v2 Compatibility Matrix

The following table catalogs every v1 feature, interface, and artifact and specifies its v2 status. A BREAKING change requires manual intervention or automated migration; a COMPATIBLE change preserves the v1 contract.

v1 Feature/Interface	Status	v2 Change	Migration Notes
runbook/v0 schema	BREAKING	Removed	Migrate to v1 first, then v2
runbook/v1 schema	BREAKING	Promoted fields	gert migrate auto-rewrites
provider/v0 schema	BREAKING	New namespace	Manual rewrite required
tool/v0 schema	BREAKING	New fields	gert migrate rewrites
CLI: gert run	COMPATIBLE	New flags added	All v1 flags preserved
CLI: gert test	COMPATIBLE	Unchanged	
CLI: gert serve	COMPATIBLE	New RPC methods	v1 methods preserved
CLI: gert validate	COMPATIBLE	v2 schema aware	
CLI: gert migrate	COMPATIBLE	v0→v1→v2 path	New -to-v2 flag
JSON-RPC: exec/*	COMPATIBLE	Additive	New methods; old methods unchanged
JSON-RPC: runbook/*	COMPATIBLE	Additive	
JSONL trace format	BREAKING	New envelope	See §10.5
Extension handshake	BREAKING	New capability fields	See §10.4
Event stream schema	BREAKING	New event kinds	Old events preserved
VS Code extension protocol	COMPATIBLE	Backward compat	v2 extension reads v1 traces
TUI adapter	COMPATIBLE	Event-based	

Key takeaways:

- All **CLI commands and flags** are backward compatible. Existing scripts using **gert run** continue to work.
- All **JSON-RPC methods** are backward compatible. The v2 **gert serve** daemon accepts v1 client requests.
- The **runbook YAML schema** has breaking changes (field promotions, renames). Automated migration is provided via **gert migrate**.
- The **tool and provider schemas** have breaking changes. Manual updates required.
- The **JSONL trace format** has breaking changes (new envelope fields). Historical trace readers must be updated.
- The **extension handshake** has breaking changes. Extension authors must adopt the v2 protocol.

11.2 Schema Migration: Runbook YAML

11.2.1 Breaking Changes Summary

The v1 → v2 runbook schema migration involves three categories of change:

1. **Field promotions:** Seven `meta.*` fields are promoted to top-level (§3.2.4).
2. **Step type renames:** `collector` → `manual`, `router` → `end`.
3. **New required fields:** `$schema`, `id` (derived from `name` if absent).

All other runbook semantics are preserved. Step execution behavior, governance primitives, evidence capture, and input/output contracts remain unchanged.

11.2.2 Before and After Example

v1 runbook (simple.runbook.yaml):

```
apiVersion: runbook/v1
```

```
meta:
```

```
  name: service-health-check
```

```
  kind: operational
```

```
  description: Check service health and restart if needed
```

```
  vars:
```

```
    service_name: "api-server"
```

```
  governance:
```

```
    allowed_commands: [systemctl, curl, jq]
```

```
flow:
```

```
  - step:
```

```
    id: check_status
```

```
    type: cli
```

```
    title: Check service status
```

```
    with:
```

```
      argv: ["systemctl", "status", "{{ .service_name }}"]
```

```
    capture:
```

```
        exit_code: exit_status

- step:
  id: prompt_restart
  type: collector
  title: Service is down. Restart?
  evidence:
    - kind: text
      name: decision

- step:
  id: restart_service
  type: cli
  when: "{{ eq .decision \"yes\" }}"
  with:
    argv: ["systemctl", "restart", "{{ .service_name }}"]

v2 runbook (migrated):

$schema: "https://schemas.gert.dev/runbook/v2.json"
apiVersion: runbook/v2

id: service-health-check
name: Service Health Check
kind: operational
description: Check service health and restart if needed

vars:
  service_name: "api-server"

governance:
  allowed_commands: [systemctl, curl, jq]

flow:
- step:
  id: check_status
  type: cli
```

```
    title: Check service status
    with:
      argv: ["systemctl", "status", "{{ .service_name }}"]
    capture:
      exit_code: exit_status

- step:
  id: prompt_restart
  type: manual
  title: Service is down. Restart?
  evidence:
    - kind: text
      name: decision

- step:
  id: restart_service
  type: cli
  when: "{{ eq .decision \"yes\" }}"
  with:
    argv: ["systemctl", "restart", "{{ .service_name }}"]
```

11.2.3 Field Mapping Table

v1 Path	v2 Path	Notes
meta.name	name	Promoted to top-level; now capitalized (human-readable)
meta.kind	kind	Promoted
meta.description	description	Promoted
meta.vars	vars	Promoted
meta.inputs	inputs	Promoted
meta.governance	governance	Promoted
meta.prose	prose	Promoted
tree	flow	Renamed (tree was v0 legacy)
step.type: collector	step.type: manual	Renamed
step.type: router	step.type: end	Renamed
tools: [string]	toolRefs: [object]	Structured (§3.5)

11.3 Automated Migration Tool

11.3.1 Command Specification

The `gert migrate` command automates runbook, tool, and provider migrations. It is extended in v2 to support v1→v2 migration.

```
# Preview migration (dry-run)
gert migrate --to-v2 --dry-run runbook.yaml

# Migrate a single runbook in-place
gert migrate --to-v2 runbook.yaml

# Migrate all runbooks in a directory tree
gert migrate --to-v2 ./runbooks/

# Output migrated YAML to stdout (for review)
gert migrate --to-v2 --output - runbook.yaml

# Check if migration is needed (exit 1 if needed, 0 if already v2)
```

```
gert migrate --to-v2 --check runbook.yaml
```

11.3.2 Flags

Flag	Type	Description
<code>-to-v2</code>	bool	Enable v1→v2 migration mode
<code>-dry-run</code>	bool	Preview changes without writing
<code>-output PATH</code>	string	Write to file or - for stdout
<code>-check</code>	bool	Exit 1 if migration needed, 0 otherwise
<code>-force</code>	bool	Overwrite existing v2 files
<code>-backup</code>	bool	Create <code>.v1.bak</code> backup before rewrite (default: true)

11.3.3 Migration Algorithm

The migration tool performs the following transformations in order:

1. **Parse input:** Read YAML with v1 parser. If `apiVersion` is already `runbook/v2`, exit 0 (no-op).
2. **Version check:** If `apiVersion` is `runbook/v0`, emit error: “v0 must be migrated to v1 first”.
3. **Add \$schema:** Insert `$schema: "https://schemas.gert.dev/runbook/v2.json"` at document top.
4. **Update apiVersion:** Change `runbook/v1` → `runbook/v2`.
5. **Derive id:** If no `meta.name`, derive from filename. Convert to kebab-case (e.g., `service_health` → `service-health`).
6. **Promote meta.* fields:** Extract `meta.name`, `meta.kind`, etc. and move to top-level. Capitalize `name` for human readability.
7. **Rename step types:** `collector` → `manual`, `router` → `end`.
8. **Rename tree:** If `tree:` is present, rename to `flow:`.
9. **Rewrite tools:** Convert shorthand `tools: [name]` to structured `toolRefs: [{name, version?}]`.

10. **Validate:** Parse the migrated YAML with the v2 parser. If validation fails, rollback and report error.
11. **Write output:** If `-output` is specified, write to that path; otherwise rewrite in-place.

11.3.4 Diff Report

The migration tool emits a structured diff report showing all changes:

Migration report for `service-health-check.runbook.yaml`:

```
[CHANGE] apiVersion: runbook/v1 → runbook/v2
[ADD]    $schema: "https://schemas.gert.dev/runbook/v2.json"
[ADD]    id: "service-health-check"
[MOVE]   meta.name → name
[MOVE]   meta.kind → kind
[MOVE]   meta.description → description
[MOVE]   meta.vars → vars
[MOVE]   meta.governance → governance
[RENAME] step "prompt_restart": type collector → manual
[OK]     3 steps validated successfully
```

Summary: 8 changes applied. No errors.

11.3.5 Error Handling

If migration fails (e.g., v2 validation error, ambiguous field conflict), the tool:

1. Rolls back any partial changes (if in-place mode).
2. Emits a structured error message with file, line, and remediation advice.
3. Exits with status 1.

Example error:

```
ERROR: Migration failed for service-health-check.runbook.yaml
Line 14: step "check_status" has invalid v2 field "with.argv"
Suggestion: v2 uses "with.command" for CLI steps. Update manually.
```

11.4 Provider and Tool Migration

11.4.1 Tool Definitions (.tool.yaml)

The v1 tool/v0 schema is replaced by tool/v2. Key changes:

1. **New required field:** `transport` (one of `stdio`, `jsonrpc`, `mcp`).
2. **Renamed field:** `executable` → `command` (aligns with runbook YAML).
3. **New field:** `capabilities` (list of required capability grants; see §4).

v1 tool definition:

```
apiVersion: tool/v0
name: kubectl
executable: /usr/local/bin/kubectl
args_template: "{{ .args }}"
```

v2 tool definition:

```
$schema: "https://schemas.gert.dev/tool/v2.json"
apiVersion: tool/v2

id: kubectl
name: kubectl
transport: stdio
command: /usr/local/bin/kubectl
args_template: "{{ .args }}"
capabilities:
  - exec:command
  - net:egress
```

Migration tool support. The `gert migrate -to-v2` command rewrites tool definitions automatically, but **requires manual review** for the `capabilities` field (the tool cannot infer required capabilities statically).

11.4.2 Provider Definitions (.provider.yaml)

The v1 `provider/v0` schema is replaced by `provider/v2`. Key changes:

1. **Namespace convention:** Provider fields now use a `provider/*` namespace to avoid name collisions with core schema fields.
2. **New resolution protocol:** v2 providers declare a `resolve_method` (one of `static`, `env`, `prompt`, `jsonrpc`).

v1 provider definition:

```
apiVersion: provider/v0
name: incident-id
source: env
env_var: INCIDENT_ID
```

v2 provider definition:

```
$schema: "https://schemas.gert.dev/provider/v2.json"
apiVersion: provider/v2

id: incident-id
name: Incident ID
resolve_method: env
provider/env:
  var_name: INCIDENT_ID
  required: true
```

Migration tool support. The `gert migrate -to-v2` command partially rewrites provider definitions but **requires manual namespace updates** for custom provider fields.

11.5 Extension Migration

11.5.1 Handshake Protocol Changes

The v2 extension handshake introduces new capability negotiation (§4.3). Extensions must update their initialization sequence to:

1. Declare supported `protocol_version` (set to "2.0").
2. Declare required `capabilities` (list of capability names).
3. Handle `CapabilityDenied` responses gracefully.

v1 handshake (JSON-RPC):

```
→ {"jsonrpc":"2.0","method":"initialize","params":{"version":"1.0"}}
← {"jsonrpc":"2.0","result":{"ready":true}}
```

v2 handshake (JSON-RPC):

```
→ {"jsonrpc":"2.0","method":"initialize","params":{"protocol_version":"2.0",
  "capabilities":["fs:read","net:egress"]}
  }}
← {"jsonrpc":"2.0","result":{"granted_capabilities":["fs:read"],
  "denied_capabilities":["net:egress"]}
  }}
```

11.5.2 Extension Compatibility Shim

To ease the transition, the v2 Extension Host provides a **v1 compatibility shim**:

- Extensions that declare `protocol_version: "1.0"` are accepted.
- The shim grants all capabilities by default (insecure, but backward-compatible).
- A deprecation warning is logged: “Extension 'foo' uses deprecated v1 protocol. Update to v2 by YYYY-MM-DD.”
- The shim will be removed in v2.1 (approximately 6 months after v2.0 release).

11.5.3 Extension Author Checklist

Extension authors must:

1. Update `protocol_version` to "2.0" in the `initialize` method.

2. Declare all required capabilities in the handshake.
3. Test against the v2 Extension Host in capability-denied scenarios.
4. Update extension manifest to declare minimum gert version (`min_gert_version: "2.0.0"`).
5. Review the capability taxonomy (§4.4) and minimize requested capabilities.

11.6 Trace Format Migration

11.6.1 Breaking Changes in JSONL Trace

The v2 trace format (§12.1) adds mandatory envelope fields. The v1 trace format had:

```
{"seq":1,"type":"run/started","timestamp":"...","data":{...}}
```

The v2 trace format requires:

```
{"event_id":"...","run_id":"...","runbook_id":"...","timestamp":"...",
  "kind":"run/started","sequence":1,"payload":{...}}
```

Field mapping:

v1 Field	v2 Field	Notes
seq	sequence	Renamed
type	kind	Renamed
data	payload	Renamed
(none)	event_id	New (UUID v4)
(none)	run_id	New (extracted from file path in v1)
(none)	runbook_id	New (from <code>meta.name</code>)

11.6.2 Trace Replay Compatibility

The v2 `gert replay` command can read v1 traces **in compatibility mode**:

1. Detect v1 format by presence of `type` field (v2 uses `kind`).

2. Synthesize missing fields: `event_id` (new UUID), `run_id` (from file path), `runbook_id` (from first event payload).
3. Map `seq` → `sequence`, `type` → `kind`, `data` → `payload`.
4. Replay as normal.

Compatibility mode will be removed in v2.1.

11.6.3 Trace Archive Migration Tool

Teams with historical v1 trace archives can migrate them to v2 format using:

```
gert trace migrate --to-v2 .runbook/runs/*/trace.jsonl
```

This rewrites all trace files in-place, adding the required v2 envelope fields.

11.7 Rollout Strategy

11.7.1 Recommended Adoption Path

Teams should adopt v2 in four phases:

Phase 1: Read-Only Validation (Week 1-2)

1. Install `gert` v2.0 alongside v1 (both can coexist).
2. Run `gert validate -v2` against existing v1 runbooks.
3. Run `gert migrate -to-v2 -dry-run -check` to identify migration scope.
4. Review migration reports; identify runbooks that require manual fixups.

Phase 2: Schema Migration (Week 3-4)

1. Migrate runbooks in a staging directory: `gert migrate -to-v2 ./runbooks/`.
2. Review migrated YAML for correctness (especially governance blocks, tool refs).
3. Run `gert validate` against migrated runbooks to confirm v2 compliance.
4. Commit migrated runbooks to version control (with `.v1.bak` backups retained).

Phase 3: Extension and Tool Migration (Week 5-6)

1. Update all `.tool.yaml` files: add `transport` and `capabilities` fields.
2. Update all `.provider.yaml` files: namespace custom fields.
3. Update custom extensions: implement v2 handshake, declare capabilities.
4. Test extensions in v2 runtime with governance policies enabled.

Phase 4: Integration Migration (Week 7-8)

1. Update integrations (CI/CD pipelines, monitoring scripts) to use v2 CLI flags.
2. Update JSON-RPC clients (if any) to handle new event kinds (§6).
3. Update trace parsers to handle v2 envelope format (§12).
4. Decommission v1 runtime once all integrations are v2-native.

11.7.2 Parallel Operation

During the migration period, v1 and v2 runtimes can coexist:

- Install v2 as `/usr/local/bin/gert2` or via version manager (`gert@2`).
- Run v1 runbooks with `gert run` (v1 binary).
- Run v2 runbooks with `gert2 run` or `gert -version 2 run`.
- The v2 runtime rejects v0 runbooks; v1 runbooks are accepted in compatibility mode.

11.7.3 Timeline

Milestone	Timeline
v2.0 release	T+0
v1 compatibility shim supported	T+0 to T+6 months
v1 handshake deprecated warning	T+3 months
v1 compatibility shim removed	T+6 months (v2.1 release)
v1 runtime end-of-life	T+12 months

11.8 v1 Compatibility Shim

11.8.1 Decision: Provide a v1 Compatibility Shim for 6 Months

The v2 parser and runtime include a **v1 compatibility mode** that accepts `runbook/v1` documents and silently normalizes them to v2 semantics. This shim is enabled by default in v2.0 and will be removed in v2.1 (approximately 6 months after v2.0 release).

Rationale:

- Lowers the migration barrier for large organizations with hundreds of runbooks.
- Allows operators to adopt v2 runtime immediately without waiting for schema migration.
- Provides a grace period for extension authors to update their code.
- Encourages migration by emitting deprecation warnings (`-strict` mode treats warnings as errors).

11.8.2 Shim Behavior

The compatibility shim:

1. Accepts `apiVersion: runbook/v1` documents.
2. Silently applies the same transformations as `gert migrate -to-v2`.
3. Emits deprecation warnings to stderr:

```
WARN: runbook 'service-health-check' uses deprecated runbook/v1 schema.  
      Run 'gert migrate --to-v2 runbook.yaml' to upgrade.  
      v1 support will be removed in gert v2.1 (2026-10-18).
```

4. If `-strict` flag is passed, treats deprecation warnings as errors (exit 1).

11.8.3 Shim Removal

The v1 compatibility shim will be removed in v2.1 (target: 6 months post-v2.0 release). After removal:

- The v2 parser rejects `runbook/v1` documents with error: “v1 schema not supported. Migrate to v2.”
- All v1 runbooks must be migrated using `gert migrate -to-v2`.
- The v1 compatibility shim code is deleted from the codebase.

11.9 Open Questions and Risks

11.9.1 Open Questions

1. **Provider field namespacing:** Should v2 enforce a strict `provider/*` namespace, or allow custom prefixes? (Recommendation: strict namespace, relaxed in v2.1 if needed.)
2. **Migration of tool discovery paths:** v1 searches `tools/` by default. Should v2 preserve this, or require explicit `-tool-dir` flags? (Recommendation: preserve default path for compatibility.)
3. **Trace archive retention:** Should `gert trace migrate` preserve v1 traces as `.v1.jsonl` backups? (Recommendation: yes, with `-no-backup` opt-out.)

11.9.2 Migration Risks

1. **Runbooks with custom `meta.*` fields:** If a v1 runbook has custom fields under `meta.*`, migration may fail. Mitigation: emit detailed error messages with field paths.
2. **Extensions relying on v1-specific behaviors:** Some extensions may depend on v1 event ordering or field names. Mitigation: provide comprehensive v2 migration guide for extension authors.
3. **Historical trace parsing:** Teams with custom trace analysis scripts will need updates. Mitigation: publish a trace format migration guide with field mappings.

11.10 Summary

Gert v2 introduces breaking changes to the runbook, tool, and provider schemas, the JSONL trace format, and the extension handshake protocol. All changes are necessary to support the new capability-based security model (§7), the event-driven architecture (§6), and the governance primitives (§11).

The migration path is designed to minimize disruption:

- **CLI and JSON-RPC contracts are backward-compatible.**
- **Schema migration is automated** via `gert migrate -to-v2`.
- **v1 compatibility shim** provides a 6-month grace period for runbook and extension migration.
- **Phased rollout** allows teams to adopt v2 incrementally over 8 weeks.

Teams should begin migration planning with Phase 1 (read-only validation) and proceed through schema, extension, and integration migration in sequence. The v1 runtime will reach end-of-life 12 months after v2.0 release, giving organizations ample time to transition.

Chapter 12

Governance and Policy

Governance is Gert’s core differentiator: allowlists, denylists, environment variable blocking, output redaction, and approval gates. This section defines the v2 governance model, specifying what policy fields survive from v1, how policy is evaluated at runtime, whether policy is host-enforced or extension-contributed, and how approval gates integrate with the event model.

Governance features are a first-class design requirement and must not be treated as optional or defer-to-extension.

12.1 Governance Primitives

The following primitives form gert’s policy vocabulary. Each is defined precisely below, with its evaluation point in the execution pipeline, its schema representation, and its trace event.

12.1.1 Command Allowlist (`allowed_commands`)

Definition. A whitelist of executable names (`argv[0]`, without path components) that may be invoked by `cli` steps and tool subprocess invocations. If the allowlist is non-empty, any command whose base name is not in the list is blocked before the subprocess is forked.

Semantics. Path-stripping is applied before matching: `/usr/bin/curl` matches the allowlist entry `curl`. Glob patterns are *not* supported in v2.0 (exact match only). The empty list (`[]`) means all commands are permitted (permissive default).

Schema.

```
meta:
  governance:
    allowed_commands: [kubectl, curl, jq, aws]
```

Evaluation point. Runtime Core, immediately before subprocess fork (after template evaluation). See §12.2.

Trace event. `governance/commandBlocked` — written when a command is denied. Fields: `stepId`, `command`, `reason`: `"not_in_allowlist"`.

12.1.2 Command Denylist (`denied_commands`)

Definition. An explicit list of executable names that are unconditionally blocked. The denylist takes precedence over the allowlist: if a command appears in both, it is denied.

Use case. Operators use the denylist to forbid dangerous commands (`rm`, `dd`, `shutdown`) without needing to enumerate all permitted commands.

Schema.

```
meta:
  governance:
    denied_commands: [rm, dd, shutdown, reboot]
```

Evaluation point. Runtime Core, same checkpoint as allowlist. Denylist check runs first.

Trace event. `governance/commandBlocked` — fields: `stepId`, `command`, `reason`: `"in_denylist"`.

12.1.3 Environment Variable Blocking (`deny_env_vars`)

Definition. A list of glob patterns matched against environment variable names. Any environment variable whose name matches a pattern is stripped from the subprocess environment before the command is executed.

Rationale. Prevents accidental or malicious exfiltration of secrets (e.g., `AWS_SECRET_ACCESS_KEY`, `GITHUB_TOKEN`) by commands in the runbook.

Schema.

```
meta:
  governance:
    deny_env_vars:
      - "*_SECRET*"
      - "*_TOKEN"
      - "AWS_*"
      - "GITHUB_*"
```

Evaluation point. Runtime Core, immediately before subprocess fork. Applied to the full process environment (not just explicitly passed vars).

Trace event. `governance/envVarBlocked` — fields: `stepId`, `varNames`: `[string]` (names of blocked variables, not values).

12.1.4 Output Redaction (redact)

Definition. A list of redaction rules. Each rule has a `pattern` (a RE2 regular expression) and a `replace` string. After a `cli` or `tool` step produces output, all redaction rules are applied to the captured stdout/stderr before the output is stored in run variables or written to the trace.

Semantics. Rules are applied in declaration order. The replace string may use RE2 backreferences (`$1`). The special replace value `[REDACTED]` is the conventional replacement for secrets.

Schema.

```
meta:
  governance:
    redact:
      - pattern: "token=[A-Za-z0-9_\\-]{20,}"
        replace: "token=[REDACTED] "
      - pattern: "password: .+"
        replace: "password: [REDACTED] "
      - pattern: "(?i)bearer\\s+[A-Za-z0-9._\\-]+"
        replace: "Bearer [REDACTED] "
```

Evaluation point. Runtime Core, immediately after subprocess/tool output is captured, before capture variables are set and before any trace write.

Trace event. `governance/redactionApplied` — fields: `stepId`, `ruleCount` (number of rules that matched, not the patterns themselves, to avoid logging the pattern that matched sensitive data).

12.1.5 Approval Gates

Approval gates are first-class step-level governance primitives. They are defined on any step type and pause execution until a minimum number of approvals from specified roles have been recorded. See §12.3 for the complete approval gate design.

12.2 Policy Evaluation Model

Policy primitives are evaluated at specific, well-defined points in the execution pipeline. The evaluation order within each checkpoint is fixed and must not be reordered by extensions.

Checkpoint	Primitives Evaluated	Phase
Parser validation	Schema structural rules	Parse
Planner validation	Import cycles, tool existence	Plan
Run pre-flight	Extension policy rules	Run init
Step pre-flight	Denylist check	Runtime Core
Step pre-flight	Allowlist check	Runtime Core
Step pre-flight	Approval gate	Runtime Core
Subprocess fork	Env var blocking	Runtime Core
Output captured	Redaction	Runtime Core
Post-step	Contract risk warning	Runtime Core

Order of precedence. Within the pre-flight checkpoint for a `cli` or `tool` step, the evaluation order is:

1. Denylist check (hard block, step fails immediately if matched).
2. Allowlist check (hard block, step fails immediately if not matched and list is non-empty).
3. Approval gate (soft pause; step is held until approvals arrive).
4. Env var blocking (applied at fork time, not a failure condition — silently strips).

A policy violation at any hard-block checkpoint writes the corresponding trace event and returns an error from `RunHandle.Next()`. The run is placed in the `policy_violation` terminal state. The operator must resolve the policy (e.g., add the command to the allowlist) and re-run; resumption from a policy violation is not supported in v2.0.

12.3 Approval Gate Design

12.3.1 Schema Declaration

Approval gates are declared in the step's `approvals` block:

```

steps:
- id: delete_database
  kind: cli
  command: "kubectl delete namespace {{ .targetNamespace }}"
  approvals:
    min: 2
    roles: ["DRI", "change-manager"]
    timeout: 24h
    on_timeout: fail # or: escalate
    escalate_to: ["senior-sre", "director-of-eng"]
    message: "Deleting {{ .targetNamespace }} is irreversible. Confirm change window."

```

12.3.2 UX Contract

When the Runtime Core reaches a step with an `approvals` block, it:

1. Writes `governance/approvalRequired` trace event (stepId, message, roles, min, timeout).
2. Emits an `ApprovalRequired` runtime event on the event channel.
3. Returns from `Next()` with a `StepResult` of kind `PendingApproval`. The step does *not* execute yet.
4. Subsequent calls to `Next()` on the same step return `ErrAwaitingApproval` until the approval quorum is met.

The adapter (VS Code extension, TUI, or `gert serve` client) is responsible for presenting the approval request to authorized users and collecting decisions.

12.3.3 Approval Decision Protocol

An approver submits a decision by calling `RunHandle.Approve(ctx, ApprovalDecision)`:

```

type ApprovalDecision struct {
  StepID  string // step to approve (must match current paused step)
  Actor   string // identity of the approver (e.g., "alice@corp.com")
  Role    string // role the approver is acting in (must be in step.approvals.roles)
  Approved bool    // true = approved, false = rejected

```

```

    Comment string    // optional justification
    At      time.Time // timestamp of decision
}

```

The runtime validates:

- The **Actor** identity is established (either from **-as** flag or, in a multi-user context, from the adapter's authenticated session).
- The **Role** is in the step's declared **approvals.roles**.
- The same actor may not approve the same gate twice.
- If **Approved = false**, the gate is immediately rejected (any single rejection vetos the gate), and the run enters the **rejected** terminal state.

When the approval count reaches **min**, the runtime writes **governance/approvalGranted** to the trace and the next call to **Next()** proceeds with normal step execution.

12.3.4 Approval Recording in the Trace

Every approval decision is written to the trace as an immutable record:

```

// governance/approvalRequired
{
  "sequence": 14,
  "event": "governance/approvalRequired",
  "stepId": "delete_database",
  "message": "Deleting prod-ns is irreversible. Confirm change window.",
  "roles": ["DRI", "change-manager"],
  "min": 2,
  "timeout": "24h",
  "at": "2026-04-18T10:00:00Z"
}

// governance/approvalDecision (written once per decision)
{
  "sequence": 15,
  "event": "governance/approvalDecision",

```



```
"stepId": "delete_database",
"actor": "alice@corp.com",
"role": "DRI",
"approved": true,
"comment": "Change window confirmed with customer.",
"at": "2026-04-18T10:05:00Z"
}

// governance/approvalGranted (written when quorum is met)
{
  "sequence": 16,
  "event": "governance/approvalGranted",
  "stepId": "delete_database",
  "approvals": ["alice@corp.com", "bob@corp.com"],
  "at": "2026-04-18T10:07:00Z"
}
```

The trace provides a complete, tamper-evident audit record of who approved what and when.

12.3.5 Timeout and Escalation

If `timeout` is set and the approval quorum is not met within the deadline:

- `on_timeout: fail`: The run enters the `timed_out` terminal state. `governance/approvalTimedOut` is written to the trace.
- `on_timeout: escalate`: The runtime emits an `ApprovalEscalated` event. If `escalate_to` roles are declared, a new `ApprovalRequired` event is emitted targeting those roles. The timeout clock restarts. Escalation is attempted once; if the escalated request also times out, the run fails.

Timeout enforcement is implemented via a `context.WithDeadline` passed to the approval wait loop. The adapter (or the `gert serve` session timer) is responsible for surfacing the timeout to the approver.

12.4 Policy-as-Code Direction

12.4.1 v2.0: Structured YAML Policy Blocks

For v2.0, gert uses **structured YAML policy blocks** embedded in the runbook's `meta.governance` section. This is the approach inherited from v1, extended with the richer primitives defined in §12.3.

Rationale for v2.0:

- The policy vocabulary is small and closed (six primitives as of v2.0). A full policy DSL would be over-engineered for this set.
- YAML policy blocks are self-contained: the runbook declares its own governance constraints without requiring external policy files or a running policy server.
- Structured blocks are straightforward to validate with JSON Schema at authoring time, giving operators early feedback in the VS Code extension.
- The existing v1 governance model (allowlists, redaction, env blocking) translates directly to structured blocks with no loss of expressiveness.

12.4.2 v2.1+: OPA Integration Hooks

Dennis's research confirms that Open Policy Agent (OPA) is the CNCF standard for policy-as-code in operational tooling. For **v2.1**, gert will introduce OPA integration at the following evaluation hooks:

```
// OPA evaluation hook interface (v2.1 target)
type PolicyEngine interface {
    // EvaluatePreStep is called before each step pre-flight.
    // The policy engine receives the full step context and returns
    // allow/deny with a reason.
    EvaluatePreStep(ctx context.Context, input PolicyInput) (PolicyDecision, error)

    // EvaluatePreRun is called once after plan construction.
    // Used for runbook-level access control (RBAC).
    EvaluatePreRun(ctx context.Context, input PolicyInput) (PolicyDecision, error)
}
```

```
type PolicyInput struct {
    Runbook  RunbookMeta
    Step     ResolvedStep
    Actor    string
    Vars     map[string]string
    Contract *contract.Contract
}

type PolicyDecision struct {
    Allow bool
    Reason string
    // For approval gates: required roles computed by policy engine
    RequiredApprovers []string
}
```

The `PolicyEngine` interface will have two implementations in v2.1:

1. **Built-in:** evaluates the structured YAML policy blocks (backward-compatible with v2.0 runbooks).
2. **OPA:** calls an OPA bundle loaded from a project policy directory (`.gert/policy/*.rego`).

The two implementations compose: the built-in engine runs first; if it allows, the OPA engine runs second. A denial from either blocks the step.

Recommendation: Do not block the v2.0 launch on OPA integration. Ship the structured YAML model first, then introduce OPA as an opt-in engine in v2.1. This allows the governance interface to be designed correctly without the complexity of Rego authoring in the initial release.

12.5 RBAC Model

12.5.1 Identity Establishment

Actor identity is established via the `-as <identity>` flag passed at invocation time. The identity string is an opaque label (conventionally an email address or a role name) recorded in the trace and used for approval gate matching.

v2.0 limitation: Identity is not authenticated. The `-as` flag is self-asserted; gert does not verify that the caller is who they claim to be. Enforcement of identity authenticity is

the responsibility of the surrounding infrastructure (CI/CD pipeline, VS Code extension SSO, MCP server authentication).

v2.1 target: An `IdentityProvider` interface will allow adapters to supply verified identity tokens (e.g., OIDC tokens from VS Code GitHub Copilot authentication), replacing self-asserted identity for governed environments.

12.5.2 Runbook-Level Access Control

Runbooks may declare a `requires_role` field to restrict who may execute them:

```
meta:
  governance:
    requires_role: "incident-responder"
    # or: requires_any_role: ["DRI", "incident-responder"]
    # or: requires_all_roles: ["DRI", "change-manager"]
```

The Runtime Core evaluates this constraint during run initialization (pre-flight, before any step executes). If the actor's identity does not match, `governance/accessDenied` is written to the trace and the run fails immediately.

12.5.3 Step-Level Role Restriction

Individual steps may declare `requires_role` to restrict who may trigger that step. This is distinct from approval gates (which require external approval) — step role restriction *prevents* a step from executing unless the actor themselves holds the required role.

```
steps:
  - id: promote_to_production
    kind: cli
    requires_role: "release-manager"
    command: "kubectl set image deployment/app app={{ .imageTag }}"
```

12.5.4 Separation of Duties

The approval gate model supports separation of duties: the `same-actor-cannot-approve` invariant (defined in §12.3) ensures that a step author cannot self-approve their own gate. In environments requiring formal change management, the actor who initiates the run and the actor(s) who approve high-risk steps must be different.

12.6 Redaction

12.6.1 Pattern-Based Redaction

The primary redaction mechanism is pattern-based: operators declare RE2 regular expressions in `meta.governance.redact`. Redaction is applied eagerly: captured output is redacted *before* any capture variable assignment or trace write.

12.6.2 Explicit Field Marking (v2.1)

For structured tool outputs (JSON objects returned by MCP tools), v2.1 will introduce explicit field marking: a tool definition may declare that a returned field contains sensitive data. The runtime will redact the field value without requiring a regex pattern.

```
# In .tool.yaml (v2.1 target)
outputs:
  - name: api_key
    sensitive: true    # value redacted from trace and captures
```

12.6.3 Redaction Guarantees

- Redaction is applied to all output paths: captured variables, trace events, and event channel payloads.
- Redaction rules are compiled (regex pre-compiled) during run initialization, not re-compiled per step.
- Redaction is applied unconditionally in all run modes (real, dry-run, replay).
- The original (pre-redaction) value is *never* written to disk in any path.

12.7 Audit Trail Requirements

The append-only JSONL trace file is gert's primary audit artifact. Every governance event is a first-class trace entry. The following governance events are guaranteed to be written:

Trace Event	Written When
<code>governance/commandBlocked</code>	A command is blocked by denylist or allowlist
<code>governance/envVarBlocked</code>	Env vars are stripped before subprocess fork
<code>governance/redactionApplied</code>	Redaction rules matched captured output
<code>governance/approvalRequired</code>	A step's approval gate is entered
<code>governance/approvalDecision</code>	An approver submits approve or reject
<code>governance/approvalGranted</code>	Approval quorum is met
<code>governance/approvalRejected</code>	An approver rejects the gate
<code>governance/approvalTimedOut</code>	Approval timeout expires
<code>governance/accessDenied</code>	RBAC check fails (run or step level)
<code>governance/contractWarning</code>	Step contract risk level is critical
<code>governance/policyViolation</code>	Generic policy violation (OPA engine, v2.1)

12.7.1 Trace Integrity

The trace file is protected by:

- **Append-only write discipline:** the trace writer never seeks backward or truncates. Each write is followed by `fsync`.
- **SHA256 evidence hash:** at run completion, the SHA256 of the entire trace file is written as the final record and also stored in a sidecar `trace.sha256` file.
- **Monotonic sequence numbers:** every trace event has a `sequence` field (incrementing integer). Gaps in sequence numbers signal corruption or tampering.

12.7.2 Compliance Alignment

These audit trail guarantees are designed to satisfy the requirements of:

- **SOC 2 Type II** — Change management controls: every command execution and approval is logged with actor identity and timestamp.
- **HIPAA** — Access controls and audit logging: env var blocking and redaction prevent PHI from appearing in traces; approval gates enforce separation of duties.
- **ITIL Change Management** — Normal/standard/emergency change paths can be modeled via `requires_role` and multi-approver gates.

12.8 Extension-Contributed Policy

Extensions may contribute governance rules through the `ContributedPolicyRules()` method of the Extension Host (see §02). These rules compose with host policy under the following invariants:

1. **Additive only.** Extension rules may add restrictions; they may never remove or override host-defined policy. An extension cannot remove a command from the denylist, relax a required role, or suppress a redaction rule.
2. **Host policy takes precedence on conflict.** If an extension rule and a host rule address the same primitive and disagree, the more restrictive interpretation applies.
3. **Extension rules are visible in the trace.** A `governance/extensionPolicyApplied` trace event is written when an extension-contributed rule is evaluated, naming the extension and rule ID.
4. **Extension rules are not trusted with approval authority.** Extensions may not grant approvals on behalf of human approvers. Approval decisions must originate from a human actor via the `RunHandle.Approve()` API.

12.8.1 Extension Policy Rule Schema

```
// PolicyRule is contributed by an extension and evaluated at pre-step checkpoint.
type PolicyRule struct {
    ID          string          // unique within the extension's namespace
    Description string
    // Evaluate returns Deny if the rule blocks this step, Allow otherwise.
    // Extensions communicate rules as static declarations (v2.0); dynamic
    // rule evaluation via the extension RPC transport is v2.1.
    StepKinds   []StepKind // step types this rule applies to (empty = all)
    CommandGlob string     // if set, only applies to commands matching this glob
    Action      PolicyAction // Deny or Warn
    Reason      string       // human-readable reason written to trace event
}
```

Extension-contributed policy is loaded during the Extension Host's `Load()` phase, before the first `Next()` call. Policy rules are static for the lifetime of a run; extensions may not add or remove rules after the run has started.

Chapter 13

Evidence, Tracing, and Resumption

Gert v2 captures a complete, durable record of every execution for audit, compliance, incident response, and operational replay. This section specifies the append-only JSONL trace format, evidence capture model, per-step snapshot contract, run resumption semantics, replay mode, and OpenTelemetry integration. Together these mechanisms ensure that every run produces a self-contained artifact that answers three questions: *what happened*, *what evidence was collected*, and *how can execution be recovered or replicated*.

13.1 Trace File Format

13.1.1 File Location and Naming

Each run produces a single trace file located at:

```
.runbook/runs/<run-id>/trace.jsonl
```

The `<run-id>` is a URL-safe string assigned at run creation time (e.g. `20260418T143022-a3f9c1`). All run artefacts are co-located under the run directory:

```
.runbook/runs/<run-id>/
  trace.jsonl           # append-only JSONL event log
  snapshots/           # per-step state checkpoints
    step-0000.json
    step-0001.json
    ...
  attachments/         # binary evidence referenced from trace events
    <sha256>.bin
```



```

...
run.yaml          # completion manifest (written on terminal event)
annotations.jsonl # operator notes (append-only)
lock              # PID lock (removed on clean exit)

```

13.1.2 JSONL Envelope

Every line in `trace.jsonl` is a self-contained JSON object with a common envelope. Consumers MUST tolerate unknown fields to allow additive schema evolution.

```

{
  "event_id": <string>,    // UUID v4 for correlation
  "sequence": <int64>,     // monotonically increasing within this run
  "kind":     <string>,    // event kind (see §12.1.3)
  "timestamp": <RFC3339>, // UTC timestamp with sub-second precision
  "run_id":   <string>,    // run identifier
  "path":     <array>,    // tree-path to the active node (may be null)
  "payload":  <object>    // kind-specific payload (see below)
}
\end{verbatim}

```

The `\texttt{sequence}` field provides a total order over events in a single run. It is assigned

`\subsection{Event Type Catalogue}`

The following event kinds are normative in v2. All user-supplied string fields (command arguments)

`\subsubsection{\texttt{run/started}}`

Emitted once, as the first event in every trace file.

```

\begin{verbatim}
"payload": {
  "runbook_id": <string>, // runbook name/path
  "user":       <string>, // actor identity
  "mode":       <string>, // "real" | "replay" | "dry-run"
  "inputs":     <object>, // resolved input values (redacted)
}

```

```

    "client":      <string>, // "cli" | "tui" | "web" | "mcp"
    "parent_run_id": <string> // omitted if not an invoke child
  }
\end{verbatim}

```

```
\subsubsection{\texttt{step/started}}
```

Emitted immediately before executing a step.

```

\begin{verbatim}
"payload": {
  "step_id":    <string>, // stable step identifier
  "step_type":  <string>, // "cli" | "manual" | "tool" | "invoke"
  "step_name":  <string>  // human-readable name from runbook YAML
}
\end{verbatim}

```

```
\subsubsection{\texttt{step/completed}}
```

Emitted after a step exits cleanly.

```

\begin{verbatim}
"payload": {
  "step_id":    <string>,
  "exit_status": <int>,      // exit code for cli steps; 0 for others
  "duration_ms": <int64>,   // wall-clock duration in milliseconds
  "captures":   <object>,   // captured output variables (redacted)
  "evidence":   [           // zero or more evidence items
    {
      "name":    <string>,
      "kind":    "text" | "checklist" | "attachment",
      "value":   <string>, // for kind=text
      "items":   <object>, // for kind=checklist
      "path":    <string>, // relative path for kind=attachment
      "sha256":  <string>, // hex SHA256 for kind=attachment
      "size":    <int64>   // bytes for kind=attachment
    }
  ]
}
\end{verbatim}

```

```
    }  
  ]  
}
```

step/failed

Emitted when a step exits with a non-zero status or returns a runtime error.

```
"data": {  
  "step_id":    <string>,  
  "error_type": <string>, // "exit_code" | "timeout" | "governance"  
                                   // | "tool_error" | "invoke_failed"  
  "error":      <string>, // human-readable message (redacted)  
  "exit_status": <int>,   // for error_type=exit_code  
  "duration_ms": <int64>  
}
```

step/skipped

Emitted when a step is not executed due to branch evaluation or iterate exhaustion.

```
"data": {  
  "step_id": <string>,  
  "reason":  "branch_not_taken" | "iterate_exhausted" | "dry_run"  
}
```

step/retrying

Emitted before each retry attempt (requires the retry policy feature).

```
"data": {  
  "step_id":    <string>,  
  "attempt":    <int>,      // attempt number (1 = first retry)  
  "max_attempts": <int>,  
  "backoff_ms":  <int64>  
}
```

tool/invoked

Emitted when the tool runtime dispatches a tool call.

```
"data": {  
  "tool_name":      <string>,  
  "transport":      "stdio" | "jsonrpc" | "mcp",  
  "correlation_id": <string>, // UUIDv4, links to tool/responded  
  "inputs":         <object>  // sanitized inputs  
}
```

tool/responded

Emitted when the tool call returns.

```
"data": {  
  "correlation_id": <string>,  
  "exit_code":      <int>,  
  "duration_ms":    <int64>,  
  "truncated":      <bool>    // true if output was capped  
}
```

manual/prompted

Emitted when a manual step displays its prompt to the operator.

```
"data": {  
  "step_id":      <string>,  
  "prompt_text":  <string>, // the prompt shown (redacted)  
  "sla_seconds":  <int>     // 0 if no SLA configured  
}
```

manual/completed

Emitted when the operator submits a response to a manual step.

```
"data": {  
  "step_id":      <string>,  
  "response":     <string>,  
  "duration_ms":  <int64>,  
  "truncated":    <bool>    // true if output was capped  
}
```

```
"response":    <string>,    // operator's text response (redacted)
"attestation": <string>,    // "operator_confirmed" | "skipped"
"approver":    <string>,    // identity of responding actor
"duration_ms": <int64>
}
```

`governance/blocked`

Emitted when a governance rule blocks execution.

```
"data": {
  "step_id":      <string>,
  "rule":         <string>,    // allowlist | denylist | env_block
                                // | output_redact | approval_gate
  "action_attempted": <string>, // description of blocked action
  "policy":       <string>     // policy name or path
}
```

`governance/approved`

Emitted when an approval gate is cleared.

```
"data": {
  "step_id":      <string>,
  "approver":     <string>,    // identity of approver
  "approval_method": <string>, // "interactive" | "mcp" | "api"
  "duration_ms":  <int64>
}
```

`run/completed`

Terminal event for a successful run.

```
"data": {
  "final_status": "passed" | "failed" | "skipped",
  "duration_ms":  <int64>,
  "step_count":   <int>,
  "passed":       <int>,
}
```

```

    "failed":      <int>,
    "skipped":    <int>
}

```

run/failed

Terminal event for a run that halted due to an unrecoverable error.

```

"data": {
  "failing_step_id": <string>,
  "error":           <string>,
  "duration_ms":     <int64>
}

```

checkpoint

Internal event written by `RunStore.SaveCheckpoint` to link a snapshot file to the trace sequence. Not user-facing but required for resumption.

```

"data": {
  "snapshot_file": <string> // e.g. "step-0003.json"
}

```

13.1.3 Crash Safety

The trace file is opened with `O_APPEND` and each event is written as a single `write(2)` call followed by `fsync`. On POSIX systems, a write of fewer than `PIPE_BUF` bytes (4096 bytes on Linux) to an `O_APPEND` file is atomic. Because the JSON encoding of a single trace event is always well under 4096 bytes for typical runbooks, partial-line corruption on crash is avoided in practice. For large payloads (e.g. tool output), the runtime truncates at a configurable limit (default 4 KB per field) before encoding.

The Go implementation:

```

// WriteTrace appends one event atomically and fsyncs.
func (d *DirRunStore) WriteTrace(event DurableEvent) error {
    p := filepath.Join(d.baseDir, "trace.jsonl")
    f, err := os.OpenFile(p, os.O_CREATE|os.O_WRONLY|os.O_APPEND, 0644)
    if err != nil {

```

```

        return fmt.Errorf("open trace: %w", err)
    }
    defer f.Close()

    data, err := json.Marshal(event)
    if err != nil {
        return fmt.Errorf("marshal event: %w", err)
    }
    data = append(data, '\n') // JSONL delimiter
    if _, err := f.Write(data); err != nil {
        return fmt.Errorf("write trace: %w", err)
    }
    return f.Sync() // durability guarantee
}

```

On recovery, any line that fails `json.Unmarshal` is silently skipped. The last clean **checkpoint** event determines the recoverable state.

13.1.4 Tamper Evidence

The v2 trace format supports optional HMAC-SHA256 signing of each event. When a signing key is configured (via `GERT_TRACE_KEY` or the workspace config), the runtime appends a **"sig"** field to the envelope containing the HMAC-SHA256 of the canonical JSON of all other fields. Verifiers can replay the trace and recompute signatures. This does not prevent deletion of events (append-only at the filesystem level provides that property) but detects modification of existing events.

13.2 Evidence Capture

13.2.1 What Constitutes Evidence

In v2, “evidence” is any artefact that attests to a step’s outcome:

- **Command output** — stdout/stderr of a `cli` step, captured and embedded in `step/completed`.
- **Manual attestations** — operator-provided text, checklist responses, and explicit confirmations from `manual` steps.

- **Tool responses** — structured JSON returned by a `tool` step, captured in `step/completed.captures`.
- **File attachments** — binary or text files explicitly attached by an operator or produced by a step, stored in the `attachments/` directory and referenced by SHA256 digest.

13.2.2 SHA256 Hashing

All file attachments are content-addressed by their SHA256 digest. The `evidence` package exposes:

```
// HashFile returns hex SHA256 and file size.
func HashFile(path string) (sha256hex string, size int64, err error)

// NewAttachmentEvidence creates an EvidenceValue with hash and size.
func NewAttachmentEvidence(path string) (*EvidenceValue, error)
```

The hash is stored in the `step/completed` event alongside the relative path under `attachments/`. Consumers can verify integrity at any time by re-hashing the referenced file.

Text evidence fields (command output, operator responses) are not individually hashed — the overall trace event integrity is provided by the optional HMAC-SHA256 event signature described in § 12.1.5.

13.2.3 File Attachments

When a step produces a binary artefact, the runtime:

1. Copies the file to `.runbook/runs/<run-id>/attachments/<sha256>.<ext>`.
2. Emits an `EvidenceValue` of kind `attachment` in the `step/completed` event with fields `path`, `sha256`, and `size`.

Attachments are deduplicated by content: if two steps produce identical file content, they share one attachment file. The trace events may reference the same `sha256` value independently.

13.2.4 Evidence Retention

Evidence is retained for the lifetime of the run directory. The workspace-level retention policy (configurable in `gert.yaml`) specifies:

- `retention.runs` — number of completed runs to retain (default: 50).
- `retention.max_age_days` — maximum age in days (default: 90).
- `retention.attachments` — whether to include attachments in retention (default: true).

The `gert gc` command enforces retention policy by removing the oldest run directories, deleting their `attachments/` content first.

13.3 Run State Snapshots

13.3.1 Checkpoint Model

A checkpoint is a complete serialisation of `RunState` written to `snapshots/step-NNNN.json` after every successfully completed step. The snapshot captures the minimum state needed to resume execution from the next step:

```
type RunState struct {
    RunID           string
    RunbookPath     string
    Mode            string           // "real" | "replay" | "dry-run"
    StartedAt       time.Time
    Actor           string
    CurrentStepIndex int           // flat-mode index (legacy compat)
    Path            TreePath      // tree-mode cursor (authoritative)
    Vars            map[string]string // resolved input bindings
    Captures        map[string]string // accumulated step outputs
    History          []*StepResult    // completed step records
    InvokeStack     []InvokeFrame    // nested invoke context stack
    IterateState    map[string]*IterateProgress
    ParallelSlots   map[string][]SlotState
    LastCheckpointSeq int64           // trace seq at checkpoint time
}
```

13.3.2 Atomic Snapshot Write

Snapshots are written atomically: the runtime marshals the state to a `.tmp` file, fsyncs it, then renames it to the final path. An incomplete snapshot (e.g. from a crash mid-write) retains the `.tmp` suffix and is skipped during recovery.

```
func (d *DirRunStore) SaveCheckpoint(state *RunState) (string, error) {
    filename := fmt.Sprintf("step-%04d.json", len(state.History))
    finalPath := filepath.Join(d.baseDir, "snapshots", filename)
    tmpPath   := finalPath + ".tmp"

    data, _ := json.MarshalIndent(state, "", " ")
    os.WriteFile(tmpPath, data, 0644)           // write to temp
    tf, _ := os.Open(tmpPath); tf.Sync(); tf.Close() // fsync temp
    os.Rename(tmpPath, finalPath)                // atomic rename
    // ... then write checkpoint trace event
}
```

13.3.3 Minimum Resumption State

The state required to resume a run from step N is exactly the snapshot at `step-NNNN.json`. It contains:

- All resolved input values (**Vars**).
- Outputs captured by completed steps (**Captures**).
- The tree cursor (**Path**) identifying which node executes next.
- Branch decisions already taken (encoded in **History** step statuses).
- Iterate progress counters and parallel slot states.
- The invoke stack for nested runbooks.

The runbook YAML itself is not stored in the snapshot; it is re-read from `RunbookPath` at resume time. Operators must not modify the runbook between the original run and resumption.

13.4 Run Resumption

13.4.1 The `gert exec -resume` Contract

`gert exec --resume <run-id>`

The runtime executes the following recovery sequence:

1. Locate `.runbook/runs/<run-id>/`.
2. Attempt to acquire the `lock` file. Fail if a live process holds it.
3. Call `RunStore.LoadLatestCheckpoint()` — scan `snapshots/` descending, skip `.tmp` files, return the newest valid JSON.
4. Re-open `trace.jsonl` for append.
5. Advance `CurrentStepIndex` by one (or advance the tree `Path` cursor past the last completed node).
6. Re-build the governance engine from the runbook’s policy block.
7. Resume the execution loop from the restored `Path`.

The resumption does *not* re-execute any step recorded in `History`. It continues from the step *after* the last checkpoint.

13.4.2 Resumability by Step Type

- **cli steps** — safe to resume. The step is re-executed from scratch. Runbook authors should ensure cli steps are idempotent (e.g. `kubectl apply` over `kubectl create`).
- **manual steps** — the operator is re-prompted. The prior response (if any) is lost. The prompt is re-displayed and the operator must re-attest.
- **tool steps** — resumable if idempotent. If the tool call was in-flight when the process crashed (i.e. a `tool/invoked` event exists with no matching `tool/responded`), the runtime re-issues the call. Authors must declare `idempotent: true` on tools where safe; others are re-executed with a warning log.

- **invoke steps** — if the child run completed (its terminal event is in the parent trace), the invoke is treated as completed. If the child run was in progress, the child is itself resumed via the nested run ID stored in the **InvokeStack**.
- **branch/iterate** — structural nodes; not executed directly. Cursor advancement handles re-entry correctly.

13.4.3 Idempotency Guidance

Runbook authors should write resumable runbooks by following these guidelines:

- Use `kubectl apply` and `terraform apply` instead of imperative create commands.
- Use `-if-not-exists` or `-create-or-replace` flags where available.
- For tool steps, declare `idempotent: true` in the tool definition when the operation is safe to call multiple times.
- For manual steps, write prompts as questions that remain valid on a second ask (“Has the database migration completed?” rather than “Confirm you have started the migration”).
- Avoid steps whose side effects depend on the current time or a random seed without an explicit override mechanism.

13.4.4 Partial Step Resumption

If a `tool/invoked` event has no matching `tool/responded` event, the tool call was interrupted mid-flight. On resumption:

1. The runtime logs a warning identifying the orphaned correlation ID.
2. If `idempotent: true`: the tool call is re-issued from the beginning.
3. If `idempotent: false` (default): the step is treated as **failed** and execution follows the normal failure path (error branch, if defined, or halt).

13.5 Replay Mode

13.5.1 Overview

```
gert exec --mode replay --scenario <scenario.yaml>
```

Replay mode executes a runbook against a set of pre-recorded command responses and evidence values. No real commands are executed and no real tools are invoked. The mode is used for:

- **Testing** — validating that a runbook’s branching logic, captures, and governance rules behave correctly for a given set of inputs and command outputs (see § 13 Testing and Acceptance).
- **Demonstration** — showing a runbook’s execution flow without real infrastructure.
- **Regression** — verifying that a refactored runbook produces the same trace as the original.

13.5.2 Scenario File Format

A scenario file is a YAML document that maps commands to recorded responses and step IDs to recorded evidence:

```
commands:
  - argv: ["kubectl", "get", "pods", "-n", "prod"]
    stdout: "NAME          READY   STATUS    RESTARTS\npod-abc    1/1     Running   0\n"
    stderr: ""
    exit_code: 0

evidence:
  check-pod-health:      # step_id
    observation:         # evidence name
      kind: text
      value: "pod is healthy, proceeding"
```

Commands are matched in order of declaration (first match wins). Unmatched commands fail the scenario unless `allow_unmatched: true` is set.

13.5.3 Replay Semantics

In replay mode:

- **cli** steps — the command executor returns the pre-recorded stdout/stderr/exit_code.
- **manual** steps — the evidence collector returns the pre-recorded evidence values instead of prompting the operator.
- **tool** steps — tool invocations are intercepted; if a matching recorded response exists, it is returned. Otherwise the step fails.
- **invoke** steps — the child runbook is executed in replay mode using the same scenario (or a nested scenario file if `scenario_file:` is specified on the invoke step).
- Governance rules — evaluated normally; replay does not bypass governance.
- Trace events — written normally; the replay trace can be compared against a golden trace for regression testing.

13.5.4 Determinism Requirement

For replay to produce the same outcome across runs, the runbook must be deterministic given fixed inputs and command outputs. Non-determinism sources to avoid:

- Timestamps in expressions (use captured values instead of `time.Now()`).
- Random identifiers in step names or captures.
- File system state that is not reproduced by the scenario.
- External API calls in tool steps that are not replayed (tool steps should be declared `idempotent: true` and covered by scenario fixtures).

13.6 OpenTelemetry Integration

13.6.1 Rationale

The JSONL trace format is optimised for durability, human readability, and offline analysis. OpenTelemetry (OTel) spans are optimised for real-time observability, distributed tracing, and integration with platforms such as Jaeger, Zipkin, Datadog, and Grafana Tempo. Both serve complementary purposes and v2 supports both simultaneously.

13.6.2 Span Hierarchy

When OTel is enabled, the runtime creates a span hierarchy that mirrors the runbook execution tree:

```
Span: gert.run           // root span, covers entire run
  Span: gert.step        // one per step executed
    Span: gert.tool.invoke // one per tool call within a step
  Span: gert.step        // next step
  ...
```

The Go implementation uses `go.opentelemetry.io/otel/trace`:

```
func (e *Engine) runWithOTel(ctx context.Context) error {
    ctx, runSpan := otel.Tracer("gert").Start(ctx, "gert.run",
        trace.WithAttributes(
            attribute.String("gert.run_id",    e.State.RunID),
            attribute.String("gert.runbook",   e.Runbook.Meta.Name),
            attribute.String("gert.actor",     e.State.Actor),
            attribute.String("gert.mode",      e.State.Mode),
        ),
    )
    defer runSpan.End()
    return e.execute(ctx)
}

func (e *Engine) executeStep(ctx context.Context, step Step) error {
    ctx, stepSpan := otel.Tracer("gert").Start(ctx, "gert.step",
        trace.WithAttributes(
            attribute.String("gert.step_id",    step.ID),
            attribute.String("gert.step_type",  step.Type),
            attribute.String("gert.step_name",  step.Name),
        ),
    )
    defer stepSpan.End()
    // ...
}
```

13.6.3 Span Attributes

The following attributes are set on all spans:

Attribute	Spans	Description
<code>gert.run_id</code>	all	Run identifier
<code>gert.runbook</code>	all	Runbook name
<code>gert.actor</code>	run	Actor identity
<code>gert.mode</code>	run	Execution mode
<code>gert.step_id</code>	step, tool	Step identifier
<code>gert.step_type</code>	step	Step type
<code>gert.tool_name</code>	tool	Tool name
<code>gert.transport</code>	tool	Transport (stdio/jsonrpc/mcp)

Trace context is propagated into tool invocations via the W3C `traceparent` header where the transport supports it (HTTP-based MCP tools). For stdio tools, context is passed as an environment variable `OTEL_TRACEPARENT`.

13.6.4 Relationship to JSONL Trace

OTel spans and the JSONL trace are complementary, not redundant:

- The JSONL trace is the authoritative durable record. It contains evidence hashes, governance events, snapshot references, and redacted output that OTel spans do not.
- OTel spans are ephemeral, push-based, and optimised for latency-sensitive observability dashboards.
- The trace event `"seq"` is embedded as the `gert.trace_seq` span attribute, enabling correlation between spans and trace events.

13.6.5 Configuration

OTel is opt-in and configured via standard OTEL environment variables:

```
OTEL_EXPORTER_OTLP_ENDPOINT=http://localhost:4317
OTEL_SERVICE_NAME=gert
OTEL_TRACES_SAMPLER=always_on
```


When `OTEL_EXPORTER_OTLP_ENDPOINT` is not set, the OTel SDK is initialised with a no-op tracer and zero overhead is incurred. There is no `-otel` flag; configuration is entirely via the environment. This ensures that CI and offline use work without modification.

13.7 Implementation Notes for Go

The key Go interfaces for this subsystem are:

```
// TraceWriter is the interface the engine uses to record events.
// Implementations must be safe for concurrent use from a single goroutine
// (the engine's execution loop); multi-goroutine safety is optional.
type TraceWriter interface {
    Write(event DurableEvent) error
    Close() error
}

// RunStore manages all artefacts for a single run.
type RunStore interface {
    RunID() string
    WriteTrace(event DurableEvent) error
    ReadTrace() ([]DurableEvent, error)
    ReadTraceSince(afterSeq int64) ([]DurableEvent, error)
    SaveCheckpoint(state *RunState) (string, error)
    LoadLatestCheckpoint() (*RunState, int64, error)
    AcquireLock() error
    ReleaseLock() error
    WriteManifest(manifest *RunManifest) error
}

// EvidenceCollector is the interface for gathering evidence at manual steps.
type EvidenceCollector interface {
    Collect(ctx context.Context, stepID string,
        prompts []Prompt) (map[string]*EvidenceValue, error)
}
```

The **DirRunStore** is the production implementation. A **MemRunStore** (storing events in a slice) is provided for unit tests where filesystem I/O is undesirable. The **ReplayEvidenceCollector** returns pre-recorded evidence for scenario tests.

Chapter 14

Adapter Contracts

Gert v2 decouples the core runtime from presentation adapters (TUI, Web, VS Code) and integration systems (CI/CD platforms, observability backends, approval workflow systems) through explicitly versioned contracts. An **adapter contract** is a stable, versioned interface that gert commits to maintain across minor and patch releases. Adapters are consumers of these contracts: they observe, render, or orchestrate gert runs, but they own no execution logic.

This chapter defines the four primary adapter surfaces exposed by gert v2: JSON-RPC execution contract, WebSocket event stream contract, tool invocation contract, and extension handshake contract.

14.1 What is an Adapter Contract?

An **adapter contract** is a versioned API specification that defines:

1. The set of RPC methods, request/response schemas, and error envelopes.
2. The stability guarantee: which fields are required, which are optional, and which may be added in future minor versions.
3. The versioning scheme: how breaking changes are communicated, and how long deprecated versions are supported.
4. The transport layer: HTTP, WebSocket, stdio JSON-RPC, etc.

Adapter contracts are the integration surface of gert. They are designed for external systems to consume. Breaking changes to a contract require a major version bump and a deprecation period.

Example consumers:

- VS Code extension (consumes JSON-RPC execution contract and WebSocket event stream) [13]
- GitLab CI runner (consumes JSON-RPC execution contract to trigger runs)
- Datadog integration (consumes event stream to emit metrics)
- Slack approval bot (consumes event stream, submits approvals via JSON-RPC)

14.2 Contract Categories

Gert v2 exposes four distinct adapter surfaces, each with its own versioning and stability guarantees.

14.2.1 JSON-RPC Execution Contract (exec/v2)

The execution contract is the primary interface for starting, monitoring, and controlling gert runs. It is served over `gert serve -stdio` (stdio JSON-RPC) or `gert serve -http` (HTTP POST to `/rpc`).

Contract version: `exec/v2`

Transport: `stdio` JSON-RPC 2.0 or HTTP POST (JSON-RPC over HTTP)

Methods:

Request schema — `exec/start`:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "method": "exec/start",
  "params": {
    "runbookPath": "./runbooks/incident-triage.yaml",
    "inputs": {
```

Method	Description
<code>exec/start</code>	Start a new runbook execution. Params: <code>runbookPath</code> , <code>inputs</code> , <code>mode</code> , <code>actorId</code> . Returns: <code>runId</code> , <code>status</code> .
<code>exec/next</code>	Advance the run by one step or until it reaches a pause point (approval, manual, error). Params: <code>runId</code> . Returns: <code>stepId</code> , <code>status</code> , <code>output</code> .
<code>exec/cancel</code>	Cancel an in-progress run. Params: <code>runId</code> , <code>reason</code> . Returns: <code>cancelled: bool</code> .
<code>exec/status</code>	Get the current status of a run. Params: <code>runId</code> . Returns: <code>status</code> , <code>currentStepId</code> , <code>progress</code> .
<code>run/list</code>	List all runs in the workspace. Params: <code>filter</code> , <code>limit</code> . Returns: <code>runs: [RunSummary]</code> .
<code>run/get</code>	Get full details of a run. Params: <code>runId</code> . Returns: <code>run: RunDetails</code> .

TABLE 14.1: JSON-RPC execution contract methods (`exec/v2`).

```

    "incident_id": "INC-12345",
    "severity": "high"
  },
  "mode": "real",           // real | dry-run | replay
  "actorId": "ops@acme.example"
}
}

```

Response schema — `exec/start`:

```

{
  "jsonrpc": "2.0",
  "id": "1",
  "result": {
    "runId": "run-abc123",
    "status": "running",
    "createdAt": "2026-04-18T12:00:00Z",
    "traceFile": "./traces/run-abc123.jsonl"
  }
}

```

```
}
```

Error envelope: All errors follow JSON-RPC 2.0 error format:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "error": {
    "code": 1001,
    "message": "run not found",
    "data": {
      "runId": "run-xyz789",
      "category": "client_error",
      "retryable": false
    }
  }
}
```

See Section 14.4 for the canonical error code registry.

Stability guarantee: Within the `exec/v2` contract:

- All required fields in request/response schemas are stable.
- Optional fields may be added in minor versions (clients MUST ignore unknown fields).
- Required fields will not be removed or renamed (breaking change → `exec/v3`).
- Error codes are stable; new codes may be added but existing codes will not change semantics.

14.2.2 WebSocket Event Stream Contract (`events/v2`)

The event stream contract allows adapters to subscribe to real-time events from a running or completed gert execution. It is served over `gert serve -http` at the `/ws` or `/events` endpoint.

Contract version: `events/v2`

Transport: WebSocket (wss:// or ws://)

Connection lifecycle:

1. Client establishes WebSocket connection to **/ws**.
2. Client sends a **subscribe** message with **runId**.
3. Server streams all events for that run in order.
4. Client may disconnect at any time; server closes gracefully.

Subscribe message:

```
{
  "action": "subscribe",
  "runId": "run-abc123",
  "sinceSequence": 0 // Optional: replay from sequence number
}
```

Event envelope: Events are transmitted as JSON objects, one per WebSocket message. The event envelope is identical to the trace event format defined in §12:

```
{
  "eventType": "step/started",
  "timestamp": "2026-04-18T12:00:00Z",
  "runId": "run-abc123",
  "stepId": "lookup_incident",
  "sequence": 42,
  "payload": {
    "stepType": "tool",
    "toolName": "acme.pd-lookup"
  }
}
```

All events carry:

- **eventType**: canonical event type (e.g., **step/started**, **governance/approval_required**).

- **timestamp**: RFC 3339 timestamp.
- **runId**: globally unique run identifier.
- **sequence**: monotonically increasing event sequence number (starts at 0).
- **payload**: event-specific data.

Reconnection protocol: If the WebSocket connection is lost, the client may reconnect and provide **sinceSequence** to resume from the last received event:

```
{
  "action": "subscribe",
  "runId": "run-abc123",
  "sinceSequence": 42  // Resume from event 43
}
```

The server will replay all events with **sequence** > 42.

Stability guarantee: Within the **events/v2** contract:

- Event envelope fields (**eventType**, **timestamp**, **runId**, **sequence**) are stable.
- New event types may be added in minor versions.
- Existing event types will not change their **eventType** string.
- Payload schemas may add optional fields; clients **MUST** ignore unknown fields.

14.2.3 Tool Invocation Contract (**tool/v2**)

The tool invocation contract defines the stdio JSON-RPC protocol that tools must implement to be invoked by gert. All tools — whether built-in, project-level, or extension-contributed — implement this contract.

Contract version: **tool/v2**

Transport: stdio JSON-RPC 2.0

Invocation request:

```
{
  "jsonrpc": "2.0",
  "id": "invocation-5",
  "method": "invoke",
  "params": {
    "action": "lookup",
    "inputs": {
      "id": "INC-12345"
    },
    "context": {
      "runId": "run-abc123",
      "stepId": "lookup_incident",
      "traceId": "trace-xyz789",
      "userId": "ops@acme.example",
      "mode": "real",
      "attempt": 1
    }
  }
}
```

Success response:

```
{
  "jsonrpc": "2.0",
  "id": "invocation-5",
  "result": {
    "outputs": {
      "incident": {
        "id": "INC-12345",
        "severity": "high",
        "status": "open"
      }
    }
  },
  "exitCode": 0,
  "telemetry": {
```

```
        "durationMs": 320,  
        "startedAt": "2026-04-18T12:00:00Z",  
        "finishedAt": "2026-04-18T12:00:00.320Z"  
    }  
}  
}
```

Error response:

```
{  
  "jsonrpc": "2.0",  
  "id": "invocation-5",  
  "error": {  
    "code": -32050,  
    "message": "incident not found",  
    "data": {  
      "category": "runtime",  
      "retryable": false,  
      "details": {  
        "incidentId": "INC-12345",  
        "statusCode": 404  
      }  
    }  
  }  
}
```

Cancellation request:

```
{  
  "jsonrpc": "2.0",  
  "id": "cancel-5",  
  "method": "cancel",  
  "params": {  
    "invocationId": "invocation-5",  
    "reason": "run-cancelled"  
  }  
}
```

MCP tool adapter: When an MCP server is used as a tool source (§5), the gert host wraps the MCP `tools/call` protocol to satisfy the tool invocation contract. The adapter:

1. Translates the `invoke` request to an MCP `tools/call` request.
2. Forwards the `context` object as `_gert_context` in the MCP request (extension to MCP protocol).
3. Translates the MCP response back to the tool invocation contract response.
4. Flattens MCP `content` arrays to a single string for capture.

Stability guarantee: Within the `tool/v2` contract:

- Request/response envelope fields are stable.
- The `context` object fields are stable; new fields may be added.
- Error code ranges are stable (see Section 14.4).
- Tools MUST ignore unknown fields in requests.

14.2.4 Extension Handshake Contract (`extension/v2`)

The extension handshake contract defines the initialization and capability negotiation protocol between the gert host and an extension process. It is defined in §4 and summarized here for completeness.

Contract version: `extension/v2`

Transport: `stdio JSON-RPC 2.0`

Initialize request:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "method": "extension/initialize",
  "params": {
```

```
"host": {
  "name": "gert",
  "version": "2.0.0",
  "apiVersion": "2.0"
},
"protocolVersion": "2.0",
"grantedCapabilities": [
  "capability/tool-registration",
  "capability/network"
],
"runContext": {
  "runId": "run-abc123",
  "mode": "real"
}
}
```

Initialize response:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "result": {
    "extension": {
      "name": "acme.incident-pack",
      "version": "1.2.0"
    },
    "protocolVersion": "2.0",
    "acknowledgedCapabilities": [
      "capability/tool-registration",
      "capability/network"
    ]
  }
}
```

Heartbeat (ping/pong):

```
Request:
{
  "jsonrpc": "2.0",
  "id": "42",
  "method": "extension/ping",
  "params": {}
}
```

```
Response:
{
  "jsonrpc": "2.0",
  "id": "42",
  "result": {
    "status": "ok"
  }
}
```

Stability guarantee: Within the `extension/v2` contract:

- The `extension/initialize` request/response schema is stable.
- New capabilities may be added; extensions **MUST** ignore unknown capabilities in `grantedCapabilities`.
- The `extension/ping` method signature is stable.
- Extension error codes (§4) are stable.

14.3 Contract Versioning

Each adapter contract has a version declared in the initial handshake or HTTP header. The version format is `{surface}/{major}` (e.g., `exec/v2`, `events/v2`). There is no minor version: any breaking change bumps the major version.

14.3.1 Version Declaration

HTTP-based contracts (JSON-RPC, WebSocket): Clients **MUST** send an `X-Gert-Contract` header on the initial HTTP request:

```
GET /ws HTTP/1.1
Host: gert.acme.example:8443
X-Gert-Contract: events/v2
```

The server responds with:

```
HTTP/1.1 200 OK
X-Gert-Contract: events/v2
```

If the client sends an unsupported contract version, the server responds with HTTP 400 and error:

```
{
  "error": {
    "code": 4001,
    "message": "unsupported contract version",
    "data": {
      "requestedVersion": "events/v3",
      "supportedVersions": ["events/v2"]
    }
  }
}
```

stdio-based contracts (tool, extension): The contract version is declared in the first message exchanged (initialize or handshake):

```
{
  "protocolVersion": "2.0"
}
```

If the protocol version is unsupported, the responder returns error code -32001 (protocol version not supported).

14.3.2 Breaking vs. Non-Breaking Changes

Non-breaking changes (minor version, no contract version bump):

- Adding optional fields to requests or responses.

- Adding new methods to the contract.
- Adding new error codes.
- Adding new event types.

Consumers **MUST** implement forward compatibility: ignore unknown fields, ignore unknown methods, ignore unknown event types.

Breaking changes (major version bump):

- Removing or renaming required fields.
- Changing the semantics of existing methods.
- Removing methods.
- Changing error code meanings.
- Incompatible transport changes (e.g., switching from JSON-RPC to gRPC).

Breaking changes require a new contract version (e.g., `exec/v2` → `exec/v3`).

14.3.3 Deprecation Policy

- When a new major contract version is released, the previous version is supported for 12 months after the new version's GA date.
- During the deprecation period, both versions are supported simultaneously.
- After 12 months, the old version is removed. Clients must upgrade.

Example timeline:

- 2026-04-01: `exec/v2` released.
- 2027-06-01: `exec/v3` released; `exec/v2` enters deprecation.
- 2028-06-01: `exec/v2` removed; only `exec/v3` supported.

14.3.4 Forward and Backward Compatibility

Forward compatibility (clients): Clients **MUST** ignore unknown fields in responses. This allows the server to add optional fields without breaking existing clients.

Example:

```
// Server adds a new optional field in exec/v2
{
  "result": {
    "runId": "run-abc123",
    "status": "running",
    "newOptionalField": "value" // Clients ignore this
  }
}
```

Backward compatibility (servers): Servers **MUST** accept requests that omit optional fields added in later minor versions. Servers **MUST NOT** require new optional fields.

Example:

```
// Client sends an old request format (before a new optional field was added)
{
  "method": "exec/start",
  "params": {
    "runbookPath": "./runbook.yaml"
    // Missing new optional field "labels"
  }
}
// Server accepts and defaults the missing field
```

14.4 Error Codes

All adapter contracts use a canonical error code registry. Error codes are integers organized by category. Codes are stable across contract versions.

Error envelope structure:


```
{
  "error": {
    "code": 2001,
    "message": "command denied: not in allowlist",
    "data": {
      "category": "governance",
      "retryable": false,
      "details": {
        "command": "rm",
        "reason": "in_denylist"
      }
    }
  }
}
```

Error code stability:

- Existing error codes will not change their meaning.
- New error codes may be added in minor versions.
- Clients MUST handle unknown error codes gracefully (fallback to generic error handling).

14.5 Contract Test Suite

The gert repository includes a contract test suite that validates adapter implementations against the canonical contracts. Adapter authors SHOULD run these tests to ensure compatibility.

14.5.1 Test Suite Structure

Contract tests are located in `testdata/contracts/`:

```
testdata/contracts/
  exec-v2/
    01-start-run.test.yaml
```

```
    02-cancel-run.test.yaml
    03-list-runs.test.yaml
events-v2/
    01-subscribe.test.yaml
    02-reconnect.test.yaml
tool-v2/
    01-invoke.test.yaml
    02-cancel.test.yaml
extension-v2/
    01-initialize.test.yaml
    02-ping.test.yaml
```

Each test case is a YAML file with:

- **description:** human-readable test case description.
- **request:** the JSON-RPC request or WebSocket message.
- **expectedResponse:** the expected response (or error).
- **preconditions:** setup required before the test (e.g., "a run with id run-abc123 exists").

14.5.2 Running Contract Tests

```
$ gert contract test --adapter=vscode
Running contract tests for adapter: vscode
✓ exec-v2/01-start-run (passed)
✓ exec-v2/02-cancel-run (passed)
× events-v2/01-subscribe (failed: missing 'sequence' field)
```

Contract test results: 2 passed, 1 failed

Adapter implementations SHOULD achieve 100% pass rate before release.

14.5.3 Contract Test CI Integration

Contract tests are part of the CI gate for all adapter implementations:

- The VS Code extension CI runs `gert contract test -adapter=vscode`.

- Third-party adapters are encouraged to use the same test suite.
- Failed contract tests block merges to the main branch.

14.6 Reference Implementations

VS Code extension. The official VS Code extension (`vscode/` in the repository) is the reference implementation of the JSON-RPC execution contract (`exec/v2`) and the WebSocket event stream contract (`events/v2`). Adapter authors should study its implementation for best practices.

Built-in tools. The built-in tools (`ext/builtin/tools/` in the repository) are the reference implementation of the tool invocation contract (`tool/v2`). They demonstrate how to handle `invoke`, `cancel`, and error responses.

MCP adapter. The MCP tool adapter (`ext/toolruntime/mcp_adapter.go`) demonstrates how to wrap an external protocol (MCP) to satisfy the tool invocation contract.

14.7 Contract Stability Guarantees

Gert v2 commits to the following stability guarantees for all adapter contracts:

1. **Semantic versioning:** major version bumps indicate breaking changes; minor/patch versions are backward compatible.
2. **12-month deprecation period:** old contract versions are supported for 12 months after a new major version is released.
3. **Additive changes only:** within a major version, only non-breaking changes (adding optional fields, new methods, new event types) are permitted.
4. **Error code stability:** error codes do not change meaning within a major version.
5. **Forward compatibility:** clients must ignore unknown fields; servers must accept omitted optional fields.

These guarantees are designed to minimize adapter churn and allow external systems to depend on gert contracts with confidence.

14.8 Contract Documentation and Tooling

14.8.1 OpenAPI Specifications

All HTTP-based contracts (JSON-RPC over HTTP, WebSocket event stream) are documented as OpenAPI 3.1 specifications [17] in `specs/contracts/`:

- `specs/contracts/exec-v2.openapi.yaml`
- `specs/contracts/events-v2.openapi.yaml`

These specifications are the source of truth for request/response schemas and are used to generate client libraries.

14.8.2 JSON Schema Validation

All JSON-RPC request/response schemas are defined as JSON Schema (Draft 2020-12) [28] in `specs/contracts/schemas/`. These schemas are used for:

- Runtime validation of requests/responses.
- Contract test validation.
- Client library generation (TypeScript, Python, Go).

14.8.3 Client Library Generation

The gert project provides code generation tools to generate client libraries from the OpenAPI and JSON Schema definitions:

```
$ gert codegen client --language=typescript --output=./clients/ts/  
Generated TypeScript client for exec/v2 and events/v2  
Files written to ./clients/ts/
```

Official client libraries are provided for:

- TypeScript (for VS Code and web adapters)
- Python (for CI/CD integrations)
- Go (for extension authors)

Code	Description	Retryable
1xxx — Execution Errors		
1001	Run not found (invalid <code>runId</code>)	No
1002	Step failed (runtime error in step execution)	Maybe
1003	Run cancelled (user or timeout cancellation)	No
1004	Run already exists (duplicate <code>runId</code>)	No
2xxx — Governance Errors		
2001	Command denied (not in allowlist or in denylist)	No
2002	Approval timeout (no approval received in time)	No
2003	Approval rejected (approver rejected the request)	No
2004	Insufficient approvals (min approval count not reached)	No
3xxx — Schema Errors		
3001	Schema validation failed (runbook does not match schema)	No
3002	Unknown step type (unrecognized <code>stepType</code>)	No
3003	Import cycle detected (circular runbook dependencies)	No
4xxx — Integration Errors		
4001	Tool not found (tool name does not resolve)	No
4002	Extension handshake failed (protocol error)	Maybe
4003	Input resolution failed (provider error or missing input)	Maybe
4004	Unsupported contract version	No
5xxx — Server Errors		
5001	Internal error (unexpected host failure)	Maybe
5002	Resource exhausted (rate limit or quota exceeded)	Yes
5003	Service unavailable (host shutting down or overloaded)	Yes

TABLE 14.2: Canonical error code registry for adapter contracts.

Chapter 15

Input Provider Framework

The input provider framework governs how gert resolves dynamic input values at run time. When a runbook declares an input with a `from:` binding, the provider framework locates the appropriate provider, dispatches a resolution request, and returns the concrete value before the first step that needs it begins.

15.1 Design Goals

- **Extensibility:** third-party systems (PagerDuty, ServiceNow, Vault, etc.) integrate as provider plugins without changes to the host binary.
- **Transparency:** every resolved value is recorded in the run trace so the operator can audit where each input came from.
- **Fallback safety:** resolution failures degrade gracefully to an interactive prompt rather than aborting the run silently.
- **Consistency with tools:** providers share the same transport layer as tools (`stdio`, `stdio-jsonrpc`, `mcp`), minimising implementation surface.

15.2 Provider Definition Schema

Each provider is declared in a `.provider.yaml` file. By convention, project-level providers live in `providers/<name>.provider.yaml`.

```
# providers/pagerduty.provider.yaml
apiVersion: provider/v2
```

```
meta:
  name: pd                                # Prefix used in from: bindings (e.g. pd.*)
  version: "1.0.0"
  description: "Resolves inputs from PagerDuty incidents and services"
  binary: pd-provider                     # Executable name (looked up in PATH)

transport:
  mode: stdio-jsonrpc                     # stdio | stdio-jsonrpc | mcp
  startup:
    argv: ["--config", "pd.yaml"]
    ready-pattern: "provider ready"
    timeout: "10s"
    shutdown-method: "shutdown"

resolution:
  method: provider/resolve                 # JSON-RPC method the host calls
  prefixes:
    - "pd."                               # All from: values starting with pd.
                                           # are dispatched to this provider

# Schema for what the provider needs in each resolution request
input-schema:
  type: object
  properties:
    bindings:
      type: object
      description: "Map of input name to from: binding string"
    context:
      type: object
      description: "Execution context (runId, userId, mode)"
  required: ["bindings"]

# Schema for what the provider returns
output-schema:
  type: object
```

```
properties:
  resolved:
    type: object
    description: "Map of input name to resolved string value"
  warnings:
    type: array
    items:
      type: string
  required: ["resolved"]

cache:
  enabled: true
  scope: run                # run | step | none
  ttl: "60s"

governance:
  requires-capabilities:
    - capability/network    # This provider needs outbound network
  env-read-patterns:
    - "PD_API_KEY"
    - "PD_*
```

15.3 Resolution Protocol

15.3.1 Resolution Flow

When the run engine encounters an input with a `from:` binding (e.g., `from: pd.incident.title`), the following steps occur:

1. **Prefix match:** the framework scans registered providers for one whose declared prefix matches the start of the binding string. The longest matching prefix wins.
2. **Provider start:** if the matched provider's process is not yet running, the host spawns it and waits for the ready signal (same mechanics as `stdio-jsonrpc` tools; see Section 6.3).

3. **Batch collection:** all inputs whose binding prefix maps to the same provider are grouped into a single resolution request to minimise round-trips.
4. **JSON-RPC call:** the host sends a `provider/resolve` request.
5. **Result merge:** resolved values are merged into the run's variable map.
6. **Fallback:** any binding that was not resolved (absent from the response's `resolved` map, or where the provider returned a warning) falls back according to the input's `fallback` field (default: interactive prompt).

15.3.2 Resolution Request Envelope

```
{
  "jsonrpc": "2.0",
  "id": "prov-1",
  "method": "provider/resolve",
  "params": {
    "bindings": {
      "incident_title": "pd.incident.title",
      "incident_severity": "pd.incident.severity",
      "service_name": "pd.service.name"
    },
    "context": {
      "runId": "run-abc123",
      "stepId": "__pre-flight__",
      "userId": "ops@acme.example",
      "mode": "real",
      "traceId": "trace-xyz"
    }
  }
}
```

15.3.3 Resolution Response Envelope

```
{
  "jsonrpc": "2.0",
  "id": "prov-1",
```

```

"result": {
  "resolved": {
    "incident_title": "Database connection pool exhausted",
    "incident_severity": "P1",
    "service_name": "payments-api"
  },
  "warnings": [],
  "telemetry": {
    "startedAt": "2026-04-18T10:00:00Z",
    "finishedAt": "2026-04-18T10:00:00.080Z",
    "durationMs": 80
  }
}
}

```

15.3.4 Resolution Error Handling

If the provider returns a JSON-RPC error, or if a binding key is absent from the `resolved` map, the host applies the input's fallback strategy:

```

# In runbook meta.inputs
inputs:
  incident_title:
    from: pd.incident.title
    description: "Active incident title"
    fallback: prompt          # prompt | fail | default
    default: "Unknown incident" # Used when fallback: default

```

- **fallback: prompt** (default): the host issues an interactive prompt for the value. In non-interactive modes (dry-run, CI) this causes the run to fail.
- **fallback: default**: the declared `default:` value is used without prompting. A warning is written to the trace.
- **fallback: fail**: the run aborts immediately with a structured error.

15.3.5 Resolution Caching

Providers may declare a cache policy in their definition. The host respects this policy:

- **scope:** `run` — the resolved value is reused for all subsequent steps in the same run that reference the same binding. This is the default for external providers.
- **scope:** `step` — the value is resolved fresh on each step that references it.
- **scope:** `none` — caching is disabled; every binding triggers a new RPC call.

Cache entries are keyed by `(runId, bindingString)`. Cached values are stored in the run's in-memory variable map only; they are not persisted across run resumptions.

15.4 Built-in Providers

The following providers are compiled into the gert host binary and are always available without a `.provider.yaml` file.

15.4.1 `env`

Reads values from environment variables.

```
inputs:
  api_key:
    from: env.ACME_API_KEY
    description: "API key for the Acme service"
```

The `env.` prefix is followed by the exact environment variable name. If the variable is not set, resolution fails and the fallback strategy is applied. The `env` provider never contacts an external process; it reads directly from the host's environment.

15.4.2 `file`

Reads values from files on the local file system.

```
inputs:
  ssh_key:
```

```
    from: "file ~/.ssh/id_rsa"
    description: "SSH private key"
  config_value:
    from: "file ./etc/gert/config.json#/database/host"
    description: "Database host from config file"
```

The path after `file.` is the file path. An optional JSON Pointer fragment (`#/path/to/field`) extracts a nested value from a JSON or YAML file. Without a fragment, the entire file contents are returned as a string.

15.4.3 prompt

The interactive prompt provider is the default resolution strategy. It is also the fallback for any failed provider resolution (unless overridden).

```
inputs:
  server_name:
    from: prompt
    description: "Server to check"
```

In `real` and `debug` modes, the host presents a terminal prompt to the operator. In `dry-run` and `replay` modes, the prompt provider reads from the scenario's `inputs.yaml` instead. In non-interactive environments where no scenario is available, the prompt provider fails with an informative error directing the operator to supply the value via `-var` or an explicit provider.

15.4.4 workspace

Reads values from the workspace configuration file (`.gert/config.yaml`).

```
inputs:
  default_region:
    from: workspace.defaults.region
    description: "AWS region from workspace config"
```

The path after `workspace.` is a dot-notation key into the workspace configuration object. This allows teams to centralise defaults without repeating them in every runbook.

15.5 Interactive Step Type Contracts

Input providers not only resolve **from:** bindings for runbook inputs, they also power the three interactive step types: **choice**, **decision**, and **collector**. When the run engine executes one of these steps, it dispatches a step-specific request to the configured input provider (or falls back to the built-in **prompt** provider).

15.5.1 Choice Step Contract

A **choice** step presents the operator with a fixed set of labeled options and stores the selected value as a variable. This is analogous to a dropdown menu or radio button group in a graphical form.

Request envelope. When the run engine executes a **choice** step, it sends a **provider/choice** JSON-RPC request to the configured provider:

```
{
  "jsonrpc": "2.0",
  "id": "choice-1",
  "method": "provider/choice",
  "params": {
    "stepId": "select-region",
    "prompt": "Select the target AWS region",
    "options": [
      { "value": "us-east-1", "label": "US East (N. Virginia)" },
      { "value": "us-west-2", "label": "US West (Oregon)" },
      { "value": "eu-west-1", "label": "Europe (Ireland)" }
    ],
    "default": "us-east-1",
    "context": {
      "runId": "run-abc123",
      "stepId": "select-region",
      "userId": "ops@acme.example",
      "mode": "real"
    }
  }
}
```

Response envelope. The provider returns the selected value:

```
{
  "jsonrpc": "2.0",
  "id": "choice-1",
  "result": {
    "selected": "us-west-2"
  }
}
```

Contract requirements.

- The `options` array must contain at least two elements.
- Each option must have both `value` (the string stored in the variable) and `label` (the human-readable text displayed to the operator).
- The `default` field is optional. If present, it must match one of the `value` fields in the `options` array.
- The provider's `selected` response must match one of the option values exactly. If the provider returns a value not in the options list, the run engine fails the step with a `choice.invalid_selection` error.
- Providers MAY render options in the order provided, or MAY sort them alphabetically by label for improved usability. The canonical order is declared in the runbook YAML.

Built-in prompt provider behaviour. The built-in `prompt` provider presents a numbered list in the terminal and waits for the operator to enter the number or value corresponding to their selection. In `dry-run` mode, it reads the value from the scenario's `inputs.yaml` or uses the `default` if provided.

15.5.2 Decision (Router) Step Contract

A `decision` step (also known as `router` in v1) presents the operator with a set of routing choices, where each choice directs execution to a different runbook or labeled section. Unlike `choice`, a `decision` is a control-flow primitive: the selected route determines which node the run engine transitions to next.

Request envelope. When the run engine executes a **decision** step, it sends a **provider/decision** JSON-RPC request:

```
{
  "jsonrpc": "2.0",
  "id": "decision-1",
  "method": "provider/decision",
  "params": {
    "stepId": "triage-severity",
    "prompt": "Select the incident severity level",
    "routes": [
      { "route": "p1-critical", "label": "P1: Critical (immediate)" },
      { "route": "p2-high", "label": "P2: High (1 hour SLA)" },
      { "route": "p3-normal", "label": "P3: Normal (24 hour SLA)" }
    ],
    "context": {
      "runId": "run-abc123",
      "stepId": "triage-severity",
      "userId": "ops@acme.example",
      "mode": "real"
    }
  }
}
```

Response envelope. The provider returns the selected route label:

```
{
  "jsonrpc": "2.0",
  "id": "decision-1",
  "result": {
    "route": "p2-high"
  }
}
```

Contract requirements.

- The **routes** array must contain at least two elements.

- Each route must have both **route** (the target node label or runbook ID) and **label** (the human-readable text).
- The provider does *not* need to know about the runbook graph structure. It only returns the selected route label. The run engine is responsible for resolving the route label to a target node and validating that the target exists.
- If the provider returns a route label not in the **routes** array, the run engine fails the step with a **decision.invalid_route** error.
- The decision result is *not* stored as a variable. It only controls execution flow. If the operator needs the decision recorded, the runbook author must add an explicit variable assignment in the target node.

Built-in prompt provider behaviour. The built-in **prompt** provider displays the routes as a numbered list and waits for the operator to select one. In **dry-run** mode, the route is read from the scenario file or the run fails if no scenario is provided.

15.5.3 Collector Step Contract

A **collector** step gathers unstructured input from the operator: free-form text, multi-line notes, file attachments, screenshots, URLs, or other evidence artifacts. The collected data is stored as named variables or artifacts in the run's state.

Request envelope. When the run engine executes a **collector** step, it sends a **provider/collect** JSON-RPC request:

```
{
  "jsonrpc": "2.0",
  "id": "collect-1",
  "method": "provider/collect",
  "params": {
    "stepId": "gather-evidence",
    "instructions": "Upload a screenshot of the error and describe the issue",
    "fields": [
      {
        "name": "screenshot",
```



```
    "type": "file",
    "label": "Error screenshot",
    "required": true,
    "accept": ["image/png", "image/jpeg"],
    "maxSizeBytes": 10485760
  },
  {
    "name": "description",
    "type": "text",
    "label": "Describe what you observed",
    "required": true,
    "multiline": true
  },
  {
    "name": "log_url",
    "type": "url",
    "label": "Link to related logs (optional)",
    "required": false
  }
],
"context": {
  "runId": "run-abc123",
  "stepId": "gather-evidence",
  "userId": "ops@acme.example",
  "mode": "real"
}
}
```

Response envelope. The provider returns the collected values and any uploaded artifacts:

```
{
  "jsonrpc": "2.0",
  "id": "collect-1",
  "result": {
    "values": {
```

```
    "description": "Database connection pool exhausted...",
    "log_url": "https://logs.acme.example/trace/xyz"
  },
  "artifacts": [
    {
      "field": "screenshot",
      "filename": "error-2026-04-18.png",
      "contentType": "image/png",
      "sizeBytes": 245631,
      "sha256": "a3c7d1f...",
      "storagePath": "artifacts/run-abc123/screenshot.png"
    }
  ]
}
```

Contract requirements.

- The `fields` array declares the form structure. Each field has:
 - `name`: the variable name where the value will be stored.
 - `type`: one of `text`, `file`, `url`, `number`, or `date`.
 - `label`: human-readable prompt text.
 - `required`: boolean flag.
 - Type-specific constraints: `accept` (MIME types for file uploads), `maxSizeBytes` (file size limit), `multiline` (for text fields).
- Providers **MUST** return all required fields in the `values` map. If a required field is missing, the run engine fails the step with a `collector.missing_required_field` error.
- For `type: file` fields, providers **MUST** upload the file to the artifact store and return metadata in the `artifacts` array. The artifact metadata includes:
 - `field`: the field name from the request.
 - `filename`: the original filename.

- `contentType`: MIME type.
 - `sizeBytes`: file size in bytes.
 - `sha256`: SHA-256 hash for integrity verification.
 - `storagePath`: path where the artifact is stored (relative to the run’s artifact root).
- File size limits are enforced by the provider. If an uploaded file exceeds `maxSizeBytes`, the provider **MUST** reject it with a `collector.file_too_large` error before storing it.
 - Supported MIME types for file uploads:
 - Images: `image/png`, `image/jpeg`, `image/gif`
 - Documents: `application/pdf`, `text/plain`, `text/markdown`
 - Archives: `application/zip`, `application/x-tar`, `application/gzip`
 - Logs: `text/plain`, `application/json`
 - Providers **MAY** support additional MIME types, but **MUST** respect the `accept` constraint if declared in the field definition.
 - The artifact storage path is provider-specific. The built-in `prompt` provider stores artifacts in `.gert/runs/<runId>/artifacts/`. External providers (e.g., S3, Azure Blob) **MAY** store artifacts in remote storage and return a signed URL or storage reference.

Built-in prompt provider behaviour. The built-in `prompt` provider displays each field as a terminal prompt. For `type: text` with `multiline: true`, it opens the operator’s configured editor (`$EDITOR`). For `type: file`, it prompts for a file path and copies the file to the local artifact store. In `dry-run` mode, all values are read from the scenario file.

15.5.4 Provider Capability Matrix

Not all providers support all interactive step types. The following table shows which step types each built-in provider supports:

Provider	Input Resolution	choice	decision	collector
prompt	Yes	Yes	Yes	Yes
env	Yes	No	No	No
file	Yes	No	No	No
workspace	Yes	No	No	No

External providers (PagerDuty, ServiceNow, Slack, etc.) MAY implement any combination of these contracts. A provider that implements **provider/choice** MUST declare this in its capability advertisement during the handshake phase:

```
# Example provider capability declaration
{
  "jsonrpc": "2.0",
  "method": "provider/initialize",
  "params": {
    "capabilities": {
      "resolution": true,
      "choice": true,
      "decision": false,
      "collector": true
    }
  }
}
```

If the run engine encounters a **choice**, **decision**, or **collector** step and the configured provider does not advertise the corresponding capability, the engine falls back to the built-in **prompt** provider.

15.6 Provider Composition

A **from:** binding may declare a priority-ordered chain of providers. The host attempts each provider in order and uses the first successful result.

```
inputs:
  api_token:
    from:
```

```
- env.ACME_API_TOKEN          # Try env var first
- vault.secret/acme/api-token # Fall back to Vault
- prompt                      # Final fallback: ask the operator
description: "API token for Acme service"
```

Chaining is evaluated left-to-right. A provider is considered to have failed if it returns an error, if the binding key is absent from the resolved map, or if the resolved value is an empty string (unless the input declares `allow-empty: true`).

15.7 Provider Lifecycle

Process model. Providers that use the `stdio-jsonrpc` or `mcp` transport are *persistent*: the host spawns each provider process once at the start of the pre-flight resolution phase and keeps it alive for the entire run. This avoids repeated startup latency for providers that need to authenticate or maintain session state (e.g., Vault token renewal, PagerDuty OAuth).

Providers using the `stdio` transport are *per-resolution*: the host spawns a new process for each resolution request.

Startup. Provider processes are started lazily on first use, following the same `ready-pattern` / timeout mechanism as `stdio-jsonrpc` tools (Section 6.3).

Shutdown. At run completion (success, failure, or cancellation) the host iterates over all running provider processes and sends the method declared in `transport.startup.shutdown-method` (default: `shutdown`). Providers are given a 5-second grace period to clean up before the host sends `SIGTERM`, then `SIGKILL` after a further 5 seconds.

Crash handling. If a provider process crashes during a resolution call, the host logs a structured `provider.crashed` trace event and applies the input's `fallback:` strategy for all bindings that were in-flight at the time of the crash.

Chapter 16

Observability and Diagnostics

Operational visibility is essential for production systems. This chapter specifies the observability model for gert v2: structured logging, metrics, distributed tracing, health probes, and diagnostics tooling. It is distinct from § 12 (Evidence, Tracing, and Resumption): **§12 specifies immutable audit records for compliance and resumption**; this chapter specifies **operational signals for operators** to answer questions like *is the system healthy?*, *what is slow?*, *where did this run fail?*, and *what is the resource utilization?*

The design follows the three-pillar observability model [5] adapted for workflow execution: metrics for aggregated performance, distributed tracing for execution flow, and structured logging for detailed context. All three are instrumented using industry standards (OpenTelemetry [18], W3C Trace Context [27], RFC 3339 timestamps [12]) to ensure interoperability with observability platforms.

16.1 Observability Model

Gert v2 applies the three-pillar observability model to workflow execution. Each pillar serves a distinct purpose:

16.1.1 Metrics: Aggregated Performance Indicators

Purpose: Track aggregate performance, throughput, error rates, and resource utilization over time.

Use Cases:

- *Alerting:* Trigger alerts when error rates exceed thresholds or latency degrades

- *Capacity Planning*: Measure run throughput and active concurrency to inform scaling decisions
- *SLO Monitoring*: Track percentile latencies (p50, p95, p99) for step execution times

Metric Types:

- **Counter**: Monotonically increasing (total runs, total errors)
- **Gauge**: Point-in-time measurement (active runs, loaded extensions)
- **Histogram**: Distribution of values (run duration, tool invocation latency)

Exposition Format: Prometheus OpenMetrics format exposed via HTTP endpoint (GET /metrics) when running in server mode (`gert serve -http`).

16.1.2 Distributed Tracing: Execution Flow Visualization

Purpose: Capture the hierarchical execution tree with timing and outcome for every run, step, and tool invocation.

Use Cases:

- *Root-Cause Analysis*: Identify which step in a failed run caused the failure
- *Latency Diagnosis*: Find which tool invocations are slow
- *Cross-System Correlation*: Link gert spans to HTTP API call traces via W3C Trace Context propagation

Implementation: OpenTelemetry SDK with OTLP exporter [18]. Spans emitted for every run (root span), step (child spans), and tool invocation (grandchild spans).

Span Hierarchy:

```
gert.run (run_id=...)
+- gert.step (step_id=backup-db, step_type=cli)
| +- gert.tool.invoke (tool=ssh, exit_code=0)
|   L- gert.governance.check (policy=allow-ssh)
+- gert.step (step_id=deploy, step_type=tool)
|   L- gert.tool.invoke (tool=kubectl, exit_code=0)
L- gert.step (step_id=verify, step_type=manual)
    L- gert.input.resolve (provider=approval-gate)
```

Trace Backends: Jaeger [7], Zipkin [29], Datadog, New Relic, AWS X-Ray (all OTLP-compatible).

16.1.3 Structured Logging: Detailed Contextual Events

Purpose: Emit detailed, machine-parseable log lines with run and step context for debugging and incident investigation.

Use Cases:

- *Debugging:* Search logs by `run_id` to see all activity for a specific run
- *Error Investigation:* View full error messages and stack traces in context
- *Compliance Audit:* Query logs for all runs by a specific actor or runbook

Log Format: JSON, one object per line, emitted to `stderr` by default.

Required Fields:

- `timestamp` (RFC 3339 [12])
- `level` (debug, info, warn, error)
- `msg` (human-readable message)
- `run_id` (when in run context)
- `step_id` (when in step context)

Aggregation: Loki, Elasticsearch, Splunk, CloudWatch Logs (any system that can parse JSON logs).

16.2 OpenTelemetry Integration

Gert v2 integrates OpenTelemetry as the standard tracing layer. All spans, metrics, and logs follow the OpenTelemetry specification [18] to ensure portability across observability backends.

16.2.1 Span Hierarchy and Attributes

OpenTelemetry spans form a hierarchical tree representing the execution structure. Each span level has required attributes for filtering and analysis.

Root Span: gert.run

The root span represents the entire run from start to completion.

Span Name: gert.run

Required Attributes:

```
runbook.id      = "deploy-app-v3"           // runbook identifier
runbook.version = "1.2.3"                   // version (if specified)
actor.id        = "alice@example.com"       // human or service principal
run.id          = "20260418T143022-a3f9c1"  // unique run ID
run.mode        = "real"                    // "real" | "replay" | "dry-run"
client.type     = "cli"                     // "cli" | "tui" | "web" | "mcp"
```

Span Status:

- OK if run outcome is **success**
- ERROR if run outcome is **failure**, **cancelled**, or **governance-blocked**

Child Span: gert.step

Each step execution produces a child span under **gert.run**.

Span Name: gert.step

Required Attributes:

```
step.id         = "backup-db"               // step identifier from YAML
step.type       = "cli"                     // "cli" | "manual" | "tool" | "invoke"
step.index      = 2                         // zero-based position in sequence
run.id          = "20260418T143022-a3f9c1"  // run correlation
step.name       = "Backup production database" // human-readable name
```

Optional Attributes:

```
step.outcome    = "success"                 // "success" | "failure" | "skipped"
step.retries    = 2                         // number of retry attempts
```

Span Events: Emit span events for key transitions:

- **governance.check.started** — governance evaluation started
- **governance.check.passed** — governance check passed

- `approval.requested` — human approval requested
- `approval.granted` — approval decision logged

Grandchild Span: `gert.tool.invoke`

Tool invocations create grandchild spans under `gert.step`.

Span Name: `gert.tool.invoke`

Required Attributes:

```

tool.name           = "kubectl"           // tool identifier
tool.transport      = "exec"              // "exec" | "ssh" | "http"
tool.exit_code      = 0                   // process exit code (exec only)
run.id              = "20260418T143022-a3f9c1" // run correlation
step.id             = "deploy"            // parent step

```

Optional Attributes:

```

tool.args           = ["apply", "-f", "deployment.yaml"] // redacted args
tool.duration_ms    = 1234                               // invocation latency

```

Grandchild Span: `gert.input.resolve`

Input provider resolutions create grandchild spans.

Span Name: `gert.input.resolve`

Required Attributes:

```

input.name          = "db_password"       // input key from runbook
provider.type       = "vault"             // provider type
provider.address    = "https://vault:8200" // endpoint (if applicable)
run.id              = "20260418T143022-a3f9c1"

```

Grandchild Span: `gert.governance.check`

Governance evaluations create grandchild spans.

Span Name: `gert.governance.check`

Required Attributes:

```

policy.name      = "allow-ssh"           // policy identifier
policy.outcome   = "allow"               // "allow" | "deny"
run.id           = "20260418T143022-a3f9c1"
step.id          = "backup-db"

```

16.2.2 Span Status and Error Handling

OpenTelemetry spans have three status codes:

- UNSET — default, no explicit status set
- OK — operation completed successfully
- ERROR — operation failed

Status Mapping:

- `gert.run` span:
 - OK if run outcome is `success`
 - ERROR if run outcome is `failure`, `cancelled`, `governance-blocked`
- `gert.step` span:
 - OK if step outcome is `success` or `skipped`
 - ERROR if step outcome is `failure`
- `gert.tool.invoke` span:
 - OK if `exit_code == 0`
 - ERROR if `exit_code != 0` or invocation crashes

When a span is marked `ERROR`, the runtime **MUST** record the error message as a span event or exception:

```

span.RecordError(err, trace.WithAttributes(
    attribute.String("error.type", "tool_invocation_failed"),
    attribute.String("error.message", err.Error()),
))

```

16.2.3 Context Propagation

Gert v2 propagates trace context across system boundaries using W3C Trace Context [27].

W3C Trace Context Headers

When a tool invokes an HTTP API, the runtime **MUST** inject W3C headers:

```
traceparent: 00-4bf92f3577b34da6a3ce929d0e0e4736-00f067aa0ba902b7-01
tracestate: gert=run_id:20260418T143022-a3f9c1
```

The `traceparent` header contains:

- `version` (00)
- `trace-id` (128-bit, globally unique)
- `parent-id` (64-bit span ID)
- `trace-flags` (sampling decision)

The `tracestate` header carries vendor-specific metadata. Gert includes `run_id` as baggage for cross-system correlation.

Baggage

The `run_id` is propagated as OpenTelemetry baggage to all child spans and external systems:

```
baggage.Set("run_id", runID)
```

This allows downstream systems to tag their own spans with the originating gert run ID, enabling end-to-end tracing across multi-system workflows.

16.2.4 OTLP Exporter Configuration

The runtime supports two trace export modes:

OTLP Export (Production)

Configure via environment variable or CLI flag:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://jaeger:4317
gert run deploy.yaml
```

Or:

```
gert run deploy.yaml --otel-endpoint=http://jaeger:4317
```

The runtime uses the OTLP/gRPC exporter by default. For HTTP, use:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://jaeger:4318
export OTEL_EXPORTER_OTLP_PROTOCOL=http/protobuf
```

Console Export (Development)

For local development without a trace backend:

```
gert run deploy.yaml --otel-console
```

This prints spans to `stderr` in a human-readable format.

16.2.5 Sampling Strategy

Gert v2 uses **parent-based sampling** with a configurable default rate:

- If an incoming request already has a trace context (e.g., gert invoked via HTTP from another traced system), **inherit the parent's sampling decision**.
- If no parent context exists, sample at the configured rate (default: 100% = always sample).

Configuration:

```
export OTEL_TRACES_SAMPLER=parentbased_traceidratio
export OTEL_TRACES_SAMPLER_ARG=0.1 # sample 10% of traces
```

For production systems with high throughput, consider 0.01 (1%) or 0.001 (0.1%) to reduce backend load.

16.3 Metrics Catalog

Gert v2 exposes the following metrics in Prometheus OpenMetrics format. All metrics are prefixed with `gert_`.

16.3.1 Run Metrics

Metric	Type	Labels	Description
<code>gert_runs_total</code>	Counter	<code>outcome</code> , <code>runbook_id</code>	Total run completions by outcome (<code>success</code> , <code>failure</code> , <code>cancelled</code>)
<code>gert_run_duration_seconds</code>	Histogram	<code>outcome</code> , <code>runbook_id</code>	End-to-end run duration (start to completion)
<code>gert_active_runs</code>	Gauge	—	Currently executing runs

TABLE 16.1: Run-level metrics

Histogram Buckets: `gert_run_duration_seconds` uses exponential buckets: 0.5, 1, 2, 5, 10, 30, 60, 120, 300, 600, 1800, 3600 (seconds).

16.3.2 Step Metrics

Metric	Type	Labels	Description
<code>gert_steps_total</code>	Counter	<code>outcome</code> , <code>step_type</code>	Total step completions by outcome and type
<code>gert_step_duration_seconds</code>	Histogram	<code>step_type</code>	Per-step duration by type (<code>cli</code> , <code>manual</code> , <code>tool</code> , <code>invoke</code>)

TABLE 16.2: Step-level metrics

Histogram Buckets: `gert_step_duration_seconds` uses buckets: 0.1, 0.5, 1, 2, 5, 10, 30, 60, 120, 300 (seconds).

16.3.3 Tool Invocation Metrics

Histogram Buckets: `gert_tool_duration_seconds` uses buckets: 0.01, 0.05, 0.1, 0.5, 1, 2, 5, 10, 30 (seconds).

Metric	Type	Labels	Description
<code>gert_tool_invocations_total</code>	Counter	<code>tool_name</code> , <code>outcome</code>	Tool invocation count by tool and outcome (success , failure)
<code>gert_tool_duration_seconds</code>	Histogram	<code>tool_name</code>	Tool invocation latency

TABLE 16.3: Tool invocation metrics

16.3.4 Approval Metrics

Metric	Type	Labels	Description
<code>gert_approval_wait_seconds</code>	Histogram	<code>step_id</code>	Time from approval request to decision (approved or rejected)
<code>gert_approvals_total</code>	Counter	<code>decision</code> , <code>step_id</code>	Total approval decisions (approved, rejected)

TABLE 16.4: Approval metrics

Histogram Buckets: `gert_approval_wait_seconds` uses buckets: 10, 30, 60, 300, 600, 1800, 3600, 7200, 14400, 28800, 86400 (seconds, up to 24 hours).

16.3.5 Extension Metrics

Metric	Type	Labels	Description
<code>gert_extension_crashes_total</code>	Counter	<code>extension_id</code>	Extension process crashes
<code>gert_extension_loaded</code>	Gauge	<code>extension_id</code>	1 if extension is loaded, 0 otherwise

TABLE 16.5: Extension metrics

16.3.6 Metrics Endpoint

When running in server mode, metrics are exposed at:

GET `http://localhost:8080/metrics`

Configure the server:

```
gert serve --http --addr=:8080
```

The response is in Prometheus text exposition format. Example:

```
# TYPE gert_runs_total counter
gert_runs_total{outcome="success",runbook_id="deploy-app-v3"} 42
gert_runs_total{outcome="failure",runbook_id="deploy-app-v3"} 3

# TYPE gert_run_duration_seconds histogram
gert_run_duration_seconds_bucket{outcome="success",runbook_id="deploy-app-v3",le="1"} 5
gert_run_duration_seconds_bucket{outcome="success",runbook_id="deploy-app-v3",le="5"} 18
gert_run_duration_seconds_bucket{outcome="success",runbook_id="deploy-app-v3",le="+Inf"} 42
gert_run_duration_seconds_sum{outcome="success",runbook_id="deploy-app-v3"} 187.4
gert_run_duration_seconds_count{outcome="success",runbook_id="deploy-app-v3"} 42

# TYPE gert_active_runs gauge
gert_active_runs 3
```

16.3.7 Push-Based Metrics (OTLP)

For environments where scraping is not feasible (e.g., ephemeral CI jobs), use OTLP metrics export:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://collector:4317
export OTEL_METRICS_EXPORTER=otlp
gert run deploy.yaml
```

The runtime pushes metrics to the OTLP endpoint at configurable intervals (default: 60 seconds).

16.4 Structured Logging

Gert v2 emits structured JSON logs to `stderr`. Every log line is a complete JSON object.

16.4.1 Log Line Format

```
{
  "timestamp": "2026-04-18T14:30:22.123456Z",
  "level": "info",
  "msg": "step execution started",
  "run_id": "20260418T143022-a3f9c1",
  "step_id": "backup-db",
  "step_type": "cli",
  "actor": "alice@example.com"
}
```

16.4.2 Required Fields

- **timestamp** — RFC 3339 format [12] with microsecond precision, UTC timezone
- **level** — one of: `debug`, `info`, `warn`, `error`
- **msg** — human-readable message describing the event

16.4.3 Contextual Fields (When Applicable)

- **run_id** — run identifier (present in all logs within run context)
- **step_id** — step identifier (present in all logs within step context)
- **step_type** — step type (`cli`, `manual`, `tool`, `invoke`)
- **actor** — user or service principal executing the run
- **tool_name** — tool identifier (for tool invocation logs)
- **error** — error message (for `error` level logs)
- **exit_code** — process exit code (for CLI step completions)

16.4.4 Log Levels

- **debug** — Detailed diagnostic information (e.g., input resolution, policy evaluation details)
- **info** — Operational milestones (e.g., step started, run completed)
- **warn** — Recoverable issues (e.g., retry attempt, fallback to default)
- **error** — Failures requiring attention (e.g., step failure, governance denial)

Configure via environment variable or CLI flag:

```
export GERT_LOG_LEVEL=debug
gert run deploy.yaml
```

Or:

```
gert run deploy.yaml --log-level=debug
```

Default: info.

16.4.5 Sensitive Data Handling

Log lines **MUST NOT** contain sensitive input values. The runtime applies the same redaction rules as the trace file (see § 12.4).

Safe:

```
{"msg": "resolved input 'db_password' from provider 'vault'"}
```

Unsafe:

```
{"msg": "resolved input 'db_password' = 'hunter2'"}
```

When logging errors that may contain sensitive data, scrub or omit the error message:

```
{"msg": "input resolution failed", "input_name": "db_password",  
  "provider": "vault", "error": "connection timeout"}
```

16.4.6 Log Output Configuration

By default, logs are written to `stderr`. To write to a file:

```
gert run deploy.yaml --log-output=file:/var/log/gert/gert.log
```

This writes logs to the file *in addition to* `stderr`.

For syslog output:

```
gert run deploy.yaml --log-output=syslog://localhost:514
```

16.4.7 Log Aggregation and Correlation

Every log line in run context includes `run_id`, enabling log aggregation by run:

Loki Query:

```
{job="gert"} | json | run_id="20260418T143022-a3f9c1"
```

Elasticsearch Query:

```
GET /gert-logs-*/_search
{
  "query": {
    "term": {"run_id": "20260418T143022-a3f9c1"}
  }
}
```

This returns all log lines for the specified run, ordered by timestamp.

16.5 Health and Readiness Endpoints

When running in server mode (`gert serve -http`), the runtime exposes HTTP endpoints for health probes. These are designed for Kubernetes liveness and readiness probes.

16.5.1 Liveness Endpoint

GET /health

Purpose: Indicates whether the gert process is alive.

Response (200 OK):

```
{
  "status": "ok",
  "version": "v2.0.0",
  "uptime_seconds": 3600
}
```

Use Case: Kubernetes liveness probe to restart crashed pods.

16.5.2 Readiness Endpoint

GET /ready

Purpose: Indicates whether the runtime is ready to accept new runs.

Response (200 OK):

```
{
  "status": "ready",
  "extensions_loaded": 3,
  "extensions_failed": 0
}
```

Response (503 Service Unavailable):

```
{
  "status": "not_ready",
  "reason": "extension 'aws-tools' failed to load",
  "extensions_loaded": 2,
  "extensions_failed": 1
}
```

Use Case: Kubernetes readiness probe to delay traffic until extensions are loaded.

16.5.3 Kubernetes Probe Configuration

```
apiVersion: v1
kind: Pod
metadata:
  name: gert-server
spec:
  containers:
  - name: gert
    image: gert:v2.0.0
    command: ["gert", "serve", "--http", "--addr=:8080"]
    livenessProbe:
      httpGet:
        path: /health
        port: 8080
      initialDelaySeconds: 5
      periodSeconds: 10
    readinessProbe:
      httpGet:
        path: /ready
        port: 8080
      initialDelaySeconds: 10
      periodSeconds: 5
```

16.6 Diagnostics Commands

Gert v2 provides CLI commands for self-diagnosis and trace inspection.

16.6.1 gert diagnose

Purpose: Run pre-flight checks to verify that the runtime environment is correctly configured.

Usage:

```
gert diagnose [--runbook=<path>]
```

Checks Performed:

1. **Runbook Schema Validation:** Parse and validate the runbook YAML against JSON Schema
2. **Tool Availability:** Verify that all declared tools (`.tool.yaml`) are executable and findable in `PATH` or `.runbook/tools/`
3. **Input Provider Connectivity:** Test that all input providers (`.provider.yaml`) can be reached (e.g., Vault is reachable, API keys are valid)
4. **Extension Loading:** Attempt to load all declared extensions and report any load failures
5. **Governance Policy Syntax:** Validate OPA policies (if policy-as-code is configured)

Output (Success):

- ✓ Runbook schema validation passed
- ✓ Tool 'kubectl' found at /usr/local/bin/kubectl
- ✓ Tool 'aws' found at /usr/local/bin/aws
- ✓ Input provider 'vault' reachable at https://vault:8200
- ✓ Extension 'aws-tools' loaded successfully
- ✓ Extension 'k8s-tools' loaded successfully

Diagnostics: PASS

Output (Failure):

- ✓ Runbook schema validation passed
- × Tool 'kubectl' not found in `PATH` or `.runbook/tools/`
- ✓ Tool 'aws' found at /usr/local/bin/aws
- × Input provider 'vault' unreachable: connection refused
- ✓ Extension 'aws-tools' loaded successfully
- × Extension 'k8s-tools' failed to load: missing dependency

Diagnostics: FAIL (3 errors)

Exit Code: 0 if all checks pass, 1 if any check fails.

16.6.2 gert trace show

Purpose: Pretty-print a completed run's trace from the JSONL file.

Usage:

```
gert trace show <run_id>
```

Output:

```
Run: 20260418T143022-a3f9c1
Runbook: deploy-app-v3
Actor: alice@example.com
Started: 2026-04-18T14:30:22Z
Completed: 2026-04-18T14:33:15Z
Duration: 2m53s
Outcome: success
```

Steps:

```
[0] backup-db (cli) - 12.3s - success
  L- tool: ssh - 12.1s - exit_code=0
[1] deploy (tool) - 45.2s - success
  L- tool: kubectl - 44.8s - exit_code=0
[2] verify (manual) - 2m10s - success
  L- approval: alice@example.com - granted
```

16.6.3 gert trace export

Purpose: Convert a JSONL trace to OpenTelemetry OTLP format for import into Jaeger or Zipkin.

Usage:

```
gert trace export <run_id> --format=otlp --output=trace.json
```

Supported Formats:

- `otlp` — OTLP JSON format
- `jaeger` — Jaeger JSON format
- `zipkin` — Zipkin JSON format

Use Case: Retrospectively analyze a run in a trace visualization tool when the runtime was not configured to export traces at execution time.

16.6.4 gert run list

Purpose: List recent runs with status.

Usage:

```
gert run list [--limit=N] [--json]
```

Output (Human-Readable):

RUN_ID	RUNBOOK	ACTOR	OUTCOME	DURATION
20260418T143022-a3f9c1	deploy-app-v3	alice@example.com	success	2m53s
20260418T140015-f3a7b2	rollback-v2	bob@example.com	failure	45s
20260418T135512-c9e8d1	backup-db	cron@system	success	3m12s

Output (JSON, for CI integration):

```
[
  {
    "run_id": "20260418T143022-a3f9c1",
    "runbook_id": "deploy-app-v3",
    "actor": "alice@example.com",
    "outcome": "success",
    "duration_ms": 173000,
    "started_at": "2026-04-18T14:30:22Z",
    "completed_at": "2026-04-18T14:33:15Z"
  },
  ...
]
```

Use Case: CI pipelines can parse JSON output to determine run status.

16.7 Integration Recipes

This section provides practical integration patterns for common observability platforms.

16.7.1 Prometheus + Grafana

Step 1: Start gert server

```
gert serve --http --addr=:8080
```


Step 2: Configure Prometheus scrape target

Add to `prometheus.yml`:

```
scrape_configs:
- job_name: 'gert'
  static_configs:
    - targets: ['localhost:8080']
  metrics_path: '/metrics'
  scrape_interval: 15s
```

Step 3: Import Grafana dashboard

Gert provides a reference Grafana dashboard JSON at `.runbook/observability/grafana-dashboard.json`.

Import it into Grafana to visualize:

- Run throughput (runs/minute)
- Success vs. failure rate
- p50, p95, p99 run duration
- Active runs (gauge)
- Step execution time by type
- Tool invocation latency

16.7.2 Jaeger (Distributed Tracing)

Step 1: Start Jaeger (all-in-one)

```
docker run -d --name jaeger \
-p 16686:16686 \
-p 4317:4317 \
jaegertracing/all-in-one:latest
```

Step 2: Configure gert to export traces

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://localhost:4317
gert run deploy.yaml
```

Step 3: View traces in Jaeger UI

Open `http://localhost:16686`, select service `gert`, and search for traces.

16.7.3 Loki + Grafana (Log Aggregation)

Step 1: Configure Promtail to scrape gert logs

Add to promtail-config.yaml:

```
scrape_configs:
  - job_name: gert
    static_configs:
      - targets:
          - localhost
        labels:
          job: gert
          __path__: /var/log/gert/*.log
    pipeline_stages:
      - json:
          expressions:
            timestamp: timestamp
            level: level
            msg: msg
            run_id: run_id
      - labels:
          level:
          run_id:
```

Step 2: Start gert with file logging

```
gert run deploy.yaml --log-output=file:/var/log/gert/gert.log
```

Step 3: Query logs in Grafana Explore

```
{job="gert"} | json | run_id="20260418T143022-a3f9c1"
```

16.7.4 Datadog / New Relic (OTLP-Compatible SaaS)

Both platforms support OpenTelemetry OTLP ingestion.

Datadog:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=https://api.datadoghq.com
export DD_API_KEY=<your-api-key>
gert run deploy.yaml
```

New Relic:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=https://otlp.nr-data.net:4317
export NEW_RELIC_API_KEY=<your-api-key>
gert run deploy.yaml
```

Consult platform documentation for exact endpoint URLs and authentication headers.

16.8 Observability for CI/CD Pipelines

Gert v2 provides machine-readable output for CI/CD integration.

16.8.1 JSON Output Mode

```
gert run deploy.yaml --output=json
```

On completion, emits a JSON summary to `stdout`:

```
{
  "run_id": "20260418T143022-a3f9c1",
  "outcome": "success",
  "step_count": 3,
  "failed_step": null,
  "duration_ms": 173000,
  "started_at": "2026-04-18T14:30:22Z",
  "completed_at": "2026-04-18T14:33:15Z"
}
```

Exit Code:

- 0 if outcome == "success"
- 1 if outcome == "failure"
- 2 if outcome == "governance-blocked"
- 3 if outcome == "cancelled"

16.8.2 GitHub Actions Example

```
– name: Run gert runbook
  id: gert_run
  run: |
    OUTPUT=$(gert run deploy.yaml —output=json)
    echo "$OUTPUT" | jq .
    RUN_ID=$(echo "$OUTPUT" | jq -r .run_id)
    OUTCOME=$(echo "$OUTPUT" | jq -r .outcome)
    echo "run_id=$RUN_ID" >> $GITHUB_OUTPUT
    echo "outcome=$OUTCOME" >> $GITHUB_OUTPUT

– name: Upload trace artifact
  if: always()
  uses: actions/upload-artifact@v3
  with:
    name: gert-trace
    path: .runbook/runs/${{ steps.gert_run.outputs.run_id }}/trace.jsonl
```

16.8.3 GitLab CI Example

```
deploy:
  script:
    – gert run deploy.yaml —output=json | tee run-output.json
    – RUN_ID=$(jq -r .run_id run-output.json)
    – echo "Run ID: $RUN_ID"
  artifacts:
    when: always
    paths:
      – .runbook/runs/
  reports:
    junit: .runbook/runs/*/trace.jsonl
```

16.9 Summary

This chapter specifies a production-grade observability model for gert v2 grounded in industry standards:

- **Three-pillar model:** Metrics (Prometheus), distributed tracing (OpenTelemetry), structured logging (JSON)
- **Span hierarchy:** `gert.run` $\rightarrow$ `gert.step` $\rightarrow$ `gert.tool.invoke` / `gert.governance.check` / `gert.input.resolve`
- **Context propagation:** W3C Trace Context headers and OpenTelemetry baggage for cross-system correlation
- **Metrics catalog:** 9 metrics covering runs, steps, tools, approvals, and extensions
- **Health endpoints:** `/health` (liveness), `/ready` (readiness), `/metrics` (Prometheus)
- **Diagnostics commands:** `gert diagnose`, `gert trace show`, `gert trace export`, `gert run list`
- **Integration recipes:** Prometheus, Grafana, Jaeger, Loki, Datadog, New Relic
- **CI/CD support:** JSON output mode with exit codes for pipeline integration

Operators gain visibility into run health, performance bottlenecks, and failure root causes. The design ensures portability across observability platforms by adhering to open standards [18, 27, 5].

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